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FILTERED LIE ALGEBROIDS AND PARABOLIC GEOMETRY

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ABSTRACT. This work is a novel exposition of the basic elements of parabolic geometry, a rich class of finite type structures generalizing conformal and projective structures. We introduce filtered Lie algebroids, a generalization of Tohru Morimoto’s filtered manifolds, and we propose abstractions of parabolic geometry based on Lie groupoids whose associated Lie algebroids are filtered. Moreover, we present a new prolongation procedure to recover the abstraction from the underlying geometric structure, using semi-direct products in the Lie groupoid category. Our construction is predicated on the observation that Lie groupoids with filtered Lie algebroids admit a semi-direct product structure whenever they also admit a Weyl gauge, an “oid” version of lifts of grading elements of a Lie algebra.

Our approach allows for the parallel treatment of infinitesimal and global aspects, the simpler infinitesimal aspect typically informing constructions in the global case. Furthermore, we establish, in the parabolic case, that the global Cartan connection is completely determined by infinitesimal data, a fact that is shared by some other, but not all, Cartan geometries.

A parabolic geometry is essentially a deformed generalized flag manifold G/H where, in previous formulations, G is assumed to be semisimple. Here G can be any Lie group whose Lie algebra is reductive.

1. INTRODUCTION

Parabolic geometry is an abstract framework that unifies a rich collection of finite-type geometric structures, such as conformal, projective, contact projective, CR,¹ almost Grassmannian, almost quaternionic, conformal Lorenzian, almost Lagrangean, Lagrangean contact, Lie contact structures, finite-type n -plane fields, and many others [12, Chapter 4]. In the first place, it provides for a unified treatment and interpretation of “curvature”, and among its important applications is the systematic derivation of invariant differential operators [29, 9, 30, 31].

These notes are an outgrowth of our attempts to tame some of the theory’s more technical aspects by approaching it from new points of view. We prove the basic existence, normalization and uniqueness theorems, culminating in a solution to the equivalence problem for a large subclass of parabolic geometries. This treatment is suitable for non-experts well versed in Lie groupoids and Lie algebroids, but we also flag points of difference and novelty for parabolic geometry experts. Our

¹non-degenerate and partially integrable [12, p.414].

results on firmly filtered Lie groupoids/algebroids and our version of prolongation are new.

1.1. The main new ideas. According to Cartan’s generalization of Klein’s El-rangen program [24], geometric structures of finite type are viewed as “symmetries deformed by curvature”. In the contemporary formulation, the object deformed is the principal bundle associated with some homogeneous space, G/H , as we describe further in 1.2 below. The parabolic geometries arise when $H \subset G$ is a parabolic subgroup and G is semisimple (or even reductive, as we will show here). Such homogeneous spaces, known as generalized flag manifolds, are in rich supply. The alternative approach promoted here is to consider deformations of action groupoids, which in the transitive case have the form $G \times M$, $M \cong G/H$. In this approach, the crucial idea of *prolongation* manifests in a most natural way.

For the sake of concreteness, consider the conformal sphere S^n , a generalized flag manifold with action groupoid $\mathrm{SO}(n+1, 1) \times S^n$. Now, differentiating conformal automorphisms, i.e., elements of $\mathrm{SO}(n+1, 1)$, we obtain a Lie groupoid morphism

$$(1) \quad \mathrm{SO}(n+1, 1) \times S^n \twoheadrightarrow \mathrm{CSO}[TS^n]$$

onto the frame groupoid $\mathrm{CSO}[TS^n]$, consisting of conformal isomorphisms between tangent spaces. The kernel of this morphism is a Lie group bundle with commutative, simply-connected fibres. By arguments recalled later, it has a natural identification with T^*S^n . A fundamental observation, not immediately obvious, but following from the general results presented here, is that (1) splits (non-canonically) delivering an identification of $\mathrm{SO}(n+1, 1) \times S^n$ with the Lie groupoid semi-direct product $\mathrm{CSO}[TS^n] \times T^*S^n$. This is significant, for in the curved case of an arbitrary conformal manifold M , where the action groupoid is gone, *a proxy for the action groupoid persists*, namely the semi-direct product $\mathrm{CSO}[TM] \times T^*M$. This is what we call the **PROLONGATION** of the conformal structure.

In the flat case, we can give the kernel T^*S^n the interpretation of all 2-jets of conformal automorphisms vanishing to first order. Relevant arguments can be found in [1]; curved analogues are offered in Section 6. This connects prolongation to the usual idea of introducing derivatives as new variables. The reader is warned, however, that prolongations appearing in the Cartan geometry literature, and prolongations in the sense of PDE’s, generally diverge in the *curved* case, and this is true here also.

To make the prolongation $\mathrm{CSO}[TM] \times T^*M$ a formal parabolic geometry, as we shall define the notion here, it suffices to equip its Lie algebroid A with a suitable $\mathfrak{so}(n+1, 1)$ -bundle structure (namely, one extending the canonical Lie algebra bundle structure on its anchor kernel) and this structure, which we call **MODEL DATA**, is identifiable with the trivial $\mathfrak{so}(n+1, 1)$ -bundle structure of the action algebroid $\mathfrak{so}(n+1, 1) \times S^n \cong A$ in the flat case. As we will see in 3.7, the Lie algebroid A of $\mathrm{CSO}[TM] \times T^*M$ has a natural filtration $A = A^{-1} \supset$

$A^0 \supset A^1 = T^*M$, and while the Lie algebra bundle structure on the associated graded $\mathrm{gr}(A)$ determined by the bracket on A is *not* modeled on $\mathfrak{so}(n+1, 1)$, there is a second, related Lie algebra bundle structure that *is*. Model data on A is constructed by finding a canonical way to lift this latter Lie algebra bundle structure from $\mathrm{gr}(A)$ to A . A “canonical” lift is one that suitably normalizes the corresponding MODEL CURVATURE, the local obstruction to the existence of $\mathrm{SO}(n+1, 1)$ -symmetry. To understand what normalization should mean, one hooks into the algebraic machinery introduced by Bertram Kostant for studying certain Lie algebra (co)homology spaces associated with a reductive Lie algebra and parabolic subalgebra.

More generally, prolongation allows us to recover, in the typical case, an abstract parabolic geometry from its “underlying geometric structure”, of which a conformal structure is but one example. In the conventional approach, the underlying structure is formalized as an “infinitesimal flag structure” [12, Definition 3.1.6, p. 248]. For reasons explained at the end of Section 8, we prefer to call the Lie groupoid version of these objects LEVI STRUCTURES.

Our version of prolongation is anticipated by the observation that natural candidates for parabolic geometries are certain Lie groupoids that *always* admit a semi-direct product structure. Indeed, splittings like the one above exist for a large class of Lie groupoids, even intransitive ones, provided they admit what we call a Weyl gauge. As a special case, we can show that the action groupoid of any generalized flag manifold is a semi-direct product (15.1, Corollary C), a fact that does not seem to have been noticed before. Weyl gauges are discussed further in 1.4 below.

Various algebraic prolongation techniques leading to the solution of the equivalence problem for parabolic geometries, which have a long and complicated history [12, Appendix A], were discovered by Tanaka [26], Morimoto [22] and Čap and Schichl [28]. The conceptually simple approach sketched above is probably the most noteworthy novelty of the present exposition. The idea, in the present context, of replacing principal bundles with Lie groupoids, dates back to Ehresmann’s 1959 paper [17].² While adopting a different approach, our work remains strongly guided by Čap and Slovák’s seminal monograph [12] and our debt to that work necessarily extends beyond those places explicitly noted.

1.2. The contemporary point of view. In the principal bundle formulation, which originates in Ehresmann’s earlier 1951 paper [16], a Cartan geometry, modeled on some homogeneous space G/H , is a smooth manifold M and a principal H -bundle P over M , equipped with: (i) A parallelism, understood as a V -valued one-form ω on P (the CONNECTION), required to be suitably compatible with the principal bundle structure; and (ii) A compatible identification of the vector space

²At this time the infinitesimal concept of a Lie algebroid had not been formalized, and this concept did not mature substantially until Jean Pradines’ work in the late 1960’s. Perhaps this explains why the groupoid approach did not immediately gain traction.

V with the infinitesimalization $\mathfrak{g}$ of G (the MODEL DATA). Combining (i) and (ii), we can view ω as $\mathfrak{g}$ -valued one-form, which is the object commonly known as a CARTAN CONNECTION. The precise definition is recalled in 5.4.

In the case of a principal H -bundle $G \rightarrow G/H$, the Maurer-Cartan form ω is a Cartan connection, and in the general case there is a notion of curvature, $d\omega - [\omega, \omega]_{\mathfrak{g}}$, which vanishes in the homogeneous case, by virtue of the Maurer-Cartan equations for G . Since it depends on the choice of model data (it involves the bracket on the Lie algebra $\mathfrak{g}$ of G) we refer to this as the MODEL CURVATURE; it's significance is that M is locally identifiable with the model G/H whenever the model curvature vanishes. (As explained later, an alternative notion of curvature, which does not depend on model data, but is nevertheless an obstruction to symmetry, will play an ancillary role in the present exposition.)

The connection and model data allow one to transfer certain structures from the model space, such as a G -invariant quadratic form on $\mathfrak{g}/\mathfrak{h}$, to structures on the underlying manifold M , for example a Riemannian metric, and this latter constitutes the “underlying geometric structure” of the Cartan geometry. In the parabolic case, the Lie algebra $\mathfrak{h}$ of H defines a filtration of $\mathfrak{g}$ which, among other things, implies that TM is non-trivially filtered in the typical case. A simple example is CR-geometry, where this filtration is defined by the contact structure $\mathcal{H}$, namely $0 =: T^0M \subset T^{-1}M := \mathcal{H} \subset T^{-2}M := TM$.

Different choices of ω may lead to the same underlying geometric structure, which leads to the first fundamental problem of the theory, how to choose ω so as to “normalize” the model curvature, for then ω (and its curvature, for example) encode basic invariants of the underlying geometry. The second fundamental problem is how to reconstruct an abstract Cartan geometry from the underlying geometric structure. Where this is possible one obtains a one-to-one correspondence between (isomorphism classes) of normalized Cartan geometries and the concrete geometries of interest, solving the “equivalence problem” for those structures. The technical tool for making this reconstruction is known as prolongation, because of its roots in the procedure in the theory of PDE's of the same name.

1.3. The new approach. In [22] Tohru Morimoto introduced a novel prolongation technique to prove existence and normalization results for Cartan connections in parabolic geometry. In it he formulates the notion of filtered manifolds, now a part of the contemporary formalism [12]. The filtration of TM for CR-geometry above is a simple example. Morimoto also demonstrates the usefulness of separating the notions of connection and model data, a theme of our own work cited shortly, and the present exposition, but not emphasized more generally.

Rather than pursuing Morimoto's original strategy, however, we synthesize an approach based on a new analysis of “firmly” filtered Lie algebroids, a natural generalization of Morimoto's filtered manifolds. In this recounting, Lie groupoids, whose corresponding infinitesimalizations are filtered Lie algebroids, replace the principal bundles of the conventional approach, along the lines intimated in 1.1.

Previous work of ours, reviewed in Section 4, shows how to view Lie algebroids or Lie groupoids as (possibly intransitive) “symmetries deformed by curvature”, by equipping them with connections *without* model data [2, 3, 4, 6, 5] (but nevertheless, and perhaps unfortunately, called Cartan connections). In the groupoid case, the connection is a multiplicatively closed horizontal n -plane field, D , and in the infinitesimal case, it is a Koszul connection, ∇ , suitably compatible with the Lie algebroid structure. The type of symmetry, if it exists, is not specified *a priori*, and the natural definition of curvature (the usual one for ∇) does not depend on a notion of model data. Here we introduce, for the first time in the “oid” context, a notion of compatible model data, and of model curvature, in which the model is specified ahead of time, as in the standard approach. With these basic ingredients we can imitate the standard theory, with some further novelties discussed below,

1.4. Weyl gauges and semi-direct products. The general considerations leading to the Lie groupoid splittings alluded to in 1.1 can be previewed here by considering the special Lie algebra case of our Theorem 10.2, a theorem applying to general Lie algebroids (globalized to Lie groupoids by Theorem 10.3):

Theorem. *Let $\cdots \supset \mathfrak{a}^{-1} \supset \mathfrak{a}^0 \supset \mathfrak{a}^1 \supset \mathfrak{a}^2 \supset \cdots$ be a filtration³ of some finite-dimensional Lie algebra $\mathfrak{a}$, and assume the graded Lie algebra $\mathrm{gr}(\mathfrak{a}^0)$, associated with $\mathfrak{a}^0$, admits a grading element E , necessarily lying in $\mathrm{gr}(\mathfrak{a})_0$. Moreover, assume that E has a lift $\epsilon \in \mathfrak{a}^0$ such that*

$$(1) \quad [\mathfrak{a}, \epsilon] \subset \mathfrak{a}^1.$$

Then $\mathfrak{a}^j \subset \mathfrak{a}$ is an ideal for all $j \geq 0$ and $\mathfrak{a}$ is isomorphic to the semi-direct product $\mathfrak{a}/\mathfrak{a}^1 \oplus_{\mathrm{ad}} \mathrm{gr}(\mathfrak{a}^1)$, the isomorphism depending on the choice of ϵ .

Recall here that an element E of a graded Lie algebra $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$ is a **GRADING ELEMENT** if $[E, X] = jX$, whenever $X \in \mathfrak{g}_j$. Note that the theorem requires only a grading element of $\mathrm{gr}(\mathfrak{a}^0)$ and not $\mathrm{gr}(\mathfrak{a})$. When the additional “invariance condition” (1) is satisfied, we call ϵ a **WEYL GAUGE**.

If $\mathfrak{a}$ is filtered and the condition about ideals is also satisfied, necessary for the existence of a Weyl gauge, we say that $\mathfrak{a}$ is **FIRMLY FILTERED**. This is a strong condition, but the right one for our general setting, as we shall see already for homogeneous spaces G/H and other examples in Section 3. The theorem above has the following corollary:

A filtered Lie algebra $\mathfrak{a}$ admits a Weyl gauge if and only if it is firmly filtered and $\mathrm{gr}(\mathfrak{a}^0)$ admits an $\mathfrak{a}$ -invariant grading element E .

The above theorem does *not* apply to the canonical filtrations associated with the parabolic subalgebras of a reductive (or semisimple) Lie algebra (see 7.1) as these cannot be firmly filtered. The point is, rather, that Weyl gauges have generalizations to filtered Lie algebroids and Lie groupoids (even intransitive ones) including the action algebroids/groupoids of generalized flag manifolds. For us,

³See 3.1 for precisely what we mean here.

a parabolic geometry is a firmly filtered, transitive Lie groupoid, together with suitably compatible model data (Definition 7.5). This model data identifies each tangent space $T_m M$ (non-canonically) with a Lie algebra quotient $\mathfrak{g}/\mathfrak{p}$, where $\mathfrak{g}$ is reductive and $\mathfrak{p} \subset \mathfrak{g}$ is parabolic. These assumptions on $\mathfrak{g}/\mathfrak{p}$ guarantee the existence of Weyl gauges, by a generalization of the italicized statement above. In the above super-simplification of the general scenario, $\mathfrak{a}/\mathfrak{a}^1$, together with its representation $\mathrm{gr}(\mathfrak{a}^1)$, play the role of the corresponding Levi structure.

The name “Weyl gauge” comes from the Weyl structures/connections in the standard theory of parabolic geometry [12], which generalize Weyl’s own connections in conformal geometry. It may be there exists a correspondence between the new semi-direct product description and the constructions in [12] but this is not obvious to us. In any case, Weyl gauges here inform our notion of prolongation (and make sense before introducing any assumptions about model data, such as semisimplicity) while in [12] prolongation is first developed without Weyl structures. In our case the semi-direct product structure of a prolongation leads to perhaps the most explicit description of parabolic geometries to date — in fact, of all transitive filtered Cartan geometries, a mild generalization (Theorem 12.6).

1.5. Infinitesimal geometry. An obvious advantage of the Lie groupoid approach is a direct relationship between global and infinitesimal representations of the geometry, namely a natural generalization of the relationship between Lie groups and Lie algebras. Indeed, our development generally begins at the infinitesimal level, where constructions are easier to understand (and how we generally discovered them) and we only globalize them later. There is even the possibility, not pursued here, of completely reducing the global theory to the infinitesimal one, using the Lie groupoid versions of Lie I and Lie II [13], provided one is happy to ignore certain topological distinctions between geometries blurred by restricting to the case of source-simply-connected Lie groupoids. For example, in such an approach, one is not concerned with the subtle differences between oriented conformal geometry, unoriented conformal geometry, and conformal spin geometry.

In the principal bundle approach, it is conventional to view something called the the adjoint tractor bundle $\mathcal{A}$, as the corresponding infinitesimal object. For example, the model curvature of a parabolic geometry is there understood as an $\mathcal{A}$ -valued two-form. This seems natural because tractor bundles are associated bundles. In the Lie groupoid point-of-view the analogue of associated bundles are Lie groupoid representations, but there is no canonical representation of a Lie groupoid on its Lie algebroid. Stated differently, the Atiyah Lie algebroid of a principal bundle is not an associated bundle; it only becomes one (is identifiable with $\mathcal{A}$) *after* choosing a Cartan connection. This essential difference between the two approaches has a related manifestation. As we recall in Section 4, a Cartan connection D on a Lie groupoid immediately infinitesimalizes to a Koszul connection ∇ on the corresponding Lie algebroid, without model data playing any

role, but it seems this connection does not appear in [12], at least not explicitly; rather the focus is on the so-called tractor connection on $\mathcal{A}$, which depends on its associated bundle structure. While the tractor connection has an analogue in the present approach, the infinitesimal Cartan connection ∇ and the adjoint tractor connection are different, except in the flat case. See 14.5(4).

1.6. Model data. Important ingredients of the present approach are the “oid” versions of classical Cartan connections, D and ∇ mentioned above. In [3] we outlined an ambitious program for constructing such Cartan connections for general geometric structures of finite type, without the need to prescribe a flat model *a priori*, using a direct PDE prolongation procedure. While some simple structures are amenable to this approach, the program has largely been unsuccessful. In the case of parabolic geometries, the specification of model data *a priori* appears to be essential: It provides a hook into the rich structure theory of reductive and semisimple Lie algebras, simplifying the normalization process, which seems intractable in the earlier PDE approach.

The notion of model data in the Lie groupoid case is different from the classical notion because it makes sense without reference to the Cartan connection D (which corresponds to the parallelism in Ehresmann’s definition). Model data identifies each tangent space $T_m M$ (non-canonically) with a Lie algebra quotient $\mathfrak{g}/\mathfrak{h}$, and if $\mathfrak{h} \subset \mathfrak{g}$ is a *separating* Lie subalgebra (true for all parabolic subalgebras when $\mathfrak{g}$ is reductive) then any compatible Cartan connection is completely determined by the model data. That is, *the Cartan geometry is completely recoverable from purely infinitesimal data*, another fact that does not appear to have been noticed before. The notion of a separating subalgebra is recalled in 5.5.

1.7. Outline. Section 3 introduces filtered Lie algebroids and the firmly filtered special case, and gives examples pertinent to the application to parabolic geometry. Section 5 generalizes the classical notion of a Cartan geometry to the intransitive case, by formulating the notion using Lie groupoid language. In 5.5 we recall the notion of separating Lie algebras, and show that Cartan geometries with “separating” model data (which include parabolic geometries) have some nice properties. Section 7 formulates the Lie groupoid abstraction of parabolic geometries, and their reductions, Levi structures, and states the main theorems on existence, normalization, uniqueness, and equivalence, whose proofs are laid out in the remaining sections.

Section 10 generalizes the notion of Weyl gauges and the semi-direct product structure theorem stated in 1.4 above. Ambiguities in the choice of gauge lead to the notion of Weyl gauge transformations, relevant to the uniqueness theorem for parabolic geometry, but these transformations are introduced in Section 7 under the guise of Lie groupoid conjugations, which are more concrete. So Section 10 concludes with a proof that these amount to same thing.

Section 4, which is logically independent of the preceding section, is mostly a summary of our aforementioned work on Cartan connections on Lie groupoids and algebroids, in the absence of model data. Model data in the “oid” context is introduced in Section 11. A crucial technical result established in Section 11 is a method for constructing model data compatible with a given Cartan connection (Proposition 11.1).

Section 12 applies the semi-direct product theorems of Section 10 to give a characterization of transitive filtered Cartan geometries, a certain generalization of parabolic geometries: All such geometries are obtained by prolonging a Levi structure and making use of the following additional data: a splitting of the underlying filtered manifold, a splitting of the anchor of the Levi structure’s Lie algebroid, and a Rho tensor, which will be familiar from the standard theory.

Section 13 introduces ψ -deformations, which are a natural collection of transformations of a Lie groupoid or Lie algebroid which we can use to relate different choices of model data, via push-forward. Unlike Weyl gauge transformations, these are necessarily *not* morphisms. They are defined by objects ψ which we view as sections of the kernel of $J^1\mathcal{G} \rightarrow \mathcal{G}$, where $J^1\mathcal{G}$ is the groupoid of one-jets of the underlying Lie groupoid $\mathcal{G}$ of the parabolic geometry, an interpretation that helped us to see how a compatible Cartan connection should transform to ensure compatibility with the ψ -deformed model data. This section concludes by establishing a relationship between ψ -deformations and Weyl gauge transformations that we need for uniqueness results.

Section ?? derives various properties of curvature, such as invariance under Lie algebroid automorphisms, and Bianchi’s identity, and lays out the algebraic machinery of Kostant [19] regarding the adjoint representation of a parabolic subalgebra’s reductive Levi subalgebra on the graded reductive Lie algebra that contains it. Actually, Kostant’s work applies in the semisimple case, but what we need from it readily generalizes to reductive Lie algebras. This is the crucial algebraic ingredient for normalizing model curvature, which can be viewed as a 2-chain in a related complex of vector bundles. Subsection 14.1 onwards is mostly a reworking of parts of [12, §3.1]. The main point is Proposition 14.2, expressing the change in model curvature upon ψ -deforming the model data, in terms of a cochain complex differential. We conclude with a brief discussion of tractor bundles, so as to relate our Cartan connections on Lie algebroids with adjoint tractor connections.

In Section 15 we provide the remaining details needed to prove the results stated in Section 3.

Audience. We assume readers are familiar with Lie groupoids and Lie algebroids — say, what is briskly summarized in the first two lectures of [13]. Other sources include [11, 21, 15] and the book [20]. In addition, parts of [5] describing the generalization of the adjoint action to Lie groupoids, may be helpful.

We circumvent a lot of Lie algebra structure theory, and obtain our generalizations to the reductive case, by adopting the very nice point of view of parabolic subalgebras in [10]. Essential parts of this description are outlined in Appendix A.

Blanket assumptions. All manifolds are smooth, which means C^∞ or real-analytic, and have connected components sharing a common dimension. With the exception of Lie groupoids, manifolds are Hausdorff. All manifolds are second countable. In particular, all transitive Lie groupoids are automatically locally trivial (source maps are both surjective and submersive) [27, Proposition A.3] and so are gauge groupoids (groupoids defined by a principal bundle).

The kernel of the anchor of a Lie algebroid, which is variously known as the isotropy, vertex bundle, or adjoint bundle, is here called the ANCHOR KERNEL.

Acknowledgements. The author is indebted to Jan Slovák and Andreas Čap for clarifying several arguments in [12]. He thanks David Calderbank for many helpful discussions, for walking the author through [10] in detail, and for invaluable encouragement. The proof of Proposition A.7 is largely due to Professor Calderbank.

2. TABLE OF COMPATIBILITY CONDITIONS

A number of compatibility conditions defined in the text are summarized in the table below. Here:

- $A^\bullet$ is a filtration of some Lie algebroid A (see 3.1).
- D is a Cartan connection on some Lie groupoid integrating A (see 4.1).
- ∇ is an infinitesimal Cartan connection on A (see 4.2).
- τ is model data on A (see 5.1).
- ϵ is a Weyl gauge for A or an integrating Lie groupoid (see 10.2 and 10.3).

	D	∇	τ	ϵ
$A^\bullet$	A^j is $\bar{D}$ -invariant $\forall j$	A^j is $\bar{\nabla}$ -invariant $\forall j$	$\tau(A^j, A^k) \subset A^{j+k}, \forall j, k$	-
D		-	C1 and c2 (see below)	-
∇			c1 and c2 (see below)	-
τ				$\text{gr}_0(\epsilon)$ is grading section of $\text{gr}_0(\tau)$

The conditions c1, c2 and C1, from 5.2, are reproduced below:

c1. τ is $\bar{\nabla}$ -invariant (where ∇ is the infinitesimalization of D in the global case).

c2. $\bar{\nabla}_Z X = \tau(Z, X)$ whenever $\#Z = 0, X \in A$.

C1. τ is $\bar{D}$ -invariant,

Here $\# : A \rightarrow TM$ denotes the anchor.

When c2 is dropped, we have WEAK COMPATIBILITY.

3. FILTERED LIE ALGEBROIDS

In this section we introduce filtered Lie algebroids and the important special case of *firmly* filtered Lie algebroids, and give examples which include observations foreshadowing aspects of parabolic geometry to be presented in later sections.

3.1. Definition. Here a filtration

$$\cdots A^{-1} \supset A^0 \supset A^1 \supset A^2 \supset \cdots$$

of some vector space or vector bundle A will always be understood to satisfy $\cap_j A^j = 0$ and $\cup_j A^j = A$. All filtrations are descending. If A is a vector bundle, then we call the filtration a **LIE ALGEBROID FILTRATION** if A is a Lie algebroid and:

F1. The induced filtration $\cdots \Gamma(A^{-1}) \supset \Gamma(A^0) \supset \Gamma(A^1) \supset \Gamma(A^2) \supset \cdots$ of the section space $\Gamma(A)$ is a filtration of Lie algebras, meaning $[X, Y] \in \Gamma(A^{j+k})$ for $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$.

F2. The kernel of the anchor contains A^0 .

A filtered Lie algebroid A is **FIRMLY FILTERED** if A^0 and the filtration of A^0 are invariant under the canonical representation of A on its anchor kernel, i.e., if $\Gamma(A^j) \subset \Gamma(A)$ is an ideal for all $j \geq 0$.

Proposition. *Let A be a filtered Lie algebroid. Then:*

- (1) *The associated graded vector bundle $\text{gr}(A)$ is a bundle of graded Lie algebras (a smooth field of graded Lie brackets).*
- (2) *If A is firmly filtered, then $\text{gr}(A)$ is, fibre-wise, the Lie algebra direct sum of its negative and non-negative parts, i.e., $\oplus_{i < 0} \text{gr}(A)_i$ and $\oplus_{i \geq 0} \text{gr}(A)_i$ are fibre-wise ideals of $\text{gr}(A)$.*

Remark. If the fibres of $\text{gr}(A)$ are pairwise isomorphic Lie algebras, then $\text{gr}(A)$ is a bona fide Lie algebra bundle, by, e.g., [20, Theorem 6.4.5]; this will be the case in all the examples discussed here.⁴

Proof. Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ be given. To prove (1), it suffices to show that

$$[fX, Y] \equiv f[X, Y] \pmod{\Gamma(A^{j+k+1})}$$

for any smooth function f on the base manifold. If $\#$ denotes the anchor, then the necessary and sufficient condition is that

$$df(\#Y)X \in \Gamma(A^{j+k+1}).$$

⁴If A is firmly filtered, transitive, and *integrable*, then $\text{gr}(A^0)$ must be a Lie algebra bundle, because the canonical representation of A on A^0 will drop to a representation of A on $\text{gr}(A^0)$, and the latter integrates to a representation of the transitive (and so locally trivial) integrating Lie groupoid $\mathcal{G}$ having simply-connected source fibres.

If $k \geq 0$ then this holds by F2. If instead $k \leq -1$, then $j + k + 1 \leq j$, implying $A^j \subset A^{j+k+1}$ and the condition is still satisfied.

Suppose A is firmly filtered. Then it is clear that $\oplus_{i < 0} \text{gr}(A)_i$ and $\oplus_{i \geq 0} \text{gr}(A)_i$ are fibre-wise subalgebras of $\text{gr}(A)$. To prove (2), it suffices to show that sections of one of these commute with sections of the other. To this end, suppose $X \in \Gamma(A^k)$ and $Y \in \Gamma(A^j)$, with $k < 0$ and $j \geq 0$. Then $j + k + 1 \leq j$, which implies $A^j \subset A^{j+k+1}$. Since A is firmly filtered, $\Gamma(A^j) \subset \Gamma(A)$ is an ideal, so that $[X, Y] \in \Gamma(A^j) \subset \Gamma(A^{j+k+1})$. It follows that $[\text{gr}_j(X), \text{gr}_k(Y)] = 0$. $\square$

Notation (Bounds on degree). Suppose A is a *graded* vector space or vector bundle. Then, we write $k \leq A$ if $A_i = 0$ for all $i < k$ and $A \leq k$ if $A_i = 0$ for all $i > k$. For example, if $A = A_{-2} \oplus A_{-1} \oplus A_0$, then we write $-2 \leq A \leq 0$. We write $|A| = k$ if $-k \leq A \leq k$ and $A_{-k} \neq 0 \neq A_k$.

If A is merely *filtered*, then we write $A \leq k$ if $\text{gr}(A) \leq k$, etc. For example, for filtered A , we write $|A| = 1$ if $\text{gr}(A) = \text{gr}(A)_{-1} \oplus \text{gr}(A)_0 \oplus \text{gr}(A)_1$ with $\text{gr}(A)_{\pm 1} \neq 0$.

The remainder of the section is devoted to examples of filtered Lie algebroids. All but the first class of examples are always firm.

3.2. Filtered Lie algebras and Lie algebra bundles. Evidently, every filtered Lie algebra, and more generally every filtered Lie algebra bundle, is a filtered Lie algebroid.

3.3. Filtered manifolds. For any Lie algebroid filtration of a tangent bundle TM , we have $TM \leq -1$, by Axiom F2. These filtrations are trivially firm. A smooth manifold M whose tangent bundle is so structured is called a **FILTERED MANIFOLD**, a notion introduced by Morimoto [22], and appearing prominently in [12]. The anchor $\# : A \rightarrow M$ of any *transitive* filtered Lie algebroid A pushes the filtration down to a filtration of TM . That is, the base of any transitive filtered Lie algebroid is a filtered manifold.⁵

3.4. Trivial filtrations. Suppose A is any Lie algebroid whose anchor kernel has constant rank (e.g., is transitive). Choosing A^0 to be this kernel, and defining $A^j = A$ for $j < 0$ and $A^j = 0$ for $j > 0$, we get a firm Lie algebroid filtration with $-1 \leq A \leq 0$ that we call the **TRIVIAL FILTRATION** of A .

3.5. Filtered Klein geometries. For a more interesting class of examples, let G be a Lie group with Lie algebra $\mathfrak{g}$ and $P \subset G$ a closed subgroup. Furthermore, suppose that $\mathfrak{g}$ admits a P -invariant filtration such that $\mathfrak{g}^0$ is the Lie algebra of P . In particular, as we recall in 7.1, a canonical filtration of this kind exists whenever G/P is a generalized flag manifold. Write $M = G/P$ and let $A = \mathfrak{g} \times M$ be the action algebroid. Then we obtain a filtration of A , as a vector bundle, by defining $A^j(m) = \text{Ad}_g(\mathfrak{g}^j) \times \{m\}$, where $g \in G$ is any element satisfying $m = g \cdot m_0$

⁵One can drop the requirement of transitivity, but only at the expense of dropping our requirement $\cap_j A_j = 0$ in the definition of a filtration.

and m_0 is the left coset of P containing the identity of G . It is not hard to see (Proposition 4.5) that A is a firmly filtered Lie algebroid (irrespective of whether $\mathfrak{g}_j \subset \mathfrak{g}$ is an ideal for any j !).

Significantly, $\text{gr}(A)$, which will be a graded Lie algebra bundle, is *not* modeled on $\text{gr}(\mathfrak{g})$ but modeled on the ordinary direct sum of the graded Lie algebras $\oplus_{i < 0} \text{gr}(\mathfrak{g})_i$, re-equipped with *negative* bracket, and $\oplus_{i \geq 0} \text{gr}(\mathfrak{g})_i$. So $\text{gr}(A)$ has *two* different, although related, Lie algebra bundle structures: the one inherited from the filtered Lie algebroid, and a “flat” one inherited from $\mathfrak{g}$. On A^0 they coincide.

3.6. Projective structures. Let $A \subset J^2(TM)$ be the abstract homogeneous PDE you solve to find the infinitesimal isometries of some projective structure on M . Then A is a subalgebroid mapped surjectively onto $J^1(TM)$ by the projection $J^2(TM) \rightarrow J^1(TM)$, whose kernel intersects A in the image A^1 of the natural injection $j: T^*M \rightarrow S^2(T^*M) \otimes TM$ defined by

$$j(Z)(X_1, X_2) = \langle Z, X_1 \rangle X_2 + \langle Z, X_2 \rangle X_1.$$

See [3, §5.9]. If we define A^0 to be the anchor kernel of A , and $A^{-1} = A$, we get a firm Lie algebroid filtration of A with $|A| = 1$. The corresponding graded Lie algebra bundle is

$$\text{gr}(A) = TM \oplus \mathfrak{gl}(TM) \oplus T^*M,$$

where TM and T^*M are commutative, $\mathfrak{gl}(TM) \oplus T^*M$ is the Lie algebra bundle semi-direct product, and TM and $\mathfrak{gl}(TM) \oplus T^*M$ are ideals, by Proposition 3.1(2).

While not immediately obvious, $\text{gr}(A)$ has a second, natural, *semisimple*, Lie algebra bundle structure, modeled on $\mathfrak{sl}(n+1, \mathbb{R})$, $n = \dim(M)$, and compatible with the grading. The bracket for this structure, which we will denote T , extends the existing bracket on $\text{gr}(A^0) = \mathfrak{gl}(TM) \oplus T^*M$ with the new relations

$$(1) \quad T(C, X) = CX, \quad T(X, Z) = j(Z)X,$$

for sections C, X and Z of $\mathfrak{gl}(TM)$, TM and T^*M respectively. Significantly, this bracket coincides with the flat one referenced in 3.5, when M is the projective sphere, viewed as a homogeneous $\text{SL}(n+1, \mathbb{R})$ -space.

As we shall see, the fact that the Lie algebra $\mathfrak{sl}(n+1, \mathbb{R})$ for the full symmetry of the homogeneous model persists in some form in the “curved” case is shared by every concrete geometric structure underlying a parabolic geometry.

3.7. Conformal structures. Let $B \subset J^1(TM)$ be the abstract homogeneous PDE you solve to find the infinitesimal isometries of a conformal structure on M [3, Section 12]. An alternative model for B is the Lie algebroid of the Lie groupoid $\text{CSO}[TM]$ of all relative conformal frames, discussed already in 1.1. In any case, B is a transitive subalgebroid whose anchor kernel $\mathfrak{cso}(TM) \subset \mathfrak{gl}(TM)$ is identifiable with the infinitesimal conformal automorphisms of tangent spaces, and B acts on TM (via restricted adjoint action, in the sense of [5], in the first case, and tautologically in the second). In particular, B acts on T^*M and, regarding T^*M as a commutative Lie algebra bundle, we construct, in the category of Lie

algebroids, the semi-direct product $A = B \oplus_{\text{ad}} T^*M$ (what we shall come to call the PROLONGATION of B). Then, if we put $A^1 = T^*M$, $A^0 = \mathfrak{cso}(TM) \oplus T^*M$ (the anchor kernel of A), and $A^{-1} = A$, then A becomes a firmly filtered Lie algebroid with $|A| = 1$. The corresponding graded Lie algebra bundle is

$$\text{gr}(A) = TM \oplus \mathfrak{cso}(TM) \oplus T^*M,$$

where $\mathfrak{cso}(TM) \oplus T^*M$ is the semi-direct product, and TM and $\mathfrak{cso}(TM) \oplus T^*M$ are ideals.

Again, there is also a semisimple Lie algebra bundle structure on $\text{gr}(A)$ compatible with the grading, in this case modeled on $\mathfrak{so}(n+1, 1)$, and coinciding with the flat one of 3.5 when M is the conformal sphere, viewed as a homogeneous $\text{SO}(n+1, 1)$ -space. After redefining the injection j above,

$$j(Z)(X_1, X_2) = \langle Z, X_1 \rangle X_2 + \langle Z, X_2 \rangle X_1 - g(X_1, X_2)g^{-1}(Z),$$

now a map $j: T^*M \rightarrow (S^2(T^*M) \otimes TM) \cap (T^*M \otimes \mathfrak{cso}(TM))$, the semisimple bracket T extends the existing bracket on $\text{gr}(A^0) = \mathfrak{cso}(TM) \oplus T^*M$, with the same additional relations (1). Here g is any metric in the conformal class.

We will see explicitly why T is a $\mathfrak{so}(n+1, 1)$ -bracket, when we take up conformal structures again in 8.3.

4. CARTAN CONNECTIONS ON LIE GROUPOIDS AND ALGEBROIDS

Suppose that G is a Lie group acting smoothly on a manifold M , possibly in a non-transitive fashion. We are interested in “deformed” versions of the action groupoid $\mathcal{G} = G \times M$, and its Lie algebroid $A = \mathfrak{g} \times M$, and to that end we identify two structures: (i) The canonical horizontal n -plane field D on $\mathcal{G}$, tangent to the foliation $G \times \{m\}$, $m \in M$ — an example of a Lie groupoid CARTAN CONNECTION, to be defined shortly; and (ii) The trivial Lie algebra bundle structure on the infinitesimalization $A = \mathfrak{g} \times M$ of $\mathcal{G}$ — an example of MODEL DATA, defined in Section 5.

Cartan connections on Lie groupoids, introduced in [3, Appendix A], generalize to the intransitive case the classical Cartan connections introduced by Ehresmann [16] and in widespread use, when the latter notion is stripped of model data. The corresponding infinitesimal notion, Cartan connections on Lie algebroids, were introduced some time earlier [2]. The relationship with the classical notion is detailed in [5]. Methods from PDE’s, such as prolongation and reduction, can be used to directly construct Cartan connections for some simple geometric structures, such as Riemannian structures, without declaring up front what the symmetric model should look like [3]. For Riemannian geometry this is very natural, because there are multiple “flat”, i.e., maximally symmetric, models: Euclidean, spherical and hyperbolic space forms. But, as mentioned in the Introduction, this approach has not been successful more generally.

As we recall here, the interpretation of curvature for Cartan connections on Lie groupoids and algebroids is an obstruction to *some* kind of (local) symmetry,

but the kind of symmetry is not preordained by the specification of a flat model a priori. In that case, what we call `MODEL CURVATURE` applies, but we will not discuss that notion until Section 5.

With the exception of Proposition 4.1, which we presume is new, this section is a summary of definitions and results presented in the cited works and elsewhere.

4.1. Cartan connections on a Lie groupoid. A `CARTAN CONNECTION` on a Lie groupoid $\mathcal{G}$ over M is an n -plane field $D \subset T\mathcal{G}$ arising from some right inverse $\mathcal{G} \rightarrow J^1\mathcal{G}$ for the canonical projection $J^1\mathcal{G} \rightarrow \mathcal{G}$, which is necessarily uniquely determined by D . Here $J^1\mathcal{G}$ denotes the Lie groupoid of one-jets of bisections of $\mathcal{G}$. Of course, we are understanding the right inverse to be a Lie groupoid morphism. This is equivalent to requiring that $D \subset T\mathcal{G}$ be multiplicatively closed, in the sense of [23, 14], but this interpretation will play no role here.

The usual adjoint action of a Lie group on its Lie algebra is generalized by an action of $J^1\mathcal{G}$ on the Lie algebroid A of $\mathcal{G}$, denoted Ad [5]. Composing with the right inverse $\mathcal{G} \rightarrow J^1\mathcal{G}$ of $J^1\mathcal{G} \rightarrow \mathcal{G}$ associated with D , we obtain a representation $\bar{D}$ of $\mathcal{G}$ on A . This representation extends the canonical one of $\mathcal{G}$ on elements of the anchor kernel of A (which also denote by Ad). We may view $\bar{D}$ as a kind of proxy for the usual adjoint action for Lie groups.

The prototype of a Cartan connection is the horizontal n -plane field D on an action groupoid $G \times M$, for a smooth action of some Lie group G on M , which need not be transitive. In this case

$$(1) \quad \bar{D}_{(g,m)}(\xi, m) = (\text{Ad}_g \xi, g \cdot m).$$

Every Cartan connection D on a *transitive* Lie groupoid can be recovered from the representation $\bar{D}$:

Proposition. *Suppose $\mathcal{G}$ is a Lie groupoid with a transitive Lie algebroid A . Then for every representation ρ of $\mathcal{G}$ on A , extending the canonical representation of $\mathcal{G}$ on the anchor kernel, there exists a unique Cartan connection D on $\mathcal{G}$ such that $\bar{D} = \rho$.*

The infinitesimal version of this result, Proposition 4.2 below, already appears in [2], and has an easy proof. The proposition above is proven in Appendix C.

4.2. Infinitesimal Cartan connections. Infinitesimalizing a Cartan connection, viewed as a Lie groupoid morphism $\mathcal{G} \rightarrow J^1\mathcal{G}$, we obtain a splitting $s: A \rightarrow J^1A$ of the exact sequence

$$(1) \quad 0 \longrightarrow T^*M \otimes A \longrightarrow J^1A \longrightarrow A \longrightarrow 0$$

in the category of Lie algebroids. Here J^1A denotes the Lie algebroid of one-jets of sections of the Lie algebroid A of $\mathcal{G}$. In the category of vector bundles, splittings s of the above sequence are in one-to-one correspondence with Koszul connections

∇ on A ; this correspondence is given by⁶

$$(2) \quad sX = J^1X + \nabla X; \quad X \in \Gamma(A).$$

When $s: A \rightarrow J^1A$ is a morphism of Lie algebroids, as is the case here, then ∇ is called a **CARTAN CONNECTION** on A [2], or an **INFINITESIMAL CARTAN CONNECTION** when we want to distinguish it from the corresponding global notion. Evidently, the notion makes sense for arbitrary Lie algebroids, not only those integrated by Lie groupoids. According to [2], ∇ is a Cartan connection on A if and only if

$$\nabla_V[X, Y] = [\nabla_V X, Y] + [X, \nabla_V Y] - \nabla_{\bar{\nabla}_X V} Y + \nabla_{\bar{\nabla}_Y V} X,$$

for all vector fields V on M and sections X, Y of A . Here $\bar{\nabla}$ denotes the associated A -connection on TM defined by

$$(3) \quad \bar{\nabla}_X V = \# \nabla_V X + [\#X, V],$$

an A -representation when ∇ is Cartan. Here $\#: A \rightarrow TM$ denotes anchor.

Given a Cartan connection ∇ on some Lie algebroid A , we can compose the corresponding splitting $s: A \rightarrow J^1A$ with the adjoint action of J^1A on A , $\text{ad}_{J^1X} Y = [X, Y]$, to obtain a self-representation $\bar{\nabla}$ of A . Explicitly, we have⁷

$$(4) \quad \bar{\nabla}_X Y = \nabla_{\#Y} X + [X, Y].$$

The anchor $\#: A \rightarrow TM$ becomes $\bar{\nabla}$ -equivariant. A TM -connection ∇ is actually Cartan if and only if, for all $X, Y \in \Gamma(A)$ and $V \in \Gamma(TM)$, we have

$$(5) \quad (\nabla_V \text{tor } \bar{\nabla})(X, Y) = \text{curv } \nabla(\#X, V)Y - \text{curv } \nabla(\#Y, V)X,$$

where

$$\text{tor } \bar{\nabla}(X, Y) := \bar{\nabla}_X Y - \bar{\nabla}_Y X - [X, Y] = \nabla_{\#Y} X - \nabla_{\#X} Y + [X, Y]$$

is the **TORSION**.

The representation $\bar{\nabla}$ of A on itself coincides with the canonical representation of A on sections lying in the anchor kernel (denoted ad in Section 10). In the transitive case, ∇ can be recovered from $\bar{\nabla}$:

Proposition ([2, Proposition 6.1]). *Suppose A is a transitive Lie algebroid. Then for every representation ρ of A on itself, extending the canonical representation of A on sections lying in the anchor kernel, there exists a unique Cartan connection ∇ on A such that $\bar{\nabla} = \rho$.*

⁶According to our sign conventions, the inclusion $T^*M \otimes A \rightarrow J^1A$ induces the following map on sections: $df \otimes X \mapsto fJ^1X - J^1(fX)$.

⁷Outside of our own work, the connections on A and TM denoted $\bar{\nabla}$ are often referred to as the associated **BASIC CONNECTIONS**.

4.3. Curvature as obstruction to symmetry. The CURVATURE of ∇ is defined as usual for Koszul connections. As mentioned above, this is not to be confused with the more classical MODEL CURVATURE, defined later in 5.6. The prototype of an infinitesimal Cartan connection is the trivial flat connection ∇ on an action algebroid $\mathfrak{g} \times M$, for some infinitesimal action of a Lie algebra $\mathfrak{g}$ on M . If we identify $\mathfrak{g}$ with the constant (∇ -parallel) sections of $\mathfrak{g} \times M$, then its bracket is $\text{tor } \bar{\nabla}(\cdot, \cdot)$. Indeed, let $\tau \in \Gamma(\wedge^2 A^* \otimes A)$ denote the trivial extension of the bracket on $\mathfrak{g}$ to a fibre-wise bracket on $A = \mathfrak{g} \times M$, and recall that the Lie algebroid bracket on $\mathfrak{g} \times M$ is given by

$$(1) \quad [X, Y] = \nabla_{\#X} Y - \nabla_{\#Y} X + \tau(X, Y).$$

Then we can write

$$(2) \quad [X, Y] = \bar{\nabla}_X Y - \bar{\nabla}_Y X - \tau(X, Y),$$

from which $\tau = \text{tor } \bar{\nabla}$ immediately follows.

Theorem ([2]). *If M is simply connected, then every Lie algebroid A equipped with a flat Cartan connection ∇ is naturally isomorphic to an action algebroid $\mathfrak{g} \times M$ (possibly intransitive) where $\mathfrak{g} \subset \Gamma(TM)$ is the Lie algebra of ∇ -parallel sections, equipped with bracket $\text{tor } \bar{\nabla}$.*

More informally, we have:

A Lie algebroid equipped with a Cartan connection is an infinitesimal symmetry deformed by curvature; the torsion of the connection is the deformed symmetry Lie bracket.

If $\text{curv } \nabla = 0$ but M is not simply connected, then A is a Lie algebra action with monodromy twist [5]. Lie groupoids with Cartan connections, whose infinitesimalizations are flat, define pseudogroups of symmetries of M , and under an additional completeness assumption, twisted Lie group actions [6, 5]. In the transitive case, one can recover a locally homogeneous structure, assuming something called “geometric closure”, if not global homogeneity up to cover [4].

Example (Riemannian geometry). Associated with every Riemannian metric is a canonical Cartan connection ∇ on the isotropy subalgebroid⁸ $A \subset J^1(TM)$ of the metric [2, 3], which is an infinitesimalization of a global connection D on the isotropy subgroupoid $\mathcal{G} \subset J^1(M \times M)$ of the metric [5]. Its significance is that the anchor induces a map $\Gamma(A) \rightarrow \Gamma(TM)$ establishing a one-to-one correspondence between ∇ -parallel sections and infinitesimal isometries (Killing fields).

Under a natural identification of A with $TM \oplus \mathfrak{so}(TM)$, the Lie bracket is given by

$$\left[\begin{pmatrix} V_1 \\ C_1 \end{pmatrix}, \begin{pmatrix} V_2 \\ C_2 \end{pmatrix} \right] = \begin{pmatrix} [V_1, V_2] \\ [C_1, C_2] + \nabla_{V_1}^{TM} C_2 - \nabla_{V_2}^{TM} C_1 + \text{curv } \nabla^{TM}(V_1, V_2) \end{pmatrix},$$

⁸This isotropy is the abstract homogeneous PDE you solve to find Killing fields, which may be identified with the Lie algebroid of the Lie groupoid $SO[TM]$ of relative orthogonal frames.

where ∇^{TM} is the Levi-Cevita connection. The Cartan connection is given by

$$\nabla_U \begin{pmatrix} V \\ C \end{pmatrix} = \begin{pmatrix} \nabla_U^M V + CU \\ \nabla_U^M C + \text{curv } \nabla^M(U, V) \end{pmatrix},$$

which has torsion

$$(3) \quad \text{tor } \bar{\nabla} \left(\begin{pmatrix} V_1 \\ C_1 \end{pmatrix}, \begin{pmatrix} V_2 \\ C_2 \end{pmatrix} \right) = \begin{pmatrix} C_1 V_2 - C_2 V_1 \\ [C_1, C_2] - \text{curv } \nabla^{TM}(V_1, V_2) \end{pmatrix}.$$

The curvature of ∇ is readily computed, vanishing precisely when M has constant sectional curvature s . From the formula for the torsion, we conclude that $\mathfrak{g}$ in the preceding theorem is the symmetry Lie algebra of the Riemannian sphere, plane or hyperbolic space, according to whether s is positive, zero, or negative.

4.4. Cartan connections compatible with a filtration. An infinitesimal Cartan connection ∇ on a filtered Lie algebroid A is **COMPATIBLE** with the filtration if the filtration is invariant under the associated representation $\bar{\nabla}$. Similarly, a Cartan connection D on a filtered Lie groupoid $\mathcal{G}$ with Lie algebroid A is **COMPATIBLE** with the filtration provided the filtration of A is invariant under the associated representation $\bar{D}$, which in turn implies compatibility of the corresponding infinitesimal Cartan connection $\bar{\nabla}$.

In addition to the examples given in 3.1, the following result justifies our focus on *firmly* filtered Lie algebroids.

Proposition. *If a transitive filtered Lie groupoid (Lie algebroid) admits a compatible (infinitesimal) Cartan connection, then it is firmly filtered.*

Proof. Suppose ∇ is a Cartan connection on a transitive filtered Lie algebroid A . Then $\bar{\nabla}$ extends the canonical representation ad of A on its anchor kernel, which contains A^0 . So $\text{ad}_X Y := [X, Y] = \bar{\nabla}_X Y$ for $X \in A$ and $Y \in A^0$. If ∇ is compatible with the filtration, then A^0 and the filtration of A^0 must therefore be ad -invariant. The groupoid case follows immediately from the infinitesimal one. $\square$

4.5. Filtered Klein geometries. We now prove some claims made in 3.5 regarding the homogeneous space $M = G/P$ described there. Let $\mathcal{G} = G \times M$ be the action groupoid, equipped with its canonical Cartan connection D , so that $\mathcal{G}$ acts on $A = \mathfrak{g} \times M$ according to 4.1(1). Since the filtration of $\mathfrak{g}$ is P -invariant, the filtration of A is $\bar{D}$ -invariant.

Since A is transitive and A^0 is the anchor kernel, we have, as vector bundles,

$$\text{gr}(A) \cong \text{gr}(TM) \oplus \text{gr}(A^0).$$

We denote the fibre-wise Lie bracket on $A = \mathfrak{g} \times M$ coming from $\mathfrak{g}$ by τ , which coincides with the torsion $\text{tor } \bar{\nabla}$, where ∇ is the infinitesimalization of D , which is just the canonical flat Koszul connection on the trivial bundle $\mathfrak{g} \times M$. We note in passing that τ is evidently $\bar{D}$ -invariant, and so $\bar{\nabla}$ -invariant. When we want

to emphasize we are using the bracket $T = \text{gr}_0(\tau)$ when viewing $\text{gr}(A)$, $\text{gr}(A^0)$ or $\text{gr}(TM)$ as Lie algebra bundles, we shall denote them by $\text{gr}^T(A)$, $\text{gr}^T(A^0)$ and $\text{gr}^T(TM)$.

Proposition. *The vector bundle filtration of $A = \mathfrak{g} \times M$ defined in 3.5 is a firm Lie algebroid filtration, and $\mathcal{G} = G \times M$ is a firmly filtered Lie groupoid. Furthermore, equipping $\text{gr}(A)$ with the corresponding Lie algebra bundle structure (Proposition 3.1), we have*

$$\text{gr}(A) = -\text{gr}^T(TM) \oplus \text{gr}^T(A^0) \quad (\text{Lie algebra bundle direct sum})$$

where $-\text{gr}^T(TM)$ is $\text{gr}^T(TM)$ equipped with negative bracket, and $T = \text{gr}_0(\tau)$.

Proof. Axiom F2 holds by construction. Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ be given. Then we claim:

- (1) $\bar{\nabla}_X Y = [X, Y] \quad \text{if } k \geq 0$
- (2) $\bar{\nabla}_X Y = \tau(X, Y) \quad \text{if } j \geq 0$
- (3) $[X, Y] = \tau(X, Y) \quad \text{if } j \geq 0 \text{ and } k \geq 0$
- (4) $[X, Y] \equiv -\tau(X, Y) \pmod{A^{j+k+1}} \quad \text{if } j < 0 \text{ and } k < 0$
- (5) $[X, Y] \in \Gamma(A^{j+k+1}) \quad \text{if } j < 0 \text{ and } k \geq 0.$

Equation (1) follows from 4.2(4) and F2. If $j \geq 0$, then Equation 4.3(1) reads $[X, Y] = -\nabla_{\#Y} X + \tau(X, Y)$, because $\#A^j = 0$ in that case. This and 4.2(4) delivers (2). Claim (3) follows immediately from the preceding two.

If $j < 0$, then $j + k + 1 \leq k$, and so $A^k \subset A^{j+k+1}$. Similarly, if $k < 0$, then $A^j \subset A^{j+k+1}$. Since the filtration is $\bar{\nabla}$ -invariant, (4) now follows from 4.3(2).

As already mentioned, if $j < 0$, then $A^k \subset A^{j+k+1}$. On the other hand, if additionally $k \geq 0$, then $[X, Y] = \bar{\nabla}_X Y$, by (1). So (5) is a consequence of the $\bar{\nabla}$ -invariance of A^k .

To establish Axiom F1 for filtered Lie algebroids, we need to show $[X, Y] \in \Gamma(A^{j+k})$. Equations (3) takes care of the case $j \geq 0$ and $k \geq 0$. Equation (4) takes care of the case $j < 0$ and $k < 0$ because $A^{j+k+1} \subset A^{j+k}$ is always true. The remaining case is $j < 0$ and $k \geq 0$ (and the case $j \geq 0$ and $k < 0$, which will follow by symmetry). But as $A^{j+k+1} \subset A^{j+k}$, this already follows from (5).

The last statement in of the theorem follows easily from (3), (4), and (5).

Since the filtration of A is $\bar{D}$ -invariant, D is compatible with the filtration, and so firmness follows from Proposition 4.4. $\square$

5. CARTAN GEOMETRIES

In this section we complete the definition of Cartan geometries à la Lie groupoids by introducing the notion of compatible model data. We recover Ehresmann's principal bundle notion as the special transitive case. Mild, verifiable, sufficient conditions for the Cartan connection to be *uniquely* determined by compatible

model data are then established in the transitive case (Theorem 5.5). Model curvature is defined, and related to the curvature of the Lie algebroid connection discussed in 4.3. It's basic properties are established.

5.1. Model data. A smooth field τ of Lie brackets on a Lie algebroid A is **MODEL DATA** if $\tau(X, Y) = [X, Y]$ whenever X and Y both lie in the kernel of the anchor of A . In particular, if A is transitive, any Lie algebra bundle structure on A extending the canonical Lie algebra bundle structure of the anchor kernel will be model data, and without further comment, in transitive cases model data will always be assumed to be of this type. We say that τ **HAS LIE ALGEBRA MODEL $\mathfrak{g}$** if every fibre of A , with the Lie algebra structure defined by restricting τ , is isomorphic to $\mathfrak{g}$.

5.2. Cartan geometries defined. An **INFINITESIMAL CARTAN GEOMETRY** on M will be a Lie algebroid A over M , equipped with an infinitesimal Cartan connection ∇ and model data τ that are **COMPATIBLE** in the sense that both these conditions hold:

c1. τ is $\bar{\nabla}$ -invariant.

c2. $\bar{\nabla}_Z X = \tau(Z, X)$ whenever $\#Z = 0$, $X \in A$.

Here $\#: A \rightarrow TM$ denotes the anchor. A **CARTAN GEOMETRY** will be a Lie groupoid $\mathcal{G}$ over M , equipped with a Cartan connection D and model data τ on its Lie algebroid A that are **COMPATIBLE**, in the sense that

C1. τ is $\bar{D}$ -invariant,

and c2 holds, where ∇ is the infinitesimalization of D .

Evidently, the infinitesimalization (A, ∇, τ) of any Cartan geometry $(\mathcal{G}, D, \tau)$ is an infinitesimal Cartan geometry. Also, it is readily verified that action groupoids are Cartan geometries and infinitesimal action algebroids are infinitesimal Cartan geometries. We refer to these geometries as the **FLAT MODELS**.

If only c1 (or C1) is known to hold, then τ is **WEAKLY COMPATIBLE** with ∇ (resp., D). For emphasis, we sometimes use “strongly compatible” as a synonym of “compatible”.

5.3. Compatibility. We now elaborate on condition c2, which seems less natural than c1 and C1. Notice first that when a Lie algebroid A is equipped with model data τ and an infinitesimal Cartan connection ∇ , then A acts on itself in *two* distinct ways: first, we have the the Lie algebroid representation $\bar{\nabla}$; second we have the representation $X \mapsto \text{ad}_X^\tau = \tau(X, \cdot)$ of A on itself as a bundle of Lie algebras. Condition c2 is simply the assertion that every element of the anchor kernel of A acts the same way in both these cases. (Elements of the kernel act on *elements* of A , not merely on sections.) As already noted, c2 holds in the flat models, which is some justification for including it. However, also:

- (1) Both C1 and c2 hold for classical (Ehresmann) Cartan geometries (Proposition 5.4).

- (2) In the transitive case, we need c2 for model curvature to be a two-form on M (see 5.6).
- (3) For so-called SEPARATING model data, C1 or c1 already imply c2 (Theorem 5.5(1)) and this will apply in many interesting examples, including parabolic geometry (Proposition A.7).
- (4) In the transitive case, and assuming c2 holds, the INFINITESIMAL ORDER of the geometry, providing a lower bound m on the semi-holonomic jet space $\bar{J}^m(TM)$ that must contain A as an embedded subalgebroid, is completely encoded in the Lie algebra model $\mathfrak{g}$ for τ , together with the subalgebra $\mathfrak{h} \subset \mathfrak{g}$ corresponding to the anchor kernel of A (Theorem 6.3).

Point (3) notwithstanding, we have the following counterexample:

Example (weak compatibility without strong compatibility). Let M be a copy of $\mathbb{R}$ and G the multiplicative group $(0, \infty)$. Then we have the trivial, transitive Lie groupoid $\mathcal{G} := M \times G \times M$ [20, Example 1.1.7], with Lie algebroid $A := TM \times \mathfrak{g}$, $\mathfrak{g} = \mathbb{R}$. If $\frac{\partial}{\partial m}$ denotes the constant vector field on M , then, by the transitivity, a Cartan connection D on $\mathcal{G}$ is defined implicitly by $\bar{D}_{(m_2, g, m_1)}(\frac{\partial}{\partial m}(m_1), \xi) = (\frac{\partial}{\partial m}(m_2), \xi)$. Viewing a section of A as a pair (u, X) of functions, $u: M \rightarrow \mathbb{R}$, and $X: M \rightarrow \mathfrak{g}$, we have $\bar{\nabla}_{(u, X)}(v, Y) = (u \frac{\partial v}{\partial m}, u \frac{\partial Y}{\partial m})$, and define $\bar{D}$ -invariant model data τ by $\tau((u, X), (v, Y)) = (0, uY - vX)$. For any $\mathfrak{g}$ -valued function X on M , $(0, X)$ is a section of the anchor kernel. But $\bar{\nabla}_{(0, X)}(u, Y) = \tau((0, X), (v, Y))$ only if $vX = 0$, which shows that c2 fails.

5.4. Classical Cartan geometries. We next show that our Cartan geometries include classical Cartan geometries as a special case (a fact we will not use elsewhere).

Let G be a Lie group with Lie subgroup H and let $\pi: P \rightarrow M$ be a left principal H -bundle. Let ω be a $\mathfrak{g}$ -valued one-form on P satisfying: (i) $\omega: T_m P \rightarrow \mathfrak{g}$ is an isomorphism at every $m \in M$ (ω is a parallism); (ii) For each $X \in \mathfrak{g}$, $\omega^{-1}(X)$ is the corresponding infinitesimal generator for the action of H on P ; and (iii) $\omega(v \cdot h) = \text{Ad}_{h^{-1}} \omega(v)$, for all $v \in T_m P$, $h \in H$ (ω is H -equivariant). Then ω is a Cartan connection in the sense of Ehresmann [16].

Take $\mathcal{G}$ to be gauge groupoid $(P \times P)/H$. The corresponding Lie algebroid, known as the Atiyah Lie algebroid, is identifiable with $A = TP/H$. On account of transitivity, and the H -equivariance of ω , a Cartan connection D on $\mathcal{G}$ is defined implicitly by $\bar{D}_{[p_2, p_1]_H}[v_1]_H = [(\omega(p_2)^{-1} \circ \omega(p_1))(v_1)]_H$, for $v_1 \in T_{p_1} P$ and $p_1, p_2 \in P$. Here $[\cdot]_H$ denotes equivalence modulo H .⁹

The H -equivariance of ω shows that model data τ on A is well-defined by

$$(1) \quad \tau([u]_H, [v]_H) = \omega(p^{-1})[\omega(p)(u), \omega(p)(v)]_{\mathfrak{g}}; \quad u, v \in T_p P.$$

⁹For a construction that avoids reference to $\bar{D}$, see [5], which already establishes a one-to-one correspondence between Cartan connections on transitive Lie groupoids and classical Cartan connections *stripped of model data*.

It follows immediately from the definition that τ is weakly compatible with D (condition C1).

The following is just a basic fact about Ehresmann-Cartan connections:

Lemma. For any H -invariant vector field U on P , and vertical vector field V on P , one has $\mathcal{L}_V(\omega(U)) = [\omega(V), \omega(U)]_{\mathfrak{g}}$.

Proof. Since $\mathcal{L}_{fV}(\omega(fU)) = f\mathcal{L}_V(\omega(U))$, for any smooth function f , it suffices to suppose V is an infinitesimal generator for the action of H . The result now follows easily from the H -equivariance of ω . $\square$

Proposition. *In the scenario just described for classical Cartan geometries, the model data τ is compatible with D .*

Proof. It remains to prove compatibility condition c2. For a section X of $A = TP/H$, let $\hat{X}$ denote the corresponding H -invariant vector field on P , so that $[\hat{X}(p)]_H = X(\pi(p))$. Then we may express (1) as $\omega(\tau(\widehat{Z}, \widehat{X})) = [\omega(\hat{Z}), \omega(\hat{X})]_{\mathfrak{g}}$, for $X, Z \in \Gamma(A)$. On the other hand, it is not too difficult to compute the infinitesimalization $\bar{\nabla}$ of $\bar{D}$: it is characterized by the identity $\omega(\widehat{\bar{\nabla}_Z X}) = \mathcal{L}_{\hat{Z}}\omega(\hat{X})$. Supposing $Z \in \Gamma(A)$ lies in the anchor kernel, $\hat{Z}$ will be vertical, and the lemma delivers $\mathcal{L}_{\hat{Z}}(\omega(\hat{X})) = [\omega(\hat{Z}), \omega(\hat{X})]_{\mathfrak{g}}$. Combining all these observations shows that $\omega(\widehat{\bar{\nabla}_Z X}) = \omega(\tau(\widehat{Z}, \widehat{X}))$, which establishes c2. $\square$

5.5. Separating model data. We call a subalgebra $\mathfrak{h} \subset \mathfrak{g}$ **SEPARATING** if it is separating for the automorphism group $\text{Aut}(\mathfrak{g})$, which is to say, that if any two Lie algebra automorphisms agree on $\mathfrak{h}$, then they must agree on all of $\mathfrak{g}$. Equivalently, the requirement is that the only automorphism restricting to the identity on $\mathfrak{h}$ is the identity element of $\text{Aut}(\mathfrak{g})$.

Similarly, we say that $\mathfrak{h} \subset \mathfrak{g}$ is **INFINITESIMALLY SEPARATING** if the only derivation vanishing on $\mathfrak{h}$ is the trivial derivation. Any separating subalgebra is automatically infinitesimally separating.

Parabolic subalgebras of reductive Lie algebras are always separating (Proposition A.7). In a semi-direct product $\mathfrak{g} = \mathfrak{h} \ltimes V$, the subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is never separating, for $(X, v) \mapsto (X, \lambda v)$ is an Lie algebroid automorphism for any scalar λ . This rules out the standard Euclidean model for Riemannian geometry. However $\mathfrak{so}(n) \subset \mathfrak{so}(n+1)$ and $\mathfrak{so}(n) \subset \mathfrak{so}(n, 1)$, the spherical and hyperbolic models for Riemannian geometry, are separating.

Model data τ on A will be **SEPARATING** if at every point $m \in M$, the anchor kernel of A sitting over m is a separating subalgebra of $A|_m$, for the Lie bracket defined by τ .

Theorem. *Let τ be separating model data on a Lie algebroid A . Then:*

- (1) *If τ is weakly compatible with an infinitesimal Cartan connection ∇ then it is strongly compatible. Similarly, if D is a Cartan connection on a Lie*

groupoid $\mathcal{G}$ integrating A , then weak compatibility of τ with D implies strong compatibility.

- (2) If A is transitive, there can exist at most one infinitesimal or global Cartan connection compatible with τ .

Separability in (2) is necessary:

Example (multiple Cartan connections compatible with model data). Let M be a parallelizable manifold of dimension n , such as $\mathbb{R}^n$. Then a Cartan connection D on the pair groupoid $\mathcal{G} = M \times M$ is a parallelism on M . The representation $\bar{D}$ of $M \times M$ on TM is parallel translation with respect to D . Let $\mathfrak{g}$ be the commutative Lie algebra over $\mathbb{R}$ of dimension n . Then all model data τ on $A = TM$, modeled on $\mathfrak{g}$, is the same and trivial: $\tau(X, Y) = 0$. It is easy to see, that τ is compatible with D for *any* choice of D .

Proof of Theorem. (1): If A is integrated by a Lie groupoid $\mathcal{G}$ and τ is weakly compatible with D , then it is weakly compatible with its infinitesimalization ∇ . So to prove (1), it suffices to consider the infinitesimal case. Fix some Z lying in the kernel $\mathfrak{h}$ of $\# : A|_m \rightarrow T_m M$. Then we may view $\bar{\nabla}_Z$ as a linear map $A|_m \rightarrow A|_m$. In fact, with respect to the bracket τ , this is a Lie algebra derivation, because τ is $\bar{\nabla}$ -invariant, assuming weak compatibility. But ad_Z^τ is also a derivation of $A|_m$, and on $\mathfrak{h}$ these two derivations coincide, by 4.2(4) and the definition of model data. Since τ is separating, $\mathfrak{h} \subset A|_m$ is a separating subalgebra, with respect to the bracket defined by τ , and so infinitesimally separating. It follows that the derivations agree on $A|_m$, which proves compatibility condition c2, and hence strong compatibility.

(2): Suppose that ∇^1 and ∇^2 are Cartan connections weakly compatible with model data τ on A , and write $\delta_X Y = \bar{\nabla}_X^1 Y - \bar{\nabla}_X^2 Y$. For each $X \in A|_m$, $m \in M$, we may regard δ_X as a linear map $A|_m \rightarrow A|_m$ and, because τ is both $\bar{\nabla}^1$ and $\bar{\nabla}^2$ -invariant (weak compatibility) it follows that δ_X is even a Lie algebra derivation, with respect to the bracket τ . Since $\bar{\nabla}^1$ and $\bar{\nabla}^2$ both extend the canonical representation of A on its anchor kernel, δ_X vanishes on $\mathfrak{h} \subset A|_m$ defined as above. By the same argument as before, $\bar{\nabla}^1$ and $\bar{\nabla}^2$ must coincide. In the transitive case, an infinitesimal Cartan connection ∇ is determined uniquely by the corresponding representation $\bar{\nabla}$, so $\nabla^1 = \nabla^2$.

Turning to the global case, suppose D^1 and D^2 are Cartan connections weakly compatible with model data τ and write $\Delta_g Y = \bar{D}_{g^{-1}}^1 \bar{D}_g^2 Y$. For each arrow $g \in \mathcal{G}$ beginning at some $m \in M$, we may regard Δ_g as a vector space automorphism $A|_m \rightarrow A|_m$ and, because τ is both $\bar{D}^1$ and $\bar{D}^2$ -invariant (weak compatibility) it follows that Δ_g is a Lie algebra automorphism, with respect to the bracket τ . Since both $\bar{D}^1$ and $\bar{D}^2$ both extend the canonical representation of $\mathcal{G}$ on the anchor kernel, $\Delta_g Z = Z$ for all Z in $\mathfrak{h} \subset A|_m$ defined as above. Since τ is separating, Δ_g must be the identity on $A|_m$, proving $\bar{D}^1 = \bar{D}^2$. In the transitive case, a Cartan

connection D is determined uniquely by the corresponding representation $\bar{D}$, so $D^1 = D^2$. $\square$

5.6. Model curvature. Consider an infinitesimal Cartan geometry (A, ∇) , which may or may not be the infinitesimalization of some global Cartan geometry. While the curvature of the infinitesimal Cartan connection ∇ is an obstruction to symmetry (Theorem 4.3) and makes sense without the prescription of any model data, MODEL CURVATURE is more amenable to normalization, as we shall see concretely in the case of parabolic geometry.

In the general case, the model curvature is a section of $\wedge^2 \mathcal{D}^* \otimes A$, where $\mathcal{D} \subset TM$ is the image of the anchor. In other words, curvature is an A -valued two-form on the orbits of A . However, for simplicity, we only define it here in the case that A is *transitive*: it is then the section κ of $\wedge^2 T^*M \otimes A$ defined by

$$\kappa(\#X, \#Y) = \text{tor } \bar{\nabla}(X, Y) - \tau(X, Y); \quad X, Y \in \Gamma(A).$$

Here $\#$ is the anchor and

$$\text{tor } \bar{\nabla}(X, Y) = \bar{\nabla}_X Y - \bar{\nabla}_Y X - [X, Y] = \nabla_{\#Y} X - \nabla_{\#X} Y + [X, Y].$$

That κ is well defined follows from 4.2(4) and compatability condition c2.

The model curvature is the Lie algebroid analogue of “curvature”, as that term is generally understood in the Cartan geometry literature.

Proposition. *The model curvature κ of a transitive infinitesimal Cartan geometry (A, ∇) determines the curvature of ∇ according to*

$$\text{curv } \nabla(U, V)X = (\bar{\nabla}_X \kappa)(U, V).$$

In particular, if $\kappa = 0$, then $\text{curv } \nabla = 0$ and $\text{tor } \bar{\nabla} = \tau$.

The consequences of $\text{curv } \nabla = 0$ have already been discussed in 4.3.

According to the proposition, κ is an obstruction to symmetry, but of the kind prescribed *a priori* by the choice of the Lie algebra model for τ . According to Theorem 4.3, the model curvature measures the difference between a “deformed” Lie bracket $\text{tor } \bar{\nabla}$, defined by the Cartan connection, and a bona fide Lie bracket, prescribed by the model data τ . This nice interpretation is obscured in the standard principal bundle approach.

Proof of proposition. For any Cartan connection ∇ on a transitive Lie algebroid, one has the easily derived identity,

$$(1) \quad \text{curv } \nabla(\#X, \#Y)Z = \bar{\nabla}_Z \text{tor } \bar{\nabla}(X, Y).$$

See, for example, [2, §5.4]. From the definition of κ ,

$$(2) \quad (\bar{\nabla}_Z \kappa)(\#X, \#Y) = (\bar{\nabla}_Z \text{tor } \bar{\nabla})(X, Y) - (\bar{\nabla}_Z \tau)(X, Y).$$

The stated identity follows from (1), (2) and the compatability condition c1. $\square$

5.7. Model curvature under automorphisms. Let (A, ∇, τ) be a transitive infinitesimal Cartan geometry. Let $f: M \rightarrow M'$ be a diffeomorphism and $q: A \rightarrow A'$ some Lie algebroid isomorphism, covering f , into some second Lie algebroid A' (necessarily transitive). It is not hard to see that the push-forward $q_*\tau$, defined by $(q_*\tau)(X, Y) = q\tau(q^{-1}X, q^{-1}Y)$, will be model data on A' .

Proposition. *The equation*

$$(1) \quad \bar{\nabla}'_X Y = q_* \bar{\nabla}_{q^*X} (q^*Y); \quad X, Y \in \Gamma(A'),$$

implicitly defines a Cartan connection ∇' compatible with the push-forward $\tau' = q_\tau$, so that (A', ∇', τ') is an infinitesimal Cartan geometry. Here $(q_*X)(m) = qX(f^{-1}(m))$ and $(q^*X)(m) = q^{-1}X(f(m))$, for $X \in \Gamma(A)$. If $\mathcal{G}$ a Lie groupoid integrating A , $\mathcal{G}'$ is a Lie groupoid integrating A' , ∇ is the infinitesimalization of a Cartan connection D on $\mathcal{G}$, and $Q: \mathcal{G} \rightarrow \mathcal{G}'$ a Lie groupoid isomorphism with infinitesimalization q , then*

$$(2) \quad \bar{D}'_g = q \circ \bar{D}_{Q^{-1}(g)} \circ q^{-1}$$

implicitly defines a Cartan connection D' on $\mathcal{G}'$ such that $(\mathcal{G}', D', \tau')$ is a Cartan geometry, whose infinitesimalization is (A', ∇', τ') . In any case, if κ is the model curvature of (A, τ, ∇) and κ' the model curvature of (A, τ', ∇') , then we obtain the

CURVATURE TRANSFORMATION RULE,

$$(3) \quad \kappa'(V_1, V_2) = q\kappa(Tf^{-1} \cdot V_1, Tf^{-1} \cdot V_2); \quad V_1, V_2 \in TM.$$

If $(\mathcal{G}, \tau, D)$ (resp., (A, τ, ∇)) is an (infinitesimal) Cartan geometry, then $(\mathcal{G}, \tau', D')$ (resp., (A, τ', ∇')) is an (infinitesimal) Cartan geometry.

Proof. To show ∇' is well-defined, we apply Proposition 4.2, which means we need to show (i) $\bar{\nabla}'$, as defined by (1), is a representation, and supposing $\#Y = 0$, that (ii) $\bar{\nabla}'_X Y = [X, Y]$. Verifying (i) is an easy computation appealing to the fact that $\bar{\nabla}$ is a representation and that q is a Lie algebroid morphism. Regarding (ii), we have

$$\bar{\nabla}'_X Y = q_* \bar{\nabla}_{q^*X} (q^*Y) = q_* [q^*X, q^*Y] = [X, Y],$$

where the second equality follows from the fact that $\bar{\nabla}_\xi \eta = [\xi, \eta]$ whenever $\#\eta = 0$ (compatibility condition c2) and that $\#(q^*Y) = f^*(\#Y)$ for any $Y \in \Gamma(A)$. It is not hard to see that compatibility of ∇ and τ establishes the compatibility of ∇' and τ' .

Regarding the global case, we prove a little more, indicating the origins of formulas (1) and (2). Note first that because Q is a groupoid morphism, we have the identity

$$(4) \quad q \circ \text{Ad}_\mu = \text{Ad}_{(J^1 Q)_\mu} \circ q; \quad \mu \in J^1 \mathcal{G},$$

where Ad is the adjoint representation of $J^1 \mathcal{G}$ on A (see [5]). For a proof of (4), suppose $\mu = J_m^1 b$, for some bisection b of $\mathcal{G}$. Then for an arbitrary element of A ,

represented by source-fibre path $t \mapsto g(t)$ beginning at m , we have

$$\text{Ad}_{J_m^1 b} \left(\frac{d}{dt} g(t) \Big|_{t=0} \right) = \frac{d}{dt} b(g(t)m) g(t) b(m)^{-1} \Big|_{t=0},$$

so that

$$\begin{aligned} (q \circ \text{Ad}_{J_m^1 b}) \left(\frac{d}{dt} g(t) \Big|_{t=0} \right) &= \frac{d}{dt} Q(b(g(t)m)) Q(g(t)) Q(b(m))^{-1} \Big|_{t=0} \\ &= \text{Ad}_{(J^1 Q)(J_m^1 b)} \frac{d}{dt} Q(g(t)) \Big|_{t=0} = \text{Ad}_{(J^1 Q)(J_m^1 b)} q \left(\frac{d}{dt} g(t) \Big|_{t=0} \right). \end{aligned}$$

and (4) is proven. If $S: \mathcal{G} \rightarrow J^1 \mathcal{G}$ is the splitting associated with D , then associated with the splitting

$$(5) \quad S' = J^1 Q \circ S \circ Q^{-1}$$

is a Cartan connection D' on $\mathcal{G}'$, and applying the identity (4) and the definition of representations $\bar{D}$ and $\bar{D}'$ associated with these connections, we obtain the formula (2). Showing that $\bar{\nabla}'$ is the infinitesimalization of $\bar{D}'$ is a simple computation. It is not hard to see that the compatability of D and τ leads to the compatibility of D' and τ' .

Regarding (3), we compute, for any $X, Y \in \Gamma(A)$,

$$\begin{aligned} \kappa'(\#X, \#Y) &= \bar{\nabla}'_X Y - \bar{\nabla}'_Y X - [X, Y] - \tau'(X, Y) \\ &= q_* \bar{\nabla}_{q^* X} (q^* Y) - q_* \bar{\nabla}_{q^* Y} (q^* X) - q_* [q^* X, q^* Y] - q_* (\tau(q^* X, q^* Y)) \\ &= q_* (\kappa(\#q^* X, \#q^* Y)) = q_* (\kappa(f^*(\#X), f^* \#Y)), \end{aligned}$$

and (3) follows. $\square$

5.8. The Bianchi identity. For any transitive infinitesimal Cartan geometry (A, ∇, τ) one has the BIANCHI IDENTITY

$$(1) \quad (\bar{\nabla}_{X_3} \kappa)(\#X_1, \#X_2) + \kappa(\kappa(\#X_1, \#X_2), \#X_3) \\ + \kappa(\# \tau(X_1, X_2), \#X_3) + \tau(\kappa(\#X_1, \#X_2), X_3) + \text{terms cyclic in 1-2-3} = 0.$$

For a proof, first note that any infinitesimal Cartan connection ∇ on a Lie algebroid, one has the Bianchi identity

$$(2) \quad (\bar{\nabla}_{X_3} \text{tor } \bar{\nabla})(X_1, X_2) + \text{tor } \bar{\nabla}(\text{tor } \bar{\nabla}(X_1, X_2), X_3) + \text{terms cyclic in 1-2-3} = 0.$$

See [3, Example 6.5]. Since $\bar{\nabla} \tau = 0$ (compatibility) it follows from the definition of κ that

$$(\bar{\nabla}_{X_3} \kappa)(\#X_1, \#X_2) = (\bar{\nabla}_{X_3} \text{tor } \bar{\nabla})(X_1, X_2)$$

holds, as do the cyclicly permuted variants thereof. Using these in (2), we obtain

$$(\bar{\nabla}_{X_3} \kappa)(\#X_1, \#X_2) + \text{tor } \bar{\nabla}(\text{tor } \bar{\nabla}(X_1, X_2), X_3) + \text{terms cyclic in 1-2-3} = 0.$$

Rearranging the definition of κ , we have $\text{tor } \bar{\nabla}(X, Y) = \kappa(\#X, \#Y) + \tau(X, Y)$, which we can use to rewrite this last equation as (1) above.

Remark. We choose to write the Bianchi identity in the form (1), as this leads to a simple “graded” form, in the case that A is a filtered Lie algebroid (Proposition 14.3).

6. CARTAN GEOMETRIES AS BUNDLES OF SEMI-HOLONOMIC JETS

In this section, not essential to the sequel, we show how to realize an infinitesimal Cartan geometry (A, ∇, τ) over M as a subalgebroid of the bundle $\bar{J}^\nu(TM)$ of ν th-order semi-holonomic jets of vector fields. Significantly, in the transitive case, ν is completely determined by the Lie algebra quotient $\mathfrak{g}/\mathfrak{h}$ on which tangent spaces of M are being modeled. Our results generalize to the curved case results we learned from [1], which in turn are based on of the classical papers [18, 25]. With the Lie algebroid formalism that we have in place, our results follow with very little effort. One can look for similar results for global Cartan geometries, whose flat versions are described in [1], but we do not pursue this here. A wrinkle in the global case is that the global notion of order (what [1] calls the GEOMETRIC ORDER) is sometimes one higher than the infinitesimal order, because of non-trivial but *discrete* kernels in what would otherwise be an embedding into the Lie groupoid $\bar{J}^\nu(M \times M)$.

In the flat, infinitesimal case (action algebroids) one can replace $\bar{J}^\nu(TM)$ with the space $J^\nu(TM)$ of bona fide holonomic jets, that is, we may view the Cartan geometry as overdetermined homogeneous PDEs for vector fields on M , and these PDE’s even have solutions, namely the infinitesimal generators of the action. Unfortunately, to extend this PDE interpretation to the curved case requires a strong condition on $\text{curv } \nabla$, as we show in Proposition 6.1 when $\nu = 2$, which covers the effective parabolic cases, as we establish later (“effective” is defined below).

Now in the scenario just mentioned, one can intersect the realization of A as a subalgebroid of $\bar{J}^2(TM)$ with $J^2(TM)$, to get a genuine PDE. Unfortunately, as one can show in conformal geometry, for example, essential information may be lost.

When a Cartan geometry is realized as a bundle of jets, certain ambiguities that arise in the choice of the connection and model data are “resolved”, as we show in Proposition 6.4, which is admittedly an advantage of the jet formalism over the abstract point which will be our principal preoccupation.

6.1. The infinitesimal order of a Cartan connection. Let E be a vector bundle over M . Between J^2E and J^1J^1E lies the bundle $\bar{J}^2(E)$ of SEMI-HOLONOMIC jets. To recall it’s definition, let $\natural$ denote a lowering of the order in any jet bundle by one. So, for example, $\natural: J^2E \rightarrow J^1E$ is the restriction of the projection $\natural: J^1J^1E \rightarrow J^1E$, etc. SPENCER’S OPERATOR $\mathcal{D}: (J^1E) \rightarrow \Gamma(T^*M \otimes E)$ is defined by $\mathcal{D}X := X - J^1(\natural X)$ so that $\mathcal{D}X = 0$ if and only if X is holonomic. One then

defines $\bar{J}^2 E$ to be the kernel of

$$\begin{aligned}\omega_1: J^1 J^1 E &\rightarrow T^*M \otimes E \\ (\omega_1 \xi) V &:= \mathcal{D}_V(\natural \xi) - \natural \mathcal{D}_V \xi.\end{aligned}$$

Here we are writing $\mathcal{D}_V X = (\mathcal{D}X)(V)$ (and one has the Liebniz identity $\mathcal{D}_V(fX) = f\mathcal{D}_V X + df(V)\natural X$). A key point of these abstractions is the following fundamental characterization of $J^2 E$ (see e.g., [3, §8.1]):

Lemma. $J^2 E$ is the kernel of the map $\omega_2: \bar{J}^2 E \rightarrow \wedge^2(T^*M) \otimes E$ defined by

$$(\omega_2 \xi)(V_1, V_2) = \mathcal{D}_{V_1} \mathcal{D}_{V_2} \xi - \mathcal{D}_{V_2} \mathcal{D}_{V_1} \xi - \natural \mathcal{D}_{[V_1, V_2]} \xi.$$

In particular, if we can show some subbundle of $\bar{J}^2 E$ is in this kernel, then we may view it as a homogeneous second order PDE for sections of E .

Now let A be a Lie algebroid with anchor $\#: A \rightarrow TM$, equipped with a Cartan connection ∇ , and let $s_\nabla: A \rightarrow J^1 A$ denote the corresponding splitting of 4.2(1). We define maps

$$\begin{aligned}\#_\nabla^1: A &\rightarrow J^1(TM), \\ \#_\nabla^2: A &\rightarrow J^1 J^1(TM)\end{aligned}$$

by $\#_\nabla^1 := J^1 \# \circ s_\nabla$ and $\#_\nabla^2 := J^1 \#_\nabla^1 \circ s_\nabla$. The reader is reminded that the prolongation $J^1 L: J^1 E \rightarrow J^1 F$ of a vector bundle morphism $L: E \rightarrow F$ covering the identity, defined by $(J^1 L)J_m^1 X := J_{L(m)}^1(LX)$, maps an element ϕ of the kernel $T^*M \otimes E$ of the projection $\natural: J^1 E \rightarrow E$ to $L \circ \phi \in T^*M \otimes F$.

Proposition. Let $A_{\text{jet}}^2 \subset J^1 J^1(TM)$ denote image of A under $\#_\nabla^2$. Then A_{jet}^2 always lies in $\bar{J}^2(TM)$, and lies in $J^2(TM)$ if and only if $\# \text{curv } \nabla(V_1, V_2)X = 0$ for all $V_1, V_2 \in \Gamma(TM)$ and $X \in \Gamma(A)$.

Proof. We have $s_\nabla X = J^1 X + \nabla X$ (see 4.2(2)) and, noting that $\mathcal{D}_V \phi = \phi(V)$ for ϕ in the kernel of $\natural$, the proof proceeds in a straightforward fashion, the second part appealing to the lemma above. $\square$

Now, declaring $\bar{J}^0 E = E$ and $\bar{J}^1 E := J^1 E$, one inductively defines the k th order semi-holonomic jet bundle $\bar{J}^k E \subset J^1 \bar{J}^{k-1} E \subset J^1 J^1 \bar{J}^{k-2} E$ as the kernel of a map $\omega_1: J^1 J^1 \bar{J}^{k-2} E \rightarrow T^*M \otimes \bar{J}^{k-2} E$, defined, as before, in terms of the Spencer operator on sections of $J^1 \bar{J}^{k-1} E$. We can then define $\#_\nabla^k: A \rightarrow J^1 \bar{J}^{k-1}(TM)$, by $J^1 \#_\nabla^{k-1} \circ s_\nabla$, arguing similarly that it's image lies in $\bar{J}^k(TM)$. We then declare the INFINITESIMAL ORDER of ∇ to be the smallest integer $\nu \leq \infty$ such that $\#_\nabla^\nu: A \rightarrow \bar{J}^\nu(TM)$ is injective.

6.2. Torsion encodes the infinitesimal order. Let $\mathfrak{a}^k(A, \nabla) := \ker \#_\nabla^k \subset A$; in particular $\mathfrak{a}^0(A, \nabla) := \ker \#$. Because $\#_\nabla^k$ is a Lie algebroid morphism, $\mathfrak{a}^k(A, \nabla)$ is a bundle of Lie algebras, but it's rank may vary from point to point. We have

$$\mathfrak{a}^0(A, \nabla) \supset \mathfrak{a}^1(A, \nabla) \supset \mathfrak{a}^2(A, \nabla) \supset \dots$$

The infinitesimal order of ∇ is evidently the smallest $\nu \leq \infty$ such that $\mathfrak{a}^\nu(A, \nabla) = 0$.

Proposition. *For each j , write $\mathfrak{a}^j := \mathfrak{a}^j(A, \nabla)$ and assume that $\mathfrak{a}^k$ has constant rank. Then at each $m \in M$,*

$$\mathfrak{a}^{k+1}|_m = \{X \in \mathfrak{a}^k|_m \mid \text{tor } \bar{\nabla}(A|_m, X) \subset \mathfrak{a}^k\}.$$

Proof. If $X \in \mathfrak{a}^k|_m$, then X has an extension, also denoted X , to a section of A lying in $\mathfrak{a}^k$, on account of the rank hypothesis (which we apparently cannot drop here, as the rank is only *upper* semicontinuous). From the formula $s_\nabla X = J^1 X + \nabla X$, it now follows easily that $X \in \mathfrak{a}^{k+1}$ if and only if $(\nabla_V X)(m) \in \mathfrak{a}^k$, for all $V \in \Gamma(TM)$, i.e., if $(\nabla_{\#Y} X)(m) \in \mathfrak{a}^k$, for all $Y \in \Gamma(A)$, by transitivity. However, on account of $\#X = 0$, we have $\nabla_{\#Y} X = [Y, X] - \text{tor } \nabla(Y, X)$, and here $[Y, X] \in \Gamma(\mathfrak{a}^k)$ because $\#^k_\nabla$ is a Lie algebroid morphism and X lies in its kernel $\mathfrak{a}^k$. The claim follows. $\square$

6.3. Infinitesimal order from algebraic considerations. Finally, we turn to a dramatic simplification of the problem of determining the infinitesimal order of a Cartan connection ∇ , in the transitive case. We obtain this whenever ∇ is compatible with some model data τ , as required in the definition of an infinitesimal Cartan geometry.

Suppose $\mathfrak{g}$ is a finite-dimensional Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ a fixed subalgebra. Following [1, 18, 25], we will write $\mathfrak{a}^0(\mathfrak{g}) := \mathfrak{h}$ and inductively define

$$\mathfrak{a}^{k+1}(\mathfrak{g}) := \{X \in \mathfrak{a}^k(\mathfrak{g}) \mid [\mathfrak{g}, X] \subset \mathfrak{a}^k(\mathfrak{g})\}.$$

The INFINITESIMAL ORDER of $\mathfrak{g}$ MODULO $\mathfrak{h}$ is the smallest $\nu \leq \infty$ such that $\mathfrak{a}^\nu(\mathfrak{g}) = 0$. While the spaces $\mathfrak{a}^k(\mathfrak{g})$ depend on the choice of $\mathfrak{h} \subset \mathfrak{g}$, we leave this implicit in the notation.

Recall that if G is a Lie group and $H \subset G$ a closed subgroup, then G acts effectively on G/H if and only if $\{1_G\}$ is the only normal subgroup of G lying within H . A pair of Lie algebras $(\mathfrak{g}, \mathfrak{h})$ with $\mathfrak{h} \subset \mathfrak{g}$ is called EFFECTIVE if $\mathfrak{g}$ has no non-zero ideal lying within $\mathfrak{h}$.

Proposition. *The infinitesimal order of $\mathfrak{g}$ modulo $\mathfrak{h}$ is finite if and only if $(\mathfrak{g}, \mathfrak{h})$ is effective.*

Proof. $\Leftarrow$: By finite-dimensionality of $\mathfrak{g}$, the sequence $\mathfrak{a}^0(\mathfrak{g}) \supset \mathfrak{a}^1(\mathfrak{g}) \supset \mathfrak{a}^2(\mathfrak{g}) \supset \dots$ stabilizes at some $\mathfrak{a}^\nu(\mathfrak{g})$, which will be an ideal of $\mathfrak{g}$ lying in $\mathfrak{h}$.

$\Rightarrow$: Suppose I is an ideal of $\mathfrak{g}$ lying in $\mathfrak{h} = \mathfrak{a}^0(\mathfrak{g})$. This anchors a simple induction showing that $I \subset \mathfrak{a}^k(\mathfrak{g})$ for all k . So if $I \neq 0$, the infinitesimal order has to be infinite. $\square$

Now suppose (A, ∇, τ) is a transitive infinitesimal Cartan geometry over a connected base M . Then (A, τ) is a Lie algebra bundle, with $\ker \#$ a Lie algebra

subbundle. So, we are led to define, for each integer $k \geq 0$, a Lie algebra subbundle $\mathfrak{a}^k(A, \tau) \subset \ker \#$, by applying the above algebraic scheme fibre-wise. Furthermore, $\mathfrak{a}^\nu(A, \tau) = 0$ if and only if $\mathfrak{a}^\nu(\mathfrak{g}) = 0$, where $\mathfrak{g}$ is an arbitrary fibre of (A, τ) and the implicitly understood subalgebra $\mathfrak{h}$ is the corresponding fibre of the anchor kernel. In this case, we say τ is **MODELED ON $\mathfrak{g}$ WITH KERNEL $\mathfrak{h}$** , and we prove:

Theorem. *Let (A, ∇, τ) be a transitive infinitesimal Cartan geometry over a connected base M , and assume the spaces $\mathfrak{a}^k(A, \nabla)$ defined in 6.1 have constant rank. Then for all $k \geq 0$, $\mathfrak{a}^k(A, \nabla) = \mathfrak{a}^k(A, \tau)$. In particular, if τ is modeled on $\mathfrak{g}$ with kernel $\mathfrak{h}$, then the infinitesimal order ν of ∇ is the infinitesimal order of $\mathfrak{g}$ modulo $\mathfrak{h}$.*

Proof. The X appearing in $\text{tor } \bar{\nabla}(A|_m, X)$ in the formula of Proposition 6.2, applying under the rank hypothesis, lies in the anchor kernel of A . This shows that $\text{tor } \bar{\nabla}(Y, X) = -\nabla_{\#Y} X + [Y, X] = -\bar{\nabla}_X Y$, by the definition of $\bar{\nabla}$, 4.2(4). But, by the compatability condition c2 (see 5.2), this is the same as $-\tau(Y, X)$, and the result immediately follows. $\square$

6.4. Jets resolve ambiguities in a geometry's specification. Later, when we consider parabolic geometry, we will construct “normal” model data τ , but the choice will only be unique up to so-called Weyl gauge transformations. These transformations are, in particular, Lie algebroid isomorphisms $q: A \rightarrow A$ covering the identity on M . For any such q , Proposition 5.7 (or a direct check) shows that $\nabla'_V X = q \nabla_V q^{-1} X$ and $\tau'(X, Y) = q\tau(q^{-1}X, q^{-1}Y)$ define a new infinitesimal Cartan geometry (A, ∇', τ') , whenever (A, ∇, τ) is a such a geometry, and these are generally distinct. However, at the level of the representation as semi-holonomic jets, there is no difference:

Proposition. *Let (A, ∇, τ) be an infinitesimal Cartan geometry with finite infinitesimal order ν , and $q: A \rightarrow A$ a Lie algebroid isomorphism covering the identity on M . Let ∇' and τ' be the push-forwards of ∇ and τ under q defined above. Then:*

- (1) *The embeddings $\#_{\nabla}^\nu$ and $\#_{\nabla'}^\nu$, defined in 6.1 have identical image $A_{\text{jet}}^\nu \subset \bar{J}^\nu(TM)$.*
- (2) *The push-forward of ∇ under $\#_{\nabla}^\nu$ to a Cartan connection on A_{jet}^ν is the same as the push-forward of ∇' under $\#_{\nabla'}^\nu$, while the push-forward of τ under $\#_{\nabla}^\nu$ to model data on A_{jet}^ν is the same as the push-forward of τ' under $\#_{\nabla'}^\nu$.*

Proof. Both parts of the proposition follow rather easily from the identity

$$(3) \quad \#_{\nabla'}^\nu \circ q = \#_{\nabla}^\nu,$$

which we prove as follow. Since q covers the identity, (3) holds if $\nu = 0$, which anchors a proof by induction. For, assuming (3) holds for some $\nu = j$, we compute,

with the help of the formula $s_{\nabla'} X = J^1 X + \nabla' X$,

$$\begin{aligned} \#_{\nabla'}^{j+1}(qX) &= (J^1 \#_{\nabla'}^j)(J^1(qX) + \nabla'(qX)) \\ &= J^1(\#_{\nabla'}^j(qX)) + \#_{\nabla'}^j \circ (q \circ \nabla X) \\ &= J^1((\#_{\nabla'}^j \circ q))(J^1 X) + (\#_{\nabla'}^j \circ q) \circ \nabla X \\ &= J^1(\#_{\nabla'}^j)(J^1 X) + \#_{\nabla'}^j \circ \nabla X \\ &= J^1(\#_{\nabla'}^j)(J^1 X + \nabla X) = J^1(\#_{\nabla'}^j)(s_{\nabla} X) = \#_{\nabla}^{j+1} X, \end{aligned}$$

which shows (3) holds when $\nu = j + 1$. $\square$

7. PARABOLIC GEOMETRIES

In this section we define parabolic geometries.

7.1. Parabolic subalgebras and parabolic filtrations. Following [10], we call a Lie subalgebra $\mathfrak{p} \subset \mathfrak{g}$ **PARABOLIC** if it contains the normalizer of a maximal nil subalgebra. Friendlier characterizations in the reductive case are offered shortly. Frequently cited examples of parabolic Lie algebras are: (i) the infinitesimal stabilizers of flags in $\mathbb{F}^n$, which are the parabolic subalgebras of $\mathfrak{gl}(\mathbb{F}, n)$, $\mathbb{F}$ an algebraically closed field; and (ii) the infinitesimal stabilizers of isotropic flags in $\mathbb{R}^{2n+k}$, for some quadratic form of signature $(n+k, n)$, which are the parabolic subalgebras of $\mathfrak{so}(n+k, n)$, $k \geq 1$. The Lie algebra $\mathfrak{so}(n)$ has only itself as a parabolic subalgebra.

Facts about parabolic subalgebras stated below without proof are detailed in Appendix A. In this subsection, $\mathfrak{g}$ is always a finite-dimensional Lie algebra over a field of characteristic zero, such as $\mathbb{R}$ or $\mathbb{C}$.

Theorem (Theorem A.3). *Suppose $\mathfrak{g}$ is **reductive** and fix a non-degenerate trace-form, which always exists. Then $\mathfrak{p} \subset \mathfrak{g}$ is parabolic if and only if its polar $\mathfrak{p}^\perp$, which does not depend on the choice of form, is nilpotent and lies in the semisimple part $[\mathfrak{g}, \mathfrak{g}]$ of $\mathfrak{g}$. In that case $\mathfrak{p}$ is self-normalizing, $\mathfrak{p}^\perp$ is an ideal of $\mathfrak{p}$, and $\mathfrak{p}/\mathfrak{p}^\perp$ is reductive.*

See (6) below and Theorem A.3 for other characterizations of parabolic subalgebras.

Remark. In fact, $\mathfrak{p}^\perp$ will be the nilpotent radical of $\mathfrak{p}$ (see A.1), and $\mathfrak{p}/\mathfrak{p}^\perp$ is known as the **REDUCTIVE LEVI QUOTIENT** of $\mathfrak{p}$. We won't need this here, but it explains the name “Levi structure” introduced later.

To connect parabolic subalgebras with filtrations, we recall a general method for constructing Lie algebra filtrations from more primitive data. For a subalgebra $\mathfrak{n} \subset \mathfrak{g}$, the map $\mathfrak{s} \mapsto \text{ad}_{\mathfrak{n}}(\mathfrak{s}) := [\mathfrak{n}, \mathfrak{s}]$ on subspaces $\mathfrak{s} \subset \mathfrak{g}$ has a right-inverse, $\mathfrak{s} \mapsto \{X \in \mathfrak{g} \mid [\mathfrak{n}, X] \subset \mathfrak{s}\}$. This motivates the construction in the following lemma:

Lemma (Proposition 2.4 of [10]). Let $\mathfrak{g}$ be an arbitrary Lie algebra and $\mathfrak{g}^0, \mathfrak{g}^1 \subset \mathfrak{g}$ subalgebras. Define two sequences of subspaces

$$\cdots \mathfrak{g}^{-3} \supset \mathfrak{g}^{-2} \supset \mathfrak{g}^{-1} \supset \mathfrak{g}^0 \quad \text{and} \quad \mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \mathfrak{g}^3 \supset \cdots$$

by inductively defining $\mathfrak{g}^j = [\mathfrak{g}^1, \mathfrak{g}^{j-1}]$ for $j \geq 2$ and $\mathfrak{g}^j = \{X \in \mathfrak{g} \mid [\mathfrak{g}^1, X] \subset \mathfrak{g}^{j+1}\}$ for $j \leq -1$. We do not require $\mathfrak{g}^0 \supset \mathfrak{g}^1$. However, if $\mathfrak{g}^1 \subset N_{\mathfrak{g}}(\mathfrak{g}^0)$, then $[\mathfrak{g}^j, \mathfrak{g}^k] \subset \mathfrak{g}^{j+k}$ for all $j, k \in \mathbb{Z}$.

In particular, if $\mathfrak{g}^1$ is an ideal of $\mathfrak{g}^0$, then the proposition applies and we call the sequence $\cdots \mathfrak{g}^{-3} \supset \mathfrak{g}^{-2} \supset \mathfrak{g}^{-1} \supset \mathfrak{g}^0 \supset \mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \mathfrak{g}^3 \supset \cdots$ the PRE-FILTRATION associated with $(\mathfrak{g}^0, \mathfrak{g}^1)$. (According to Definition 3.1, it is not a Lie algebra filtration in our sense unless it additionally descends to 0 and ascends to $\mathfrak{g}$.)

Proposition (Proposition A.4, Corollary A.4, Propositions A.6, A.7, A.8, Theorem A.8). *Fix a non-degenerate trace-form on a reductive Lie algebra $\mathfrak{g}$. Then:*

- (1) *For any parabolic subalgebra $\mathfrak{p} \subset \mathfrak{g}$, the pre-filtration of $\mathfrak{g}$ associated with $(\mathfrak{p}, \mathfrak{p}^\perp)$, which is independent of the choice of form, is a bona fide Lie algebra filtration with $|\mathfrak{g}| = m$ for some $m \geq 0$ (see Notation 3.1). This is the CANONICAL FILTRATION associated with $\mathfrak{p}$.*
- (2) *The associated graded $\text{gr}(\mathfrak{g})$ admits a unique grading element $E \in [\text{gr}(\mathfrak{g}), \text{gr}(\mathfrak{g})]$. In particular, $\mathfrak{g}$ and $\text{gr}(\mathfrak{g})$ are isomorphic Lie algebras (by arguments recalled in 10.1).*
- (3) *Modulo $\mathfrak{p}$, the infinitesimal order of $\mathfrak{g}$, in the sense of 6.3, is 2, assuming $\text{gr}(\mathfrak{g})$ contains no non-zero ideals lying in $\text{gr}(\mathfrak{g})_0$. It is infinite otherwise.*
- (4) *Every parabolic subalgebra $\mathfrak{p}$ of $\mathfrak{g}$ is separating, in the sense of 5.5.*
- (5) *Every parabolic subalgebra $\mathfrak{p}$ of $\mathfrak{g}$ contains $Z(\mathfrak{g})$.*
- (6) *A subalgebra $\mathfrak{p}$ is parabolic if and only if there exists a grading of $\mathfrak{g}$ such that $Z(\mathfrak{g}) \subset \mathfrak{g}_0$ and $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$. In that case the corresponding filtration, defined by $\mathfrak{g}^j := \bigoplus_{i \geq j} \mathfrak{g}_i$, is the canonical filtration associated with $\mathfrak{p}$ if and only if one of the following equivalent conditions hold:*
 - $\mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \mathfrak{g}^3 \supset \cdots$ is the lower central series of $\mathfrak{g}^1$.
 - $\mathfrak{g}_1$ generates $\bigoplus_{j \geq 1} \mathfrak{g}_j = \mathfrak{g}^1$.
 - $\mathfrak{g}_{-1}$ generates $\bigoplus_{j \leq -1} \mathfrak{g}_j$.

For any parabolic $\mathfrak{p} \subset \mathfrak{g}$, a grading satisfying all these conditions exists.

In a filtered reductive Lie algebra $\mathfrak{g}$, the subalgebra $\mathfrak{g}^0$ need not be parabolic, even if $Z(\mathfrak{g})$ is trivial ($\mathfrak{g}$ is semisimple): Construct a filtration of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$ with $-1 \leq \mathfrak{g} \leq 0$ by defining $\mathfrak{g}^{-1} = \mathfrak{g}$, and letting $\mathfrak{g}^0$ be the one-dimensional subalgebra of diagonal matrices (which cannot be parabolic, because it does not contain its Killing polar $(\mathfrak{g}^0)^\perp$).

There exist graded semisimple Lie algebras for which the associated filtration is not the canonical filtration associated with the necessarily parabolic subalgebra $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$. For let $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$ and let $\{X_{-\alpha}, H_\alpha, X_\alpha\}$ be the standard basis. Then the grading of $\mathfrak{g}$ by the eigenspaces of ad_{H_α} , with the corresponding eigenvalues $0, \pm 2$ defining the degrees, has the stated property.

Property (5) implies that the infinitesimal order of $\mathfrak{g}$, modulo a parabolic subalgebra $\mathfrak{p}$, cannot be finite unless $\mathfrak{g}$ is semisimple (apply Proposition 6.3).

Definition. An arbitrary filtration of a reductive Lie algebra $\mathfrak{g}$ will be called **PARABOLIC** if $\mathfrak{p} := \mathfrak{g}^0$ is parabolic, and the filtration is the canonical filtration associated with $\mathfrak{p}$.

7.2. Parabolic filtrations and trace-forms. Later we will want to fix a non-degenerate trace-form on a parabolically filtered, reductive Lie algebra by first fixing such a trace-form on the associated graded. For this we need the following result.

Proposition. *If $\mathfrak{g}$ is parabolically filtered, then each trace-form B on $\mathrm{gr}(\mathfrak{g})$ (or non-zero scalar multiple thereof) has a NATURAL LIFT $B^\wedge$ to a (scalar multiple of a) trace-form on $\mathfrak{g}$, non-degenerate if B is non-degenerate. Here naturality means that for any filtration-preserving isomorphism $q: \mathfrak{g} \rightarrow \mathfrak{h}$ between parabolically filtered reductive Lie algebras, we have $q_*(B^\wedge) = (\mathrm{gr}_0(q)_*B)^\wedge$, where $q_*b(X, Y) := b(q^{-1}X, q^{-1}Y)$.*

The natural lift is defined by identifying $\mathfrak{g}$ and $\mathrm{gr}(\mathfrak{g})$ using a splitting defined by some lift $\epsilon \in \mathfrak{g}^0$ of the grading element $E \in [\mathrm{gr}(\mathfrak{g}), \mathrm{gr}(\mathfrak{g})]$ (as described later in 10.1) and by showing the particular choice of ϵ does not matter. Details are provided in A.5.

7.3. Filtered Lie groupoids. We will call a Lie groupoid $\mathcal{G}$ over M **FILTERED** if its associated Lie algebroid A is filtered. Now $\mathcal{G}$ acts on elements of the anchor kernel of A , and so on elements of A^0 ; if A is firmly filtered then this action of $\mathcal{G}$, denoted Ad , will *infinitesimally* preserve A^0 and its filtration, but need not do so globally. However, if it does, then $\mathcal{G}$ itself is said to be **FIRMLY FILTERED**.

7.4. Generalized flag manifolds. Suppose that G is a Lie group with reductive Lie algebra $\mathfrak{g}$, and $P \subset G$ a parabolic subgroup. By definition, this means P has a parabolic Lie algebra $\mathfrak{p} \subset \mathfrak{g}$ and that the associated canonical filtration of $\mathfrak{g}$ is P -invariant under Ad (it is automatically $\mathfrak{p}$ -invariant under ad). The quotient $M = G/P$ is known as a **GENERALIZED FLAG MANIFOLD**. The projective and conformal spheres are examples. Generalized flag manifolds constitute the flat models of parabolic geometries.

The P -invariance of the canonical filtration leads to a firm filtering of the action algebroid $A = \mathfrak{g} \times M$, as described in 3.5. It is easy to see that the action groupoid $\mathcal{G} = G \times M$ is also firmly filtered.

7.5. Parabolic geometries defined. Suppose A is a filtered Lie algebroid. Then model data τ on A is **COMPATIBLE** with the filtration if this is also a filtration with respect to τ , i.e., $\tau(X, Y) \in A^{j+k}$ whenever, $X \in A^j$ and $Y \in A^k$ have a common base point.

A PARABOLIC GEOMETRY is a transitive, firmly filtered, Lie groupoid $\mathcal{G}$ over some smooth manifold M , with associated Lie algebroid denoted A , and filtration-compatible model data τ on A , such that:

- P1.** The Lie algebra model for τ is reductive.
- P2.** A^0 is the kernel of the anchor of A .
- P3.** With respect to τ , the filtration of A is fibre-wise parabolic, in the sense of Definition 7.1.
- P4.** There exists a Cartan connection D on $\mathcal{G}$ compatible with τ .

An (infinitesimal) Cartan connection D (resp., ∇) is COMPATIBLE with the filtration if the filtration is $\bar{D}$ -invariant (resp., $\bar{\nabla}$ -invariant).

Proposition.

- (1) Assuming M is connected, it suffices to check P3 holds over a single point $m \in M$.
- (2) D is necessarily compatible with the filtration of A .
- (3) In P4, it suffices there exists a Cartan connection D that is **weakly** compatible with τ (i.e., τ is $\bar{\nabla}$ -invariant).
- (4) D is uniquely determined by τ .

Proof. For (1), see 15.2, and for (2), see Corollary A of 15.1. By P1, P2, and P3, and Proposition 7.1(4), the model data τ is separating. So (3) and (4) follow from Theorem 5.5. $\square$

Conclusion (4) means that a parabolic geometry is unambiguously a Cartan geometry, in the sense of 5.2.

An INFINITESIMAL PARABOLIC GEOMETRY consists of a transitive, firmly filtered Lie algebroid A , and filtration-compatible model data τ , satisfying the same conditions given above, but with the Cartan connection D in P4 replaced with an infinitesimal Cartan connection ∇ . An infinitesimal analogue of the proposition holds, except that for (1) we need to assume A is integrable. Evidently, every parabolic geometry has a corresponding infinitesimalization.

Parabolic geometries are generalized by FILTERED CARTAN GEOMETRIES, introduced in Section 12.

7.6. Regularity. Let A be any *transitive* filtered Lie algebroid. In that case the anchor $\# : A \rightarrow TM$ induces an isomorphism $\oplus_{i < 0} \text{gr}(A)_i \cong \text{gr}(TM)$, and model data τ compatible with the filtration is said to be REGULAR if the induced bracket on $\text{gr}(TM)$ is minus one times the canonical bracket (the one induced by the Jacobi-Lie bracket on vector fields).¹⁰ It is not too hard to show that the canonical model data associated with any filtered Klein geometry, and in

¹⁰The minus sign is connected to the convention adopted here, and often in the Lie groupoid literature, that brackets on the Lie algebras of Lie groups are defined by *right*-invariant vector fields; most of the parabolic geometry literature, including [12], adopts the opposite convention.

particular the canonical model data associated with a generalized flag manifold, is regular (Proposition 4.5).

A parabolic geometry is REGULAR if the associated model data τ is regular.

8. LEVI STRUCTURES

The concrete manifestations of parabolic geometries — their “underlying geometric structures” — are typically obtained by applying a reduction procedure. Formally, reduction leads to the definition of Levi structures, several examples of which are given in this section.

8.1. Reduction. Let $(\mathcal{G}, \tau)$ be a parabolic geometry and D the compatible Cartan connection. The proof of the following fact, given later in 15.3, is elementary:

Proposition. *Let $\mathcal{G}^1 \subset \mathcal{G}$ denote the source-connected subgroupoid integrating A^1 . Then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups; it is also a normal subgroupoid of $\mathcal{G}$, so that $\mathcal{G}/\mathcal{G}^1$ is a Lie groupoid, also transitive.*

Remark. It follows from Remark 7.1 that A^0/A^1 is fibre-wise reductive, so that $\mathcal{G}/\mathcal{G}^1$ has reductive vertex groups, suitably understood.

Since we suppose $\mathcal{G}$ is firmly filtered, we have a grading-preserving representation, Ad , of $\mathcal{G}$ on $\text{gr}(A^0)$. Moreover, it is easy to see that $\mathcal{G}^1$ acts trivially here (Proposition 10.3(2)). So Ad descends to a representation of $\mathcal{H} := \mathcal{G}/\mathcal{G}^1$ on $\text{gr}(A^0)$. To see that this extends to a representation on $\text{gr}(A)$, we first observe:

Lemma. For any $k \in \mathcal{G}^1$, $\bar{D}_k$ maps A^j to A^{j+1} .

Proof. Since $\mathcal{G}^1$ has connected source fibres, it suffices to prove $\bar{\nabla}_Z$ maps A^j to A^{j+1} , whenever $Z \in A^1$. This follows from compatibility condition c2. $\square$

Since D is compatible with the filtration of A , the representation $\bar{D}$ of $\mathcal{G}$ on A descends to a grading-preserving representation of $\mathcal{G}$ on $\text{gr}(A)$, and by the lemma, $\mathcal{G}^1$ is acting trivially. So $\bar{D}$ drops to a representation of $\mathcal{H}$ on $\text{gr}(A)$, denoted here by R^D . This representation extends the aforementioned representation of $\mathcal{H}$ on $\text{gr}(A^0)$, because D extends Ad .

The observations of the preceding paragraph carry over to the canonical, infinitesimal representation, ad , of A on A^0 , delivering a canonical representation of $B := A/A^1$ on $\text{gr}(A^0)$ and an extension R^∇ of this representation to one of B on $\text{gr}(A)$, which are simply infinitesimalizations of the global ones: $R^\nabla = dR^D$.

If we write $T := \text{gr}_0(\tau)$, then $(\text{gr}(A), T)$ is a graded Lie algebra bundle with the same reductive Lie algebra model as (A, τ) , by Proposition 7.1(2), which shows, moreover, that $\text{gr}(A^1) = \oplus_{i \geq 1} \text{gr}(A)_i$ must be orthogonal to the parabolic $\text{gr}(A^0) = \oplus_{i \geq 0} \text{gr}(A)_i$, for any non-degenerate trace-form for T . By transitivity, we have $\oplus_{i \leq -1} \text{gr}(A)_i \cong \text{gr}(TM)$, so we obtain a representation of $\mathcal{H}$ on $\text{gr}(TM)$. On the other hand, we can use a non-degenerate trace-form for T to identify $\text{gr}(A^1)$ with $(\oplus_{i \leq -1} \text{gr}(A)_i)^* \cong \text{gr}(TM)^* \cong \text{gr}(T^*M)$ (see Lemma A.2(11)). Under this

identification, the representation R^D of $\mathcal{H}$ on $\text{gr}(T^*M)$ will be the dual of the one on $\text{gr}(TM)$, if we suppose that the trace-form is R^D -invariant, which (with the help of transitivity) can always be arranged, on account of the R^D -invariance of T , which follows from the $\bar{D}$ -invariance of τ . Finally, we notice that $A^0/A^1 = B^0$, where B^0 is the anchor kernel of B .

In summary, we have

$$(1) \quad \text{gr}(A) \cong \text{gr}(TM) \oplus B^0 \oplus \text{gr}(T^*M),$$

supporting the reductive Lie algebra bundle structure T (c.f. 3.6 and 3.7) and with $\mathcal{H}$ acting on each component.

Remark. By our preceding Remark, B^0 is a fibre-wise reductive Levi subalgebra of the parabolic $B^0 \oplus \text{gr}(T^*M)$.

8.2. Levi structures. A LEVI STRUCTURE is a certain triple $(\mathcal{H}, T, R)$. Specifically, it consists of a filtered, transitive, Lie groupoid $\mathcal{H}$, whose filtered Lie algebroid B has anchor kernel B^0 and satisfies $B \leq 0$ (see Notation 3.1), together with a reductive Lie algebra bundle structure T on

$$(1) \quad \text{gr}(TM) \oplus B^0 \oplus \text{gr}(T^*M)$$

and a representation R of $\mathcal{H}$ on the same, such that:

- (2) R preserves the obvious grading of (1).
- (3) R extends the canonical representation of $\mathcal{H}$ on the anchor B^0 of its infinitesimalization B , while its restrictions to representations on $\text{gr}(TM)$ and $\text{gr}(T^*M)$ are dual.
- (4) T is R -invariant.
- (5) T extends the canonical Lie algebra bundle structure on B^0 , i.e., $T(C, C') = [C, C']$, while T and R are compatible, in the sense that $T(C, X) = dR_C X$ and $T(C, Z) = -(dR_C)^* Z$. Here $C, C' \in \Gamma(B^0)$, $X \in \Gamma(\text{gr}(TM))$, and $Z \in \Gamma(\text{gr}(T^*M))$.
- (6) The grading of (1) is compatible with T , and, under iterations of the bracket T , $\text{gr}(TM)$ is fibre-wise generated by $\text{gr}(TM)_{-1}$.
- (7) The natural pairing of $\text{gr}(TM)$ and $\text{gr}(T^*M)$ coincides with that induced by some non-degenerate R -invariant trace-form form for T , or some non-zero real multiple thereof.
- (8) With respect to T , the fibre-wise centre of (1) lies in B^0 .

Notice that (8) and Proposition 7.1(6) ensure that $B^0 \oplus \text{gr}(T^*M)$ is fibre-wise T -parabolic; (6) ensures that the filtration defined by the obvious grading of (1) is the associated canonical filtration associated with $B^0 \oplus \text{gr}(T^*M)$. (By Remark 7.1, B^0 is a fibre-wise reductive Levi subalgebra, and in particular, the vertex groups of $\mathcal{H}$ are necessarily reductive.) By Proposition 7.1(2) there will be a unique fibre-wise grading section $E \in \Gamma(B^0)$; the uniqueness and (4) ensure that E is R -invariant.

It is not required that T restrict to the canonical Lie algebra bundle structure on $\text{gr}(TM)$ (here TM is getting the filtration pushed down by the anchor of B). However, if this restriction is the *negative* of the canonical one, then the geometry is **REGULAR**. In that case, one can replace T in condition (6) with the canonical bracket, and a filtered manifold M satisfying the condition is called **FUNDAMENTAL**. The structure of a fundamental filtered manifold is completely determined by the distribution $\mathcal{D} = T^{-1}M$.

Proposition. *Let $\mathcal{H}$, T and B assume the meanings of 8.1, and let $R := R^D$ be the grading-preserving representation of $\mathcal{H}$ on $\text{gr}(TM) \oplus B^0 \oplus \text{gr}(T^*M)$ described there. Then $(\mathcal{H}, T, R)$ is a Levi structure called the **LEVI REDUCTION** of $(\mathcal{G}, \tau)$.*

Proof. Our observations under 8.1 establish everything that is not obvious, with the exception of property (5), which follows from compatability condition c2. $\square$

If B in the definition of a Levi structure is an arbitrary Lie algebroid (not necessarily integrated by some Lie groupoid $\mathcal{H}$) and R now denotes some *infinitesimal* action of B on (1), and we replace dR in (5) with R , then the preceding conditions define an **INFINITESIMAL LEVI STRUCTURE** and, as we will see later, the immediately preceding proposition has an infinitesimal analogue.

8.3. Conformal structures. Let $\mathcal{E}[w]$ denote the line bundle of densities on M of weight $-w/n$, $n = \dim(M)$, and write $TM[w] = TM \otimes \mathcal{E}[w]$. Then we may view a conformal structure (of pure signature) as an inner product $\mathbf{g}$ on the bundle $TM[-1]$, which induces the “index lowering/raising” isomorphisms $X \mapsto X^\flat: TM[w] \rightarrow TM[w-2]$ and $Z \mapsto Z^\sharp: TM[w] \rightarrow TM[w+2]$. We may extend $\mathbf{g}$ to a real quadratic form $(\cdot, \cdot)$ on

$$W := \mathcal{E}[1] \oplus TM[-1] \oplus \mathcal{E}[-1]$$

of signature $(n+1, 1)$, by declaring $\mathcal{E}[1] \perp TM[-1] \perp \mathcal{E}[-1]$ and $(Z, X) = \langle Z, X \rangle := Z(X)$, for $X \in \Gamma(TM)$ and $Z \in \Gamma(T^*M)$. Next, define a linear vector bundle isomorphism ρ , from

$$(1) \quad T^*M \oplus \mathfrak{so}(TM) \oplus TM$$

onto $\mathfrak{so}(W)$, the $\mathfrak{so}(n+1, 1)$ -bundle of fibre-wise isometries of $(\cdot, \cdot)$, by declaring

$$\begin{aligned} \rho(A + \text{aid}_{TM})(s, Y, t) &= (as, \hat{A}Y, -at) \\ \rho(X)(s, Y, t) &= (-\langle X^\flat, Y \rangle, tX, 0) \\ \rho(Z)(s, Y, t) &= (0, -sZ^\sharp, \langle Z, Y \rangle), \end{aligned}$$

for sections A , a , X and Z of $\mathfrak{so}(TM)$, $\mathbb{R} \times M$, TM and T^*M respectively. Here $\hat{A}(sX) := sA(X)$, for $sX \in \Gamma(TM[-1])$. In matrix notation we can write this as

$$\rho(s, A + \text{aid}_{TM}, Z) = \begin{bmatrix} a & -X^\flat & 0 \\ -Z^\sharp & \hat{A} & X \\ 0 & Z & -a \end{bmatrix}.$$

We equip (1) with the obvious grading, concentrated in degrees $-1, 0, 1$, and, to make ρ into a representation of (1) on W , we are led to define the grading-compatible bracket T on (1) generated by the relations

$$\begin{aligned} T(C, X) &= CX, & T(C, Z) &= -C^*Z, & T(C_1, C_2) &= [C_1, C_2], \\ T(X, Z) &= \langle Z, X \rangle \text{id}_{TM} + Z \otimes X - X^\flat \otimes Z^\sharp, \end{aligned}$$

where, to view $X^\flat \otimes Z^\sharp$ as an endomorphism of TM , a contraction of densities is implied. This bracket and the one in 3.7 are the same (we have $\mathbf{g} = g \otimes s_g^2$, where $s_g \in \Gamma(\mathcal{E}[1])$ is the density taking value one on g -orthonormal bases) and makes vector bundle (1) into an $\mathfrak{so}(n+1, 1)$ -bundle.

Since grading components of index i and j are necessarily B_ρ -orthogonal if $i+j \neq 0$ (Proposition A.2(7)) the following readily computed formulas sufficiently describe the trace-form B_ρ :

$$B_\rho(C_1, C_2) = 2 \text{trace}(C_1 C_2), \quad B_\rho(X, Z) = 2 \langle Z, X \rangle.$$

This shows that B_ρ is non-degenerate, and we satisfy (7), by adopting $B_\rho/2$ as our form.

We let $\mathcal{H} = \text{CSO}[TM]$, whose filtered Lie algebroid B has already been described in 3.7. Letting R denote the obvious grading-preserving representation of $\mathcal{H}$ on (1), we obtain a Levi structure $(\mathcal{H}, T, R)$.

In the next example the base manifold M is non-trivially filtered:

8.4. n -plane fields with growth vector $(n, n(n+1)/2)$. Consider, on a smooth manifold of dimension $n(n+1)/2$, an n -plane field $\mathcal{D} \subset TM$ that is maximally non-integrable. That is, at every $m \in M$, local sections of $\mathcal{D}$ and their iterated Jacobi-Lie brackets, generate $T_m M$. Unlike contact structures, such an n -plane field is a geometric structure of finite type, admitting a classical normalized Cartan connection, as Bryant already showed when $n = 3$ [7, 8].

The quotient $TM/\mathcal{D}$ and $\wedge^2 \mathcal{D}$ have the same rank, and, by the assumption of maximal non-integrability, the morphism $V_1 \wedge V_2 \mapsto [V_1, V_2] \bmod \mathcal{D}: \wedge^2 \mathcal{D} \rightarrow TM/\mathcal{D}$ is surjective, and whence an isomorphism. Taking $T^{-1}M = \mathcal{D}$ and $T^{-2}M = TM$, we make TM into a filtered Lie algebroid with $-2 \leq TM \leq -1$ and, on account of the isomorphism, have the identification $\text{gr}(TM) \cong \wedge^2 \mathcal{D} \oplus \mathcal{D}$ with grading-compatible bracket characterized by

$$(1) \quad [X_1, X_2] = X_1 \wedge X_2; \quad X_1, X_2 \in \Gamma(\mathcal{D}).$$

To construct a regular Levi structure from this data, equip

$$W := \mathcal{D} \oplus (\mathbb{R} \times M) \oplus \mathcal{D}^*$$

with a real quadratic form $(\cdot, \cdot)$, of signature $(n+1, n)$, by declaring $\mathcal{D}$ and $\mathcal{D}^*$ to be isotropic, $\mathcal{D} \perp (\mathbb{R} \times M) \perp \mathcal{D}^*$, $(Z, X) = Z(X)$, and $(f_1, f_2) = f_1 f_2$; here

X lies in $\mathcal{D}$, Z in $\mathcal{D}^*$ and f_1, f_2 are smooth real functions. Next, define a linear vector bundle isomorphism ρ , from

$$(2) \quad \wedge^2 \mathcal{D} \oplus \mathcal{D} \oplus \mathfrak{gl}(\mathcal{D}) \oplus \mathcal{D}^* \oplus \wedge^2 \mathcal{D}^* \cong \text{gr}(TM) \oplus \mathfrak{gl}(\mathcal{D}) \oplus \text{gr}(T^*M)$$

onto $\mathfrak{so}(W)$, the $\mathfrak{so}(n+1, n)$ -bundle of fibre-wise isometries of $(\cdot, \cdot)$, by declaring

$$\begin{aligned} \rho(C)(Y, f, W) &= (CY, 0, -C^*W) \\ \rho(X)(Y, f, W) &= (fX, -W(X), 0) \\ \rho(X_1 \wedge X_2)(Y, f, W) &= (W(X_2)X_1 - W(X_1)X_2, 0, 0) \\ \rho(Z)(Y, f, W) &= (0, Z(Y), -fZ) \\ \rho(Z_1 \wedge Z_2)(Y, f, W) &= (0, 0, Z_1(Y)Z_2 - Z_2(Y)Z_1). \end{aligned}$$

Here C is a section of $\mathfrak{gl}(\mathcal{D})$, X, X_1, X_2, Y are sections of $\mathcal{D}$ and Z, Z_1, Z_2, W are sections of $\mathcal{D}^*$. In matrix notion we can write this as

$$\rho(X_1 \wedge X_2, X, C, Z, Z_1 \wedge Z_2) = \begin{bmatrix} C & X & -X_1 \wedge X_2 \\ Z & 0 & -X^{**} \\ Z_1 \wedge Z_2 & -Z^{**} & -C^* \end{bmatrix},$$

where on the right, $X_1 \wedge X_2, Z_1 \wedge Z_2, X^{**}$ and Z^{**} are interpreted, in the obvious way, as sections of $\text{Hom}(\mathcal{D}^*, \mathcal{D})$, $\text{Hom}(\mathcal{D}, \mathcal{D}^*)$, $\text{Hom}(\mathcal{D}^*, (\mathbb{R} \times M))$ and $\text{Hom}((\mathbb{R} \times M), \mathcal{D}^*)$ respectively. We equip (2) with the obvious grading, concentrated in degrees $-2, -1, 0, 1, 2$, and, to make ρ into a representation of (2) on W , we are led to define the grading-compatible bracket T on (2) generated by the relations

$$\begin{aligned} T(X_1, X_2) &= -X_1 \wedge X_2, & T(C, X) &= CX, \\ T(C, X_1 \wedge X_2) &= CX_1 \wedge X_2 + X_1 \wedge CX_2, \\ T(Z_1, Z_2) &= Z_1 \wedge Z_2, & T(C, Z) &= -C^*Z, \\ T(C, Z_1 \wedge Z_2) &= -C^*Z_1 \wedge Z_2 - Z_1 \wedge C^*Z_2, \\ T(C_1, C_2) &= [C_1, C_2], & T(X, Z) &= Z \otimes X, \\ T(X_1 \wedge X_2, Z_1 \wedge Z_2) &= Z_1(X_1)Z_2 \otimes X_2 + Z_2(X_2)Z_1 \otimes X_1 \\ &\quad - Z_2(X_1)Z_1 \otimes X_2 - Z_1(X_2)Z_2 \otimes X_1, \\ T(X_1 \wedge X_2, Z) &= Z(X_1)X_2 - Z(X_2)X_1, & T(Z_1 \wedge Z_2, X) &= Z_2(X)Z_1 - Z_1(X)Z_2. \end{aligned}$$

This makes (2) into a $\mathfrak{so}(n+1, n)$ -bundle. In passing, we note that $\mathfrak{gl}(\mathcal{D}) \oplus \mathcal{D}^* \oplus \wedge^2 \mathcal{D}^*$ is the stabilizer of the maximally isotropic subbundle $\mathcal{D}^* \subset W$.

Since grading components of index i and j are necessarily B_ρ -orthogonal if $i+j \neq 0$ (Proposition A.2(7)) the following readily computed formulas sufficiently

describe the trace-form B_ρ :

$$\begin{aligned} B_\rho(C_1, C_2) &= 2 \operatorname{trace}(C_1 C_2) \\ B_\rho(Z, X) &= 2Z(X) \\ B_\rho(Z_1 \wedge Z_2, X_1 \wedge X_2) &= 2Z_1(X_1)Z_2(X_2) - 2Z_1(X_2)Z_2(X_1). \end{aligned}$$

This shows that B_ρ is non-degenerate, and we satisfy (7), by adopting $B_\rho/2$ as our form.

The frame groupoid $\mathcal{H}$ of $\mathcal{D}$ acts on each summand in (2) in an obvious way, and hence on (2) in a grading-preserving way — we denote this representation by R . The filtration of TM pulls back, via anchor, to a trivially firm filtration of the Lie algebroid B of $\mathcal{H}$, with the additional understanding that B^0 is the anchor kernel, and $B^j = 0$ for positive j . The triple $(\mathcal{H}, R, T)$ is now easily seen to be Levi structure. Note we have made choices in the definition of ρ that ensure the sign in the first relation for T is opposite to that in (1), to ensure regularity.

Unfortunately for us, the analogue of a *global* Levi structure is called an “infinitesimal flag structure” in [12]. Moreover, from the present point of view, “infinitesimal flag structure” seems a name more apt for the action algebroids described in 3.5, specialized to the parabolic case. To avoid confusion, we eschew the term altogether.

9. EXISTENCE, UNIQUENESS, AND EQUIVALENCE

We now state the main theorems concerning the existence and uniqueness of parabolic geometries with given Levi reduction, which culminate in an equivalence between certain categories of parabolic geometries and Levi structures. Proofs of these results are taken up in subsequent sections, as indicated by references to later restatements.

9.1. Existence: prolonging a Levi structure to a parabolic geometry. Our proof of the following basic existence theorem is new:

Theorem (Theorem 15.4). *Every Levi structure $(\mathcal{H}, T, R)$ is, up to isomorphism, the reduction of some parabolic geometry $(\mathcal{G}, \tau)$. If the Levi structure is regular, then the parabolic geometry can be chosen to be regular.*

Morphisms of Levi structures are defined in 15.9.

The model data τ is not uniquely defined by the Levi structure, but determined up to choices of a suitable tensor $P \in \Gamma(\operatorname{gr}(T^*M) \otimes \operatorname{gr}(T^*M))$ called the RHO TENSOR, splitting of the anchor $\#: B \rightarrow TM$, and splitting of the filtration of TM . Here B is the Lie algebroid of $\mathcal{H}$. Suitable normalization of this data is addressed in 9.3 below.

The key idea in the theorem’s proof is to observe that every parabolic geometry $(\mathcal{G}, \tau)$ has a semi-direct product structure, which follows from the guaranteed

existence of Weyl gauges, a notion introduced in Section 10. Specifically, the representation R^D of $\mathcal{G}/\mathcal{G}^1$ on $\mathrm{gr}(T^*M)$, defined by the τ -compatible Cartan connection D , integrates to an action of $\mathcal{G}/\mathcal{G}^1$ on $\exp(\mathrm{gr}(T^*M))$, the Lie group bundle with simply-connected nilpotent model that integrates the nilpotent Lie algebra bundle $\mathrm{gr}(T^*M)$; and one obtains, after fixing a gauge, an isomorphism

$$\mathcal{G} \cong \mathcal{G}/\mathcal{G}^1 \times_{R^D} \exp(\mathrm{gr}(T^*M)),$$

where $\times_{R^D}$ indicates semi-direct product. This follows from Theorem 10.3, given later.

To recover the underlying Lie groupoid $\mathcal{G}$ associated with some parabolic geometry having $(\mathcal{H}, T, R)$ as reduction, we accordingly define $\mathcal{G} = \mathcal{H} \times_R \exp(\mathrm{gr}(T^*M))$, which we call the **PROLONGATION** of the Levi structure. The more technical task is to reconstruct suitable model data τ such that $\mathrm{gr}_0(\tau) = T$. In fact, we state and prove a more general result, Theorem 12.6, characterizing arbitrary transitive filtered Cartan geometries. This result includes an explicit description of τ and a compatible Cartan connection D , in terms of the aforementioned Rho tensor P and splittings.

9.2. Weyl gauge transformations. The “inner” automorphisms of an arbitrary Lie groupoid are those given by conjugation by bisections. When $\mathcal{G}$ is firmly filtered, a special role is played by those inner automorphisms generated by bisections of $\mathcal{G}^1 \subset \mathcal{G}$, the bundle of simply-connected nilpotent Lie groups integrating the totally intransitive Lie subalgebroid $A^1 \subset A$, where a bisection is the same thing as a section. We call such automorphisms **WEYL GAUGE TRANSFORMATIONS**, for reasons explained in 10.4. By Proposition 8.1, there is, for each such transformation, some $Z \in \Gamma(A^1)$ such that each arrow $g \in \mathcal{G}$, beginning at some $m \in M$, is mapped to $\exp(Z(gm))g \exp(-Z(m))$. This conjugation has derivative

$$X \mapsto e^{-\mathrm{ad}_Z} X := X - [Z, X] + \frac{1}{2!}[Z, [Z, X]] - \cdots,$$

and morphisms of a filtered Lie algebroid A of this form will be called **INFINITESIMAL WEYL TRANSFORMATIONS**.¹¹ Both global and infinitesimal Weyl gauge transformations cover the identity of the base M , and $e^{-\mathrm{ad}_Z}$ is filtration-preserving, by F1. The corresponding map $\mathrm{gr}_0(q): \mathrm{gr}(A) \rightarrow \mathrm{gr}(A)$ is the identity. Evidently, in a global parabolic geometry, every infinitesimal Weyl transformation is integrated by a unique global Weyl gauge transformation.

Example. Let A be a filtered Lie algebroid with $|A| = 1$. Then the infinitesimal Weyl gauge transformations have the form $X \mapsto X + [Z, X]$, for some $Z \in \Gamma(A^1)$.

Proposition. *If $(\mathcal{G}, \tau)$ is a (regular) parabolic geometry, and τ' the push-forward of τ under a Weyl gauge transformation Q , then $(\mathcal{G}, \tau')$ is another (regular) parabolic geometry with the same Levi reduction.*

¹¹The minus sign enters here because of our convention that right-invariant vector fields define the Lie bracket on the Lie algebra of a Lie group.

Proof. If τ is model data then τ' is model data, regular if τ' is. This is elementary. Since $\text{gr}_0(q) = \text{id}$, we have $\text{gr}_0(\tau') = \text{gr}_0(\tau)$. Suppose D is the unique Cartan connection compatible with τ . Then Proposition 5.7 shows that the push-forward D' of D defined by formula 5.7(2) is a Cartan connection compatible with τ' , and so is the unique such Cartan connection. From the formula, the general form of a Weyl gauge transformation given above, and Lemma 8.1, it is not too hard to see that $R^{D'} = R^D$. $\square$

9.3. Normalization of curvature. We next address certain freedom in the choice of model data τ in Theorem 9.1. We show that we can choose τ so as to “normalize” the model curvature κ , without changing the underlying Levi structure $(\mathcal{H}, R, T)$, which we now regard as fixed. The model curvature κ , defined in 5.6, is a section of $\wedge^2 T^*M \otimes A$. However, to describe normalization, we need a different representation of κ . To that end, we fix some convenient multiple of an R -invariant non-degenerate trace-form on $\text{gr}(A) \cong \text{gr}(TM) \oplus B \oplus \text{gr}(T^*M)$. Such a form exists, by Axiom 8.2(7). Moreover, by the R -invariance and transitivity, this form is completely determined by its restriction to a single fibre of $\text{gr}(A)$. Applying Proposition 7.2, our bilinear form has a natural lift to a non-degenerate form b^τ on the Lie algebra bundle (A, τ) , for any choice of τ with $\text{gr}_0(\tau) = T$, and the R -invariance of the form on $\text{gr}(A)$ translates to $\bar{D}$ -invariance of b^τ (and hence $\bar{\nabla}$ -invariance) where D is the unique Cartan connection on $\mathcal{G}$ compatible with τ (and ∇ its infinitesimalization).

Since the filtration of A is parabolic (Axiom P3), we have $A^1 = (A^0)^\perp$, where we may take $\perp$ with respect to b^τ , and this leads to an identification $(A/A^0)^*$ with A^1 . But, by transitivity, $A/A^0 \cong TM$, because A^0 is the anchor kernel (Axiom P2), and we conclude $T^*M \cong A^1$. In this way, we may view κ as a section of $\wedge^2 A^1 \otimes A$.

Now, as the bracket τ makes the vector bundle A into a representation of the Lie algebra bundle A^1 , $\wedge^2 A^1 \otimes A$ appears in a natural chain complex

$$(1) \quad 0 \xleftarrow{\partial^\tau} A \xleftarrow{\partial^\tau} A^1 \otimes A \xleftarrow{\partial^\tau} \wedge^2 A^1 \otimes A \xleftarrow{\partial^\tau} \dots$$

Over each point $m \in M$, this is the classical chain complex associated with the representation ad^τ of the Lie algebra $A^1|_m$ on $A|_m$. Following [12], we will refer to ∂^τ as the KOSTANT CODIFFERENTIAL. Basic facts about such complexes immediately carry over to the case of Lie algebra bundle representations and we use such facts without further comment. By definition (c.f., [12, p. 262]),

$$(2) \quad \begin{aligned} \partial^\tau(Y_1 \wedge \dots \wedge Y_{n+1} \otimes X) &= \sum_{i=1}^{n+1} (-1)^i Y_1 \wedge \dots \wedge \widehat{Y}_i \wedge \dots \wedge Y_{n+1} \otimes \tau(Y_i, X) \\ &\quad + \sum_{i < j} (-1)^{i+j} \tau(Y_i, Y_j) \wedge Y_1 \wedge \dots \wedge \widehat{Y}_i \wedge \dots \wedge \widehat{Y}_j \wedge \dots \wedge Y_{n+1} \otimes X, \end{aligned}$$

where hats indicate omission. For example, $\partial^\tau(Y_1 \wedge Y_2 \otimes X) = Y_1 \otimes \tau(Y_2, X) - Y_2 \otimes \tau(Y_1, X) - \tau(Y_1, Y_2) \otimes X$. Incidentally, appealing to compatibility Axiom c2 (see 5.2), we can express ∂^τ in terms of the representation $\bar{\nabla}$ instead,

$$\begin{aligned} \partial^\tau(Y_0 \wedge \cdots \wedge Y_n \otimes X) &= \sum_{i=1}^{n+1} (-1)^{i+1} Y_1 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \wedge Y_{n+1} \otimes \bar{\nabla}_{Y_i} X \\ &\quad + \sum_{i < j} (-1)^{i+j} [Y_i, Y_j] \wedge Y_1 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \wedge \widehat{Y_j} \wedge \cdots \wedge Y_{n+1} \otimes X, \end{aligned}$$

but emphasize ∂^τ is not a differential operator, only an algebraic one.

We say model data τ in a parabolic geometry is **NORMAL** if $\partial^\tau \kappa = 0$, where κ is the curvature of τ . Our proof of the following is an adaption of that of [12, Propositions 3.1.10, 3.1.13]:

Theorem (Theorem 15.6). *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry. Then there exists **normal**, model data τ' such that $(\mathcal{G}, \tau')$ is a regular parabolic geometry with the same Levi reduction.*

9.4. Harmonic curvature. In a regular parabolic geometry with normal model data τ , the homology class $\kappa_H := [\kappa] \in H_2(A^1, A)$ is called the **HARMONIC CURVATURE**. While it seems the harmonic curvature contains less information, it is nevertheless a complete obstruction to flatness, as the corollary below shows.

It is not too hard to see that the filtration of A induces a filtration of $H_k(A^1, A)$. However, according to (2) below, something much stronger and useful is true:

Theorem (Proposition 14.1, Proposition 15.7). *Let κ be the model curvature of a regular and normal parabolic geometry $(\mathcal{G}, \tau)$, and let $k \geq 0$ be any integer. Then:*

- (1) *In a natural representation of $\mathcal{G}$ on $H_k(A^1, A)$ defined by the compatible connection D , the subgroupoid $\mathcal{G}^1$ acts trivially, i.e., $H_k(A^1, A)$ is a representation of $\mathcal{G}/\mathcal{G}^1$.*
- (2) *$H_k(A^1, A)$ admits a canonical **grading** compatible with its natural filtration, allowing harmonic curvature to be unambiguously split into grading components, $\kappa_H = \sum_{i \geq 1} (\kappa_H)_i$.*
- (3) *If κ is homogeneous of degree $l \geq 1$ and $(\kappa_H)_l = 0$, then κ is homogeneous of degree $l + 1$.*

One says κ is **HOMOGENEOUS OF DEGREE l** if κ can be written as a finite sum of terms of the form $X \wedge Y \otimes Z$ where X, Y, Z lie in A^j, A^k, A^m , respectively, with $j + k + m \leq l$; equivalently, if we view κ as a section of $\wedge^2 T^*M \otimes A$, if $\kappa(X, Y)$ lies in A^{j+k+l} whenever X and Y lie in A^j and A^k respectively. See also [12, §3.1.1, p. 235].

We obtain the following by applying (3) inductively:

Corollary (c.f. Theorem 3.1.12, p. 265, [12]). *In a normal, regular parabolic geometry, $\kappa_H = 0$ implies $\kappa = 0$, and hence also $\text{curv } \nabla = 0$.*

The harmonic curvature κ_H is a section of $H_2(A^1, A)$, which is a representation of a transitive groupoid, and can accordingly be understood as an associated bundle; its splitting into grading components can be understood by studying a single fibre, which will be a representation of the (reductive) vertex group of $\mathcal{G}/\mathcal{G}^1$. More generally, each fibre of $H_j(A^1, A)$, $j \geq 0$, is of the form $H_j(\mathfrak{g}^1, \mathfrak{g})$, where $\mathfrak{g}$ is the reductive Lie algebra model for the Lie algebra bundle structure τ , which gets a canonical filtration, because A is parabolically filtered; the subalgebra $\mathfrak{g}^0$ will be parabolic, its orthogonal (with respect to any non-degenerate trace-form) is $\mathfrak{g}^1$. Fixing a lift ϵ of the unique grading element $E \in \text{gr}(\mathfrak{g})$, we can write $\mathfrak{g}$ as a graded Lie algebra, $\mathfrak{g}_{-k} \oplus \cdots \oplus \mathfrak{g}_k$, $k \geq 0$ (Proposition 7.1(2)). This allows us to introduce the cohomology $H^j(\mathfrak{g}_-, \mathfrak{g})$, where $\mathfrak{g}_- = \mathfrak{g}_{-k} \oplus \cdots \oplus \mathfrak{g}_{-1}$. (This space, in turn, is the model for the bundle of cohomologies for a cochain complex $\wedge^\bullet \text{gr}(T^*M) \otimes \text{gr}(A)$ that will enter in the proof of the uniqueness result described next.) According to (a mild generalization) of Kostant's version of Hodge theory in this setting [19] (see also Proposition 14.4), there is an identification $H_j(\mathfrak{g}^1, \mathfrak{g}) \cong H^j(\mathfrak{g}_-, \mathfrak{g})$, which does not depend on the choice of ϵ , and the grading of the space on the right gives a filtration-compatible grading of the space on the left, corresponding exactly to the grading in (2) (but for which we will give a direct argument).

In the semisimple case, Kostant's theorem, restated in [12, Theorem 3.3.5, p. 351], computes the cohomology $H^j(\mathfrak{g}_-, \mathfrak{g})_l$, for each $l \in \mathbb{Z}$, for complex $\mathfrak{g}$; some additional arguments extend the result to the real case. We refer the reader to [12] for details we do not reproduce here. It seems one ought to be able to generalize this computation to the reductive case, but we have not attempted this.

9.5. Uniqueness up to Weyl gauge transformations. According to the next result, normal model data is uniquely determined, up to Weyl gauge transformations, provided a Lie cohomological condition is satisfied. Our proof of the following is an adaption of that of [12, Proposition 3.1.14]:

Theorem (Theorem 15.8). *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry and let $\mathfrak{g}$ be a Lie algebra model for τ , identified with a graded reductive Lie algebra, as detailed at the end of 9.4. Assume $H^1(\mathfrak{g}_-, \mathfrak{g})^1 = 0$. Then if $(\mathcal{G}, \tau')$ is second parabolic geometry with $\text{gr}_0(\tau') = \text{gr}_0(\tau)$ and both τ and τ' are normal, then $\tau' = q_*\tau$ for some infinitesimal Weyl gauge transformation q .*

Here $H^1(\mathfrak{g}_-, \mathfrak{g})^1 = \oplus_{l \geq 1} H^1(\mathfrak{g}_-, \mathfrak{g})_l$.

Remark. The non-uniqueness is really an artifact of our abstract formalism (and the same holds in the principal bundle approach). Passing to a representation of the Cartan geometry as a bundle of semi-holonomic jets, the ambiguity in the choice of τ disappears (Proposition 6.4).

9.6. The equivalence problem. The parabolic geometries over M form a category if we define a morphism of parabolic geometries $(\mathcal{G}, \tau)$ and $(\tilde{\mathcal{G}}, \tilde{\tau})$ over M

to be a Lie groupoid morphism $Q: \mathcal{G} \rightarrow \tilde{\mathcal{G}}$ covering the identity, whose infinitesimalization $q = dQ: A \rightarrow \tilde{A}$ is filtration-preserving and pushes τ to $\tilde{\tau}$. One can similarly make the collection of Levi structures over M into a category; for details see 15.9. As for parabolic geometries, we can similarly identify a graded reductive Lie algebra model $\mathfrak{g}$ for the Lie algebra bundle structure T of a Levi structure $(\mathcal{H}, R, T)$. The Lie algebra $\mathfrak{g}$ we associate with a Levi reduction can be taken to be as the same Lie algebra $\mathfrak{g}$ we use for the parabolic geometry that we are reducing. In a parabolic geometry the condition

$$(1) \quad H^1(\mathfrak{g}_-, \mathfrak{g})^1 = 0$$

holds if and only if it holds for every geometry in the same isomorphism class, and the same is true for Levi structures.

Combining Theorems 9.1, 9.3 and 9.5, and appealing to the fact that prolongation is a functor (Theorem 15.9), we establish the following solution to the equivalence problem for certain Levi structures:

Theorem (Theorem 15.10). *The map that sends a regular normal parabolic geometry over M to its corresponding Levi reduction, induces a bijection between: (i) isomorphism classes of regular normal parabolic geometries over M satisfying (1); and (ii) isomorphism classes of regular Levi structures over M satisfying (1).*

10. WEYL GAUGES AND SEMI-DIRECT PRODUCT STRUCTURE

We now return to the general study of filtered Lie algebroids introduced in Section 3, our main goal being the generalization of Theorem 1.4 to Lie algebroids and Lie groupoids. The idea of the proof of Theorem 1.4 is to show that the stabilizer $B \subset \mathfrak{a}$ of ϵ is a complement for $\mathfrak{a}^1$ in $\mathfrak{a}$, and that B preserves the grading of $\mathfrak{a}^1$ determined by ϵ , under the adjoint representation. We suppress details, as we will provide them for the generalizations.

The groupoids and algebroids in this section are not required to be transitive and the only class of Lie groups or algebras playing any special role are nilpotent ones. In particular, nothing here needs to be reductive or semisimple.

10.1. Splittings of a filtered Lie algebra. If $\mathfrak{g}$ is a finite-dimensional filtered Lie algebra, then the existence of a grading element $E \in \text{gr}(\mathfrak{g})$ has strong consequences for the Lie algebra. For, choosing any $\epsilon \in \mathfrak{g}^0$ such that $\text{gr}_0(\epsilon) = E$, one can show, using induction and $\cap_j \mathfrak{g}^j = 0$, that the maps

$$X \mapsto [\epsilon, X] - jX: \mathfrak{g}^j \rightarrow \mathfrak{g}^{j+1}$$

are well-defined projections, whose kernels constitute a grading of $\mathfrak{g}$ for which ϵ is a grading element; this in turn gives rise to an isomorphism $\Phi_\epsilon: \text{gr}(\mathfrak{g}) \rightarrow \mathfrak{g}$ which must in fact be a Lie algebra isomorphism. When we alter the choice of ϵ , the isomorphism Φ_ϵ changes in a way made precise in (2) below.

We call $\Phi_\epsilon: \text{gr}(\mathfrak{g}) \rightarrow \mathfrak{g}$ an *E-splitting* of the filtration. With E and ϵ fixed as above, there is evidently a one-to-one correspondence between E -splittings and the affine space $\epsilon + \mathfrak{g}^1$, because $\Phi_{\epsilon'}(E) = \epsilon'$ for all $\epsilon' \in \epsilon + \mathfrak{g}^1$.

Now $\mathfrak{g}^1$ is a nilpotent Lie algebra, integrated by a simply connected nilpotent Lie group, which we denote by $\exp(\mathfrak{g}^1)$; nilpotency ensures the exponential map is a diffeomorphism. Part (1) of the lemma below shows that $\exp(\mathfrak{g}^1)$ acts freely and transitively on the set of E -splittings by constructing a concrete realization G^1 . A proof of this known result appears in Appendix B.

Lemma. Let $\epsilon \in \mathfrak{g}^0$ be a lift of some grading element $E \in \text{gr}(\mathfrak{g})$. Let G^0 be *any* Lie group integrating $\mathfrak{g}^0$, and let $G^1 \subset G^0$ be the connected Lie subgroup with Lie algebra $\mathfrak{g}^1$. Then:

- (1) Under the adjoint action, G^1 acts freely and transitively on the affine space $\epsilon + \mathfrak{g}^1 \subset \mathfrak{g}^0$, and consequently on E -splittings of $\mathfrak{g}$. In particular, G^1 is simply connected (i.e., $G^1 \cong \exp(\mathfrak{g}^1)$) so that the infinitesimal adjoint action of $\mathfrak{g}^1$ on $\mathfrak{g}$ has a unique globalization $\text{Ad}: G^1 \rightarrow \text{Aut}(\mathfrak{g})$ extending the adjoint action just mentioned.
- (2) For any $\epsilon_1, \epsilon_2 \in \epsilon + \mathfrak{g}^1$, we have $\Phi_{\epsilon_2} = \text{Ad}_g \circ \Phi_{\epsilon_1}$, where $g \in G^1$ is the unique element for which $\epsilon_2 = \text{Ad}_g \epsilon_1$.

10.2. Weyl gauges for filtered Lie algebroids. A WEYL GAUGE for a filtered Lie algebroid A will be any section ϵ of A^0 such that $E = \text{gr}_0(\epsilon)$ is a fibre-wise grading section of the bundle of Lie algebras $\text{gr}(A^0)$, and

$$(1) \quad [X, \epsilon] \in A^1 \text{ for all } X \in A.$$

If A is firmly filtered, then the canonical action of A on its anchor kernel determines an action ad of A on $\text{gr}(A^0)$. Equation (1) then holds if and only if E is ad -invariant. That is, in the firm case, Weyl gauges exist if and only if an ad -invariant fibre-wise grading section of $\text{gr}(A^0)$ exists, and any lift to A^0 of such a section is a Weyl gauge. In fact, according to the corollary stated below all Weyl gauges arise in this way.

Note that we do not require that E be a fibre-wise grading section of $\text{gr}(A)$, only of $\text{gr}(A^0)$ (but see also 11.2).

Theorem. Let A be a filtered Lie algebroid admitting a Weyl gauge ϵ , and let $B \subset A$ denote the isotropy of ϵ under the canonical representation of A on its anchor kernel, i.e., the kernel of the map $X \mapsto [X, \epsilon]: A \rightarrow A^1$. Then:

- (2) A is firmly filtered, and in particular, $A^1 \subset A$ is an ideal (i.e., $\Gamma(A^1) \subset \Gamma(A)$ is an ideal).
- (3) A is the internal semi-direct product of B and A^1 .
- (4) The grading $A^0 = \bigoplus_{i \geq 0} \mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of B . In particular, the splitting $\Phi_\epsilon: \text{gr}(A^1) \rightarrow A^1$ intertwines the representation of $A/A^1 \cong B$ on $\text{gr}(A^1)$ with the representation of B on A^1 , so that A is the external semi-direct product of A/A^1 and $\text{gr}(A^1)$.

- (5) Under the identification $A \cong A/A^1 \oplus_{\text{ad}} \text{gr}(A^1)$ of (4), the Weyl gauge ϵ takes the form $\epsilon = (E, 0)$, for some section E of $A^0/A^1 \subset A/A^1$ that commutes under Lie algebroid bracket with all sections of A/A^1 ; and $\text{ad}_E Y = jY$ for any $Y \in \text{gr}(A^0)_j$.

Note that the identification $A \cong A/A^1 \oplus_{\text{ad}} \text{gr}(A^1)$ transfers to $A/A^1 \oplus_{\text{ad}} \text{gr}(A^1)$ the following filtration: $(A/A^1 \oplus_{\text{ad}} \text{gr}(A^1))^i = (A^i/A^0) \oplus_{\text{ad}} (\oplus_{j \geq \max\{1, i\}} \text{gr}(A)_j)$.

Example (projective geometry). In example 3.6 the Lie algebroid $A \subset J^2(TM)$ acts on TM by projecting onto $J^1(TM)$ and applying the usual adjoint action, $\text{ad}_{J^1V} W = [V, W]$. This gives us associated representations of A on $\mathfrak{gl}(TM)$ and T^*M . On each component of $\text{gr}(A^0) = \mathfrak{gl}(TM) \oplus T^*M$ the canonical action of A is given by these representations, which shows that $E = -\text{id} \in \mathfrak{gl}(TM) \subset \text{gr}(A^0)$ is ad-invariant. Consequently, any lift ϵ of $-\text{id}$ through $A^0 \rightarrow A^0/A^1 = \mathfrak{gl}(TM)$ is a Weyl gauge for A . According to the theorem, each such gauge defines an identification of A with the semi-direct product $J^1(TM) \oplus_{\text{ad}} T^*M$.

Proof of theorem. The isotropy $B \subset A$ is a subalgebroid, by Jacobi's identity. We first prove

$$(6) \quad A = B \oplus A^1,$$

as vector bundles. It suffices to show that the restriction r of $X \mapsto [X, \epsilon]$ to a map $A^1 \rightarrow A^1$ is an isomorphism. However, by F1, this restriction is filtration-preserving, and the corresponding map $\text{gr}(A^1) \rightarrow \text{gr}(A^1)$ sends $\xi \in \text{gr}(A^1)_j$ to $j\xi$, because $\text{gr}_0(\epsilon)$ is a fibre-wise grading section of $\text{gr}(A^0)$. As this is obviously an isomorphism, r must be too.

Next we prove the grading $A^0 = \oplus_{i \geq 0} \mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of B . To that end, suppose $Y \in \Gamma(\mathfrak{a}_j)$ for some $j \geq 0$ and let $X \in \Gamma(B)$ be arbitrary. The Jacobi identity for the bracket on A gives us

$$[X, [\epsilon, Y]] = [[X, \epsilon], Y] + [\epsilon, [X, Y]].$$

On the other hand, applying ad_X to both sides of the identity $[\epsilon, Y] = jY$, we have

$$[X, [\epsilon, Y]] = j \text{ad}_X Y.$$

Combining these two results, we obtain

$$[[X, \epsilon], Y] + [\epsilon, \text{ad}_X Y] = j \text{ad}_X Y.$$

But $[X, \epsilon] = 0$, because $X \in \Gamma(B)$. We conclude that $\text{ad}_X Y \in \Gamma(\mathfrak{a}_j)$.

Having proven, in particular, that B preserves the filtration of A^0 , to establish (2) it suffices, in view of (6), to show that the filtration of A^0 is invariant under the canonical action of itself, which follows immediately from F1.

Claim (3) follows from (2) and (6). The first part of (4) has already been addressed, and the rest now follows easily. Part (5) is now also clear. $\square$

Corollary. A filtered Lie algebroid A admits a Weyl gauge if and only if it is firmly filtered and $\text{gr}(A^0)$ admits an ad-invariant fibre-wise grading section E .

Proof. Given a firmly filtered Lie algebroid A with ad-invariant fibre-wise grading section E of $\text{gr}(A^0)$, any lift of E to a section $\epsilon \in \Gamma(A^0)$ is immediately seen to be a Weyl gauge. The converse statement follows from the theorem. $\square$

If A is a filtered Lie algebroid, then A^1 , which lies in the anchor kernel of A , is a bundle of nilpotent Lie algebras, and is accordingly integrated by a bundle of simply-connected nilpotent Lie groups, denoted $\exp(A^1)$.

Remark. By simple-connectivity, the adjoint action of A^1 on A^0 integrates to an action Ad of $\exp(A^1)$ on A^0 . We have the identity $\text{Ad}_{\exp(Z)} X = e^{-\text{ad}_Z} X$, for any $Z \in \Gamma(A^1)$ and $X \in \Gamma(A^0)$, but the right-hand side makes sense for any $X \in \Gamma(A)$.

Proposition. *Let A be a filtered Lie algebroid admitting a Weyl gauge, and $\mathcal{G}$ any Lie groupoid integrating A . Then the source-connected Lie subgroupoid $\mathcal{G}^1 \subset \mathcal{G}$ integrating A^1 is a bundle of simply-connected nilpotent Lie groups isomorphic to $\exp(A^1)$.*

Proof. We prove the first part of the conclusion, as the second part follows from the first, by Lie I for Lie groupoids. By Axiom F2, A^0 and A^1 lie in the kernel of the anchor of A , and so both are bundles of Lie algebras (totally intransitive). The source-connectedness of $\mathcal{G}^1$ now implies that $\mathcal{G}^1$ is totally intransitive (its source and target fibres coincide) so that these fibres are connected Lie groups. These Lie groups are subgroups of the fibres of the source-connected subgroupoid $\mathcal{G}^0 \subset \mathcal{G}$ integrating A^0 (also a bundle of Lie groups). With a Weyl gauge ϵ in hand, we may apply Lemma 10.1(1) fibre-wise to show that these groups are in fact simply-connected and nilpotent. $\square$

10.3. Filtered Lie groupoids and Weyl gauges. Let $\mathcal{G}$ be a filtered Lie groupoid over M . The filtration need not be firm; see 7.3 for definitions. We call $\epsilon \in \Gamma(A)$ a WEYL GAUGE for $\mathcal{G}$ if it is a Weyl gauge for A , and if for every arrow $g \in \mathcal{G}$ with source $m \in M$ we have

$$(1) \quad \text{Ad}_g \epsilon(m) - \epsilon(gm) \in A^1(gm),$$

where gm denotes the target of g . If $\mathcal{G}$ is firmly filtered, then this condition holds if and only if $E = \text{gr}_0(\epsilon)$ is $\mathcal{G}$ -invariant under Ad , in which case the requirement 10.2(1) is automatic.

Proposition. *Let $\mathcal{G}$ be a filtered Lie groupoid with Lie algebroid A and $\mathcal{G}^1 \subset \mathcal{G}$ the source-connected Lie subgroupoid integrating A^1 .*

- (2) *Under Ad , $\mathcal{G}^1$ acts trivially on $\text{gr}(A^0)$.*
- (3) *If $\mathcal{G}$ admits a Weyl gauge then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups isomorphic to $\exp(A^1)$.*

Proof. Part (2) holds because A^1 already acts trivially on $\text{gr}(A^0)$ by F1, and $\mathcal{G}^1$ has connected source-fibres. Part (3) is an immediate corollary of Proposition 10.2. $\square$

To navigate a mild obstruction to obtaining semi-direct products in the groupoid case, we will say the filtration of $\mathcal{G}$ is **WEAKLY FIRM** if A^1 is $\mathcal{G}$ -invariant under Ad . Evidently a firm filtration of $\mathcal{G}$ is weakly firm. (This notion will not be important for our later applications to parabolic geometry.)

Remark. Suppose $\mathcal{G}$ is filtered and admits a Weyl gauge ϵ . Then if $\mathcal{G}$ has connected source fibres, then we claim it is weakly firm. Indeed, ϵ will be a Weyl gauge for the Lie algebroid A of $\mathcal{G}$, so that $A^1 \subset A$ must be an ideal, by Theorem 10.2(2). Our claim follows now from the connectedness assumption.

Suppose $\mathcal{G}$ is firmly filtered, so that $\mathcal{G}/\mathcal{G}^1$ acts on $\text{gr}(A^0)$, by (2). This action, also denoted Ad , preserves the Lie algebra bundle bracket. Now, let $\exp(\text{gr}(A^1))$ denote the totally intransitive Lie groupoid with simply-connected fibres that integrates $\text{gr}(A^1)$, a Lie algebra bundle with nilpotent Lie algebra model. Then the action just mentioned integrates to an action of $\mathcal{G}/\mathcal{G}^1$ on $\exp(\text{gr}(A^1))$, also denoted Ad , preserving the totally intransitive Lie groupoid structure, i.e., it maps each (nilpotent) Lie group fibre isomorphically onto another Lie group fibre. We can accordingly define the semi-direct product $\mathcal{G}/\mathcal{G}^1 \times_{\text{Ad}} \exp(\text{gr}(A^1))$, and have the following globalization of Theorem 10.2.

Theorem. *Let $\mathcal{G}$ be a weakly firm, filtered Lie groupoid with Lie algebroid A . Define $\mathcal{G}^1 \subset \mathcal{G}$ as in the proposition above, and suppose $\mathcal{G}$ admits a Weyl gauge ϵ . Then, if $\mathcal{H} \subset \mathcal{G}$ denotes the isotropy subgroupoid of ϵ under the canonical representation of $\mathcal{G}$, then:*

- (4) $\mathcal{G}^1 \subset \mathcal{G}$ is normal,
- (5) $\mathcal{G} = \mathcal{G}^1 \mathcal{H}$ is the internal semi-direct product of $\mathcal{H}$ and $\mathcal{G}^1$,
- (6) The grading $A^0 = \bigoplus_{i \geq 0} \mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of $\mathcal{H}$.
- (7) $\mathcal{G}$ is firmly filtered.
- (8) As a Lie groupoid, $\mathcal{G}$ is identifiable with the external semi-direct product $\mathcal{G}/\mathcal{G}^1 \times_{\text{Ad}} \exp(\text{gr}(A^1))$. Specifically, let $\Phi_\epsilon: \text{gr}(A^1) \rightarrow A^1$ denote the splitting associated with ϵ , and let $\tilde{\Phi}_\epsilon: \exp(\text{gr}(A^1)) \rightarrow \mathcal{G}^1$ be the morphism of totally intransitive Lie groupoids that integrates Φ_ϵ . For $g \in \mathcal{G}/\mathcal{G}^1$, define $g^\mathcal{H} \in \mathcal{H}$ by the requirement $g^\mathcal{H} \mathcal{G}^1 = g$. Then the map

$$(g, k) \mapsto \tilde{\Phi}_\epsilon(k) g^\mathcal{H}: \mathcal{G}/\mathcal{G}^1 \times_{\text{Ad}} \exp(\text{gr}(A^1)) \rightarrow \mathcal{G}$$

is a Lie groupoid isomorphism.

Proof. Since we assume $\mathcal{G}$ is weakly firm, $A^1 \subset A$ is invariant under the canonical representation Ad of $\mathcal{G}$ on $\mathcal{G}^1$. But, by the proposition, the exponential map for $\mathcal{G}^1$ is a diffeomorphism, and we will have $g \exp(Z) g^{-1} = \exp(\text{Ad}_g Z)$ for any arrow $g \in \mathcal{G}$ beginning at the base point of each $Z \in A^1$. This proves (4) because the filtration is weakly firm.

To prove (5) we now show each arrow $g \in \mathcal{G}$, beginning at some $m \in M$, can be written uniquely as $g = kh$, with $k \in \mathcal{G}^1$ and $h \in \mathcal{H}$. Indeed, from (1) we

have $\text{Ad}_g \epsilon(m) \in \epsilon(gm) + A^1|_{gm}$. But, according to Lemma 10.1(1), the Lie group $\mathcal{G}_{gm}^1$ (the source=target fibre of $\mathcal{G}^1$ at $gm \in M$) acts freely and transitively on the affine space $\epsilon(gm) + A^1|_{gm}$. Therefore there exists a unique $k \in \mathcal{G}^1$ such that $\text{Ad}_g \epsilon(m) = \text{Ad}_k \epsilon(gm)$. But this means $\text{Ad}_{k^{-1}g} \epsilon(m) = \epsilon(gm) = \epsilon(k^{-1}gm)$, i.e., $k^{-1}g \in \mathcal{H}$.

To prove (6), suppose $Y \in \Gamma(\mathfrak{a}_j)$ for some $j \geq 0$ and let $h \in \mathcal{H}$ be an arbitrary arrow beginning at some $m \in M$. Since the bracket on A^0 is $\mathcal{G}$ -invariant under Ad , the grading identity $[\epsilon, Y](m) = jY(m)$ gives us $[\text{Ad}_h \epsilon(m), \text{Ad}_h Y(m)] = j \text{Ad}_h Y(m)$. From the definition of $\mathcal{H}$, we conclude that $[\epsilon(hm), \text{Ad}_h Y(m)] = j \text{Ad}_h Y(m)$, establishing that $\text{Ad}_h Y(m) \in \mathfrak{a}_j$, as required.

In view of the fact that $\mathcal{G} = \mathcal{G}^1 \mathcal{H}$, and having established (6), it suffices, in proving (7), to show that the filtration of A^0 is invariant under the action of $\mathcal{G}^1$. But A^0 is already invariant under the action of A^1 by F1. Since $\mathcal{G}^1$ is source-connected, the filtration of A^0 is therefore invariant under the action of $\mathcal{G}^1$.

Part (8) is a rather straightforward consequence of the other parts, whose proof is omitted. $\square$

Corollary. *A weakly firm, filtered Lie groupoid $\mathcal{G}$ with Lie algebroid A admits a Weyl gauge if and only if it is firmly filtered and $\text{gr}(A^0)$ admits an Ad -invariant fibre-wise grading section E .*

Proof. Given an Ad -invariant fibre-wise grading section E of $\text{gr}(A^0)$, any lift to a section of A^0 is a Weyl gauge for $\mathcal{G}$, as is immediately verified. The converse statement follows from the theorem. $\square$

10.4. Weyl gauge transformations. Let A be a firmly filtered Lie algebroid over M and ϵ a Weyl gauge lifting some grading section E of $\text{gr}(A^0)$. Then, applying Theorem 10.2, we obtain an isomorphism $\phi_\epsilon: (A/A^1) \oplus_{\text{ad}} \text{gr}(A^1) \xrightarrow{\sim} A$. A new Weyl gauge ϵ' , lifting E , determines a second isomorphism $\phi_{\epsilon'}: (A/A^1) \oplus_{\text{ad}} \text{gr}(A^1) \xrightarrow{\sim} A$, so that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}: A \rightarrow A$ is a filtration-preserving Lie algebroid automorphism covering the identity on M , and mapping the gauge ϵ to ϵ' . We may construct automorphisms of a firmly filtered Lie groupoid in a similar way, using Theorem 10.3.

Theorem. *The automorphisms of firmly filtered Lie algebroids and Lie groupoids described above are precisely the Weyl gauge transformations (conjugations) defined in 9.2.*

Proof of the infinitesimal case. Let $\epsilon, \epsilon', \phi_\epsilon$ and $\phi_{\epsilon'}$ have the meanings above. Then it is not hard to see, from the statement of Theorem 10.2, that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ is the unique Lie algebroid isomorphism φ covering the identity such that:

- (1) $\varphi(B_\epsilon) = B_{\epsilon'}$, where B_ϵ denotes the isotropy of ϵ under the canonical representation of A on its anchor kernel.
- (2) $\varphi(X) \equiv X \pmod{A^1}$, for all $X \in \Gamma(A)$.

- (3) $\varphi(Y) = \Phi_{\epsilon'} \circ \Phi_{\epsilon}^{-1}(Y)$ for all $Y \in \Gamma(A^1)$, where $\Phi_{\epsilon}: \text{gr}(A^1) \rightarrow A^1$ denotes the splitting determined by ϵ .

Now, a fibre-wise application of Lemma 10.1(1) show that there exists $Z \in \Gamma(A^1)$ such that $\epsilon' = \text{Ad}_{\exp(Z)} \epsilon = e^{-\text{ad}_Z} \epsilon$. See also Remark 10.2. To show $\phi_{\epsilon'} \circ \phi_{\epsilon}^{-1}$ is a Weyl gauge transformation, it suffices to show that the Weyl gauge transformation $\varphi := e^{-\text{ad}_Z}$ satisfies the above conditions. But (1) follows immediately, by construction, because $e^{-\text{ad}_Z}$ is a Lie algebroid morphism; (2) follows from the expansion $e^{-\text{ad}_Z} X := X - [Z, X] + \frac{1}{2!}[Z, [Z, X]] - \dots$ and the fact that $\Gamma(A^1) \subset \Gamma(A)$ is an ideal (A is firmly filtered); while (3) follows from the fibre-wise application of Lemma 10.1(2).

It remains to show *every* Weyl gauge transformation $e^{-\text{ad}_Z}$ is of the form $\phi_{\epsilon'} \circ \phi_{\epsilon}^{-1}$ for some ϵ and ϵ' . But the above argument already shows this, if we choose ϵ to be any Weyl gauge and $\epsilon' := e^{-\text{ad}_Z}(\epsilon)$. $\square$

Proof of the global case. Let $\mathcal{G}$ be a firmly filtered Lie groupoid with Lie algebroid A , and let ϵ and ϵ' have the meanings above. Applying Theorem 10.3, we obtain Lie groupoid isomorphisms $\phi_{\epsilon}, \phi_{\epsilon'}: \mathcal{G}/\mathcal{G}^1 \times_{\text{Ad}} \exp(\text{gr}(A^1)) \rightarrow \mathcal{G}$. From the statement of the theorem, it is not too hard too see that $\phi_{\epsilon'} \circ \phi_{\epsilon}^{-1}$ is the unique Lie groupoid isomorphism φ covering the identity such that:

- (4) $\varphi(\mathcal{H}_{\epsilon}) = \mathcal{H}_{\epsilon'}$, where $\mathcal{H}_{\epsilon}$ denotes the isotropy of ϵ under the canonical representation of $\mathcal{G}$ on the anchor kernel of A .
- (5) $\varphi(g)\mathcal{G}^1 \equiv g\mathcal{G}^1$, for all $g \in \mathcal{G}$.
- (6) $d\varphi(Y) = \Phi_{\epsilon'} \circ \Phi_{\epsilon}^{-1}(Y)$ for all $Y \in \Gamma(A^1)$, where $\Phi_{\epsilon}: \text{gr}(A^1) \rightarrow A^1$ denotes the splitting determined by ϵ .

In convincing herself of this, the reader will keep in mind that $\mathcal{G}^1 \cong \exp(A^1)$ and $\exp(\text{gr}(A^1))$ are source simply-connected, so that a groupoid morphism having one of these as domain is determined completely by its infinitesimalization.

Now choose $Z \in \Gamma(A^1)$ as in the infinitesimal argument, so that $\epsilon' = \text{Ad}_{\exp(Z)} \epsilon = e^{-\text{ad}_Z} \epsilon$. To show that $\phi_{\epsilon'} \circ \phi_{\epsilon}^{-1}$ is a Weyl gauge transformation, it suffices to show that the Weyl gauge transformation $\varphi := K_{\exp(Z)}$ satisfies the above conditions, where $K_{\exp(Z)}: \mathcal{G} \rightarrow \mathcal{G}$ denotes conjugation by the (bi)section $\exp(Z)$. To this end, suppose $h \in \mathcal{G}$ is an arrow beginning at some $m \in M$ and that $h \in \mathcal{H}_{\epsilon}$. Then $\text{Ad}_h \epsilon(m) = \epsilon(hm)$. Applying $\text{Ad}_{\exp(Z)}$ to both sides, we conclude that $\text{Ad}_{K_{\exp(Z)}(h)} \epsilon'(m) = \epsilon'(m)$, showing $K_{\exp(Z)}(h) \in \mathcal{H}_{\epsilon'}$. We can reverse the roles of ϵ and ϵ' and repeat the argument, and conclude that $K_{\exp(Z)}(\mathcal{H}_{\epsilon}) = \mathcal{H}_{\epsilon'}$, i.e., (4) holds. The normality of $\mathcal{G}^1$ (Theorem 10.3(4)) means $g\mathcal{G}^1 = \mathcal{G}^1 g$, for any $g \in \mathcal{G}$. If $g \in \mathcal{G}$ is an arbitrary arrow, beginning at some $m \in M$, we compute, appealing to normality twice,

$$\begin{aligned} K_{\exp(Z)}(g)\mathcal{G}^1 &= \mathcal{G}^1 \exp(Z(gm))g \exp(-Z(m)) \\ &= \mathcal{G}^1 g \exp(-Z(m)) = g \exp(-Z(m))\mathcal{G}^1 = g\mathcal{G}^1, \end{aligned}$$

which proves (5). Since $dK_{\exp(Z)} = e^{-\text{ad}_Z}$, (6) already follows from the argument given in the infinitesimal case.¹²

To show every Weyl gauge transformation $K_{\exp(Z)}$ is of the form $\phi_{\epsilon'} \circ \phi_{\epsilon}^{-1}$ for some ϵ and ϵ' , take ϵ to be any Weyl gauge, $\epsilon' := e^{-\text{ad}_Z}(\epsilon)$, and repeat the preceding argument. $\square$

11. MORE ON MODEL DATA

In 11.1 we establish a core technical result on constructing Cartan connections compatible with given model data τ . Transitivity will be an essential assumption.

When a Weyl gauge ϵ is compatible with model data in an appropriate sense, this leads to gradings of the underlying Lie algebroid A , and not just of A^0 , which we describe in 11.2.

11.1. Constructing Cartan connections compatible with model data. In Section 12 we shall need the following important result, reducing, in the *transitive* case, the problem of constructing Cartan connections compatible with given model data, to the problem of constructing representations of a suitable subalgebroid/subgroupoid.

Proposition. Infinitesimal case. *Let A be a transitive Lie algebroid that can be written as vector bundle sum $A = B \oplus A'$, where $B \subset A$ is a subalgebroid and $A' \subset A$ an ideal lying in the kernel of the anchor $\#: A \rightarrow TM$. Let τ be model data on A and suppose there exists a representation ρ^B of B on A , with respect to which τ is invariant, and such that ρ^B extends the canonical representation of $B \subset A$ on the anchor kernel of A :*

$$(1) \quad \rho_X^B Z = [X, Z] \text{ whenever } \#Z = 0, Z \in \Gamma(A), X \in \Gamma(B).$$

Then there exists a unique Cartan connection ∇ on A , necessarily weakly compatible with τ , such that

$$(2) \quad \bar{\nabla}_{X+Y} W = \rho_X^B W + \tau(Y, W); \quad X \in \Gamma(B), Y \in \Gamma(A'), W \in \Gamma(A).$$

If τ is compatible with a filtration of A , and ρ^B preserves the filtration, then ∇ is also compatible with the filtration. If, additionally,

$$(3) \quad \rho_Z^B W = \tau(Z, W) \text{ whenever } \#Z = 0, Z \in \Gamma(B), W \in \Gamma(A),$$

then τ is strongly compatible with ∇ .

Global case. *Suppose $\mathcal{G}$ is a Lie groupoid, with Lie algebroid A , that can be written $\mathcal{G} = \mathcal{G}'\mathcal{H}$ with $\mathcal{G}' \cap \mathcal{H} = M$ (groupoid of identity elements), where $\mathcal{G}' \subset \mathcal{G}$ is a totally intransitive, normal, Lie subgroupoid with simply-connected fibres. Let τ be model data on A and suppose there is a representation $\rho^{\mathcal{H}}$ of $\mathcal{H}$*

¹²The source of the minus sign is our convention of using right-invariant vector fields to define Lie brackets, so that $dK_{\exp(Z)}X = \text{Ad}_{J^1 \exp(Z)}X = e^{-\text{ad}_Z}X$, $X \in \Gamma(A)$.

on A , with respect to which τ is invariant, and such that $\rho^{\mathcal{H}}$ extends the canonical representation of $\mathcal{H} \subset \mathcal{G}$ on the anchor kernel of A :

$$(4) \quad \rho_h^{\mathcal{H}} Z = \text{Ad}_h Z \text{ whenever } \#Z = 0, h \in \mathcal{H}.$$

Then, there exists a unique Cartan connection D on $\mathcal{G}$, necessarily weakly compatible with τ , such that

$$(5) \quad \bar{D}_{kh} W = \text{Ad}_k^{\tau} \rho_h^{\mathcal{H}} W,$$

for each arrow $h \in H$ beginning at some $m \in M$, $k \in \mathcal{G}^1|_{hm}$ and $W \in A|_m$. Here Ad^{τ} is unique representation of $\mathcal{G}'$ on A integrating the infinitesimal representation ad^{τ} of the Lie algebroid A' of $\mathcal{G}'$ on A defined by $\text{ad}_Y^{\tau} W = \tau(Y, W)$. If τ is compatible with a filtration of A , and ρ^B preserves the filtration, then D is also compatible with the filtration.

Note that Ad^{τ} is well-defined on account of the hypothesis of simple-connectivity.

Proof. Infinitesimal case. Under the hypotheses of the theorem, define an A -connection ρ on A using the right-hand side of (2):

$$\rho_{X+Y} W = \rho_X^B W + \tau(Y, W); \quad X \in \Gamma(B), Y \in \Gamma(A'), W \in \Gamma(A).$$

By Proposition 4.2, to show there exists a Cartan connection ∇ satisfying (2), it suffices to show that ρ is actually a representation, and, secondly, that ρ extends the canonical representation of A on sections of its anchor kernel:

$$(6) \quad \rho_W Z = [W, Z] \text{ whenever } \#Z = 0, W \in \Gamma(A).$$

If $W \in \Gamma(B)$, then (6) holds, by hypothesis (1). If $W \in \Gamma(A')$ and $\#Z = 0$, then $\rho_W Z = \tau(W, Z) = [W, Z]$, because both W and Z lie in the anchor kernel and τ is model data. Since $A = B \oplus A'$, this proves (6).

Since the restriction of $X \mapsto \rho_X$ to B is ρ^B , a representation, and the restriction of $Y \mapsto \rho_Y$ to A' is $Y \mapsto \tau(Y, \cdot)$, also a representation, to show ρ is itself a representation, it suffices to prove that

$$\rho_X \rho_Y W - \rho_Y \rho_X W - \rho_{[X, Y]} W = 0, \quad W \in \Gamma(A),$$

whenever $X \in \Gamma(B)$ and $Y \in \Gamma(A')$. But in that case, $[X, Y] \in \Gamma(A')$, because $A' \subset A$ is an ideal, and we compute

$$\begin{aligned} \rho_X \rho_Y W - \rho_Y \rho_X W - \rho_{[X, Y]} W &= \rho_X^B(\tau(Y, W)) - \tau(Y, \rho_X^B W) - \tau([X, Y], W) \\ &= \rho_X^B(\tau(Y, W)) - \tau(Y, \rho_X^B W) - \tau(\rho_X^B Y, W), \quad \text{by (1)} \\ &= 0, \quad \text{by the } \rho^B\text{-invariance of } \tau. \end{aligned}$$

Proof of the remaining claims in infinitesimal case is straightforward and omitted.

Global case. Define a map $(g, W) \mapsto \rho_g W \in A$, from the submanifold of all $(g, W) \in \mathcal{G} \times A$ such that g is an arrow beginning at the base point of W , using the right-hand side of (5):

$$\rho_{kh} W = \text{Ad}_k^{\tau} \rho_h^{\mathcal{H}} W; \quad k \in \mathcal{G}', h \in \mathcal{H}.$$

To show that there exists a unique Cartan connection D on A satisfying (5), it suffices to show that ρ is a representation, and that ρ extends the canonical representation of $\mathcal{G}$ on the anchor kernel of A :

$$(7) \quad \rho_g Z = \text{Ad}_g Z \text{ whenever } \#Z = 0$$

and $g \in \mathcal{G}$ is an arrow beginning at the base point of Z .

By the hypothesis (4), (7) already holds for $g \in \mathcal{H}$. Since $\mathcal{G} = \mathcal{G}'\mathcal{H}$, to prove (7) in general, it suffices to prove

$$(8) \quad \text{Ad}_k^\tau Z = \text{Ad}_k Z \text{ whenever } \#Z = 0, k \in \mathcal{G}'.$$

In fact, since $\mathcal{G}'$ has connected fibres, it suffices to prove the infinitesimalization,

$$\text{ad}_Y^\tau Z = [Y, Z] \text{ whenever } \#Z = 0, Y \in A'.$$

But this is true because both Y and Z here lie in the anchor kernel, $\text{ad}_Y^\tau Z := \tau(Y, Z)$, and τ is model data.

Since $\rho^\mathcal{H}$ and $\rho^{\mathcal{G}^1}$ are already representations, to show ρ itself is a representation, it suffices to show, for any arrow $h \in \mathcal{H}$ beginning at some $m \in M$, that

$$\rho_h^\mathcal{H} \rho_k^{\mathcal{G}^1} \rho_{h^{-1}}^\mathcal{H} W = \rho_{hkh^{-1}}^{\mathcal{G}^1} W,$$

where $k \in \mathcal{G}^1|_m$ and $W \in A|_{hm}$ are arbitrary. In fact, since $\mathcal{G}^1$ has connected fibres, it is enough to show the two Lie group homomorphisms $\mathcal{G}^1|_m \rightarrow \text{Aut}(A|_m)$ below have the same derivative:

$$\begin{aligned} k &\mapsto \rho_h^\mathcal{H} \rho_k^{\mathcal{G}^1} \rho_{h^{-1}}^\mathcal{H} \\ k &\mapsto \rho_{hkh^{-1}}^{\mathcal{G}^1}. \end{aligned}$$

The readily computed derivatives are:

$$\begin{aligned} X &\mapsto \rho_h^\mathcal{H} \text{ad}_X^\tau \rho_{h^{-1}}^\mathcal{H}(\cdot) = \rho_h^\mathcal{H} \tau(X, \rho_{h^{-1}}^\mathcal{H}(\cdot)) \\ \text{and } X &\mapsto \text{ad}_{\text{Ad}_h X}^\tau(\cdot) = \tau(\text{Ad}_h X, \cdot) = \tau(\rho_h^\mathcal{H} X, \cdot), \quad \text{by (4).} \end{aligned}$$

Equality of these derivatives follows from the $\rho^\mathcal{H}$ -invariance of τ .

Proof of the remaining claims of the proposition is elementary and omitted. $\square$

11.2. Gradings associated with model data. Let us say that model data τ is COMPATIBLE with a Weyl gauge $\epsilon \in \Gamma(A^0)$ if $E = \text{gr}_0(\epsilon)$ is a fibre-wise grading element of $\text{gr}^T(A)$, $T = \text{gr}_0(\tau)$, and not just of $\text{gr}(A^0) = \text{gr}^T(A^0)$.

Let A be a filtered Lie algebroid admitting a Weyl gauge ϵ . Then, according to Theorem 10.2, we have an associated grading $A^0 = \oplus_{i \geq 0} \mathfrak{a}_i$ that is invariant under the action of the isotropy subalgebroid $B \subset A$ of ϵ . If ϵ is compatible with model data τ , that is itself compatible with the filtration of A , then this grading extends to a grading $A = \oplus_i \mathfrak{a}_i$, where $X \in \mathfrak{a}_j$ if and only if $\tau(\epsilon, X) = jX$. This defines a splitting isomorphism $\Phi_\epsilon: \text{gr}^T(A) \rightarrow A$.

Proposition. Infinitesimal case. *In the scenario above, let ∇ be any infinitesimal Cartan connection weakly compatible with τ . Then:*

- (1) The grading $A = \oplus_i \mathfrak{a}_i$ is invariant under the representation $\bar{\nabla}$ of B on A .
- (2) For any $X \in \Gamma(\mathfrak{a}_k)$, $k \geq 0$, and $Y \in \Gamma(\mathfrak{a}_j)$, $j \in \mathbb{Z}$, one has $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j \oplus \mathfrak{a}_{j+k})$.

Global case. If $\mathcal{G}$ is a weakly firm, filtered Lie groupoid with Lie algebroid A , ϵ is a Weyl gauge for $\mathcal{G}$ compatible with the model data τ , and D is a Cartan connection on $\mathcal{G}$ compatible with τ , then:

- (3) The grading $A = \oplus_i \mathfrak{a}_i$ associated with ϵ is invariant under the representation $\bar{D}$ of the isotropy subgroupoid $\mathcal{H} \subset \mathcal{G}$ of ϵ .

The reader is warned that B does not generally align itself with the complement $\oplus_{i \leq 0} \mathfrak{a}_i$ of $A^1 = \oplus_{i \geq 1} \mathfrak{a}_i$ in A defined by the grading. We may only conclude that B is the graph of some linear function $\oplus_{j \leq 0} \mathfrak{a}_j \rightarrow A^1$ that vanishes on $\mathfrak{a}_0$, which is necessarily contained in B .

Corollary. If a (weakly firm) filtered Lie algebroid (resp., Lie groupoid) admits a Weyl gauge compatible with some model data τ , then any Cartan connection weakly compatible with τ is compatible with the filtration.

Proof of corollary. If ∇ is the Cartan connection in the Lie algebroid case, then $\bar{\nabla}$ -invariance of the filtration of A follows from (1) and (2), and the semi-direct product structure of A (Theorem 10.2(3)).

For the Lie groupoid case and a global Cartan connection D , we know the filtration is $\bar{\nabla}$ -invariant, where ∇ is the infinitesimalization of D , from the Lie algebroid case just proven. In particular, continuing in the notation of the proposition, the restriction of $\bar{\nabla}$ to a representation of A^1 on A is filtration-preserving. Therefore, the restriction of $\bar{D}$ to a representation of $\mathcal{G}^1$ is filtration-preserving, because $\mathcal{G}^1$ is source-connected. Full $\bar{D}$ -invariance of the filtration follows from this fact, (3), and the semi-direct product structure of $\mathcal{G}$ (Theorem 10.3(5)). $\square$

Proof of proposition. Our proofs of (1) and (2) are variations on an argument appearing in the proof of Theorem 10.2. Indeed, suppose that $Y \in \Gamma(\mathfrak{a}_j)$ for some j and let $X \in \Gamma(B)$ be arbitrary. Then, applying $\bar{\nabla}_X$ to both sides of the identity $\tau(\epsilon, Y) = jY$, and appealing to the $\bar{\nabla}$ -invariance of τ (weak compatibility), we have

$$\tau(\bar{\nabla}_X \epsilon, Y) + \tau(\epsilon, \bar{\nabla}_X Y) = j \bar{\nabla}_X Y.$$

But $\bar{\nabla}_X \epsilon = [X, \epsilon]$, because ϵ is in the anchor kernel, while $[X, \epsilon] = 0$, because $X \in \Gamma(B)$. We conclude that $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j)$. This proves (1).

Turning to (2), write $\bar{\nabla}_X Y = \sum_i W_i$ for some $W_i \in \Gamma(\mathfrak{a}_i)$, which implies

$$\tau(\epsilon, \bar{\nabla}_X Y) = \sum_i i W_i.$$

On the other hand, $\tau(\epsilon, Y) = jY$, which, after applying $\bar{\nabla}_X$ to both sides, and appealing to the $\bar{\nabla}$ -invariance of τ , becomes

$$\tau(\bar{\nabla}_X \epsilon, Y) + \tau(\epsilon, \bar{\nabla}_X Y) = j \sum_i W_i.$$

As X and ϵ are sections of A^0 , we have $\bar{\nabla}_X E = [X, \epsilon] = \tau(X, \epsilon) = -kX$. From this and the two preceding equations we deduce

$$\sum_i (i - j)W_i + k\tau(Y, X) = 0.$$

Since $A = \oplus_i \mathfrak{a}_i$ is a grading with respect to τ , the last term on the left-hand side is a section of $\Gamma(\mathfrak{a}_{j+k})$. Projecting onto each component $\mathfrak{a}_i$ of the grading $A = \oplus_i \mathfrak{a}_i$, we obtain $W_i = 0$ for all i , with the possible exceptions $i = j, j + k$, which shows $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j \oplus \mathfrak{a}_{j+k})$.

The proof of (3) is similar to that of (1) above, following from the $\bar{D}$ -invariance of τ , the definition of the spaces $\mathfrak{a}_j$, and the fact that $\mathcal{H}$ fixes ϵ . $\square$

12. FILTERED CARTAN GEOMETRIES

A Cartan geometry $(\mathcal{G}, D, \tau)$ with Lie algebroid A will be called a **FILTERED CARTAN GEOMETRY** if $\mathcal{G}$ is firmly filtered, in the sense of 7.3, and if $\mathcal{G}$ admits a Weyl gauge compatible with τ (11.2). There is no requirement that τ be reductive or semisimple, and such notions will play no role in the current section. However, Proposition 15.1, stated later, shows that every parabolic geometry is a filtered Cartan geometry.

An infinitesimal Cartan geometry (A, ∇, τ) is an **INFINITESIMAL FILTERED CARTAN GEOMETRY** if A is firmly filtered and admits a τ -compatible Weyl gauge.

In this section, we provide a general, explicit, description of *transitive*, filtered Cartan geometries, by exploiting theorems on semi-direct product structure established in Section 10. In this description, we specify the additional data needed to recover a transitive Cartan geometry from its (abstract) Levi reduction. While we are mainly interested in the special parabolic case, the extra abstraction introduced here emphasizes Weyl gauges as the essential actor in the resulting characterization.

12.1. Abstract Levi structures: the infinitesimal case. The reduction procedure described for parabolic geometries in 8.1 generalizes to transitive filtered Cartan geometries. We begin with the infinitesimal case.

Let (A, ∇, τ) be a transitive infinitesimal filtered Cartan geometry and assume A^0 is the anchor kernel. Since A is firmly filtered, the canonical representation ad of A on A^0 , in which A^1 is acting trivially, descends to a representation ad of $B := A/A^1$ on $\text{gr}(A^0)$. This canonical representation is extended by a representation of B on $\text{gr}(A)$, induced by the representation $\bar{\nabla}$ of A on itself, and we denote this

by R^∇ . If we write $\mathfrak{a}_j := \text{gr}(A)_j$ for $j > 0$, then using the transitivity, $\text{gr}(A)$ can be identified with

$$\mathbf{S0}. \text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j).$$

In addition to the grading-preserving representation R^∇ on the above, we have the Lie algebra bundle bracket $T := \text{gr}_0(\tau)$ which, on account of the compatibility Axiom c2, shows that

$$(1) \quad T(C, V) = R_C^\nabla V, \quad T(C, Z) = R_C^\nabla Z, \quad T(C, C') = [C, C'],$$

whenever $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\text{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$. The last equation holds because τ extends the canonical bracket on A^0 . Note $R_C^\nabla Z$ does not actually depend on ∇ , because it depends only on the canonical representation of A on A^1 .

Motivated by the above, an **ABSTRACT INFINITESIMAL LEVI STRUCTURE** will be a 4-tuple $(B, \oplus_{j>0} \mathfrak{a}_j, R, T)$, where:

- (2) B is a transitive, filtered Lie algebroid over M with $B \leq 0$ and with B^0 as its anchor kernel.
- (3) $\oplus_{j>0} \mathfrak{a}_j$ is a graded vector bundle over M , with $\mathfrak{a}_j = 0$ for all sufficiently large j .
- (4) R is a grading-preserving representation of B on $\mathbf{S0}$ that extends the canonical action of B on B^0 .
- (5) T is an R -invariant Lie algebra bundle structure on $\mathbf{S0}$ compatible with the obvious grading and satisfying

$$T(C, V) = R_C V, \quad T(C, Z) = R_C Z, \quad T(C, C') = [C, C'],$$

whenever $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\text{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$.

- (6) As a graded Lie algebra bundle, $\mathbf{S0}$ admits an R -invariant grading section, necessarily of the form $(0, E, 0)$, for some $E \in \Gamma(B^0)$, where E is required to commute with arbitrary sections of B .

It is not required that T restrict to the canonical Lie algebra bundle structure on $\text{gr}(TM)$. If this restriction is the *negative* of the canonical one, then the Levi structure is **REGULAR**.

According to the immediately preceding observations, associated with every transitive infinitesimal filtered Cartan geometry is an infinitesimal Levi structure, called its **LEVI REDUCTION**, regular if the filtered Cartan geometry is regular.

12.2. Abstract Levi structures: the global case.

An **ABSTRACT LEVI STRUCTURE** is a 4-tuple $(\mathcal{H}, \oplus_{j>0} \mathfrak{a}_j, R, T)$ where:

- (1) $\mathcal{H}$ is a transitive, filtered Lie groupoid over M with Lie algebroid B , whose anchor kernel is B^0 , and with $B \leq 0$.
- (2) $\oplus_{j>0} \mathfrak{a}_j$ is a graded vector bundle over M , with $\mathfrak{a}_j = 0$ for all sufficiently large j .
- (3) R is a grading-preserving representation of $\mathcal{H}$ on $\mathbf{S0}$ that extends the canonical action of $\mathcal{H}$ on B^0 .

- (4) T an R -invariant Lie algebra bundle structure on S_0 compatible with the grading, and such that

$$T(C, V) = dR_C V, \quad T(C, Z) = dR_C Z, \quad T(C, C') = [C, C'],$$

for all $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\text{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$.

- (5) As a graded Lie algebra bundle, S_0 admits an R -invariant grading section, necessarily of the form $(0, E, 0)$, for some $E \in \Gamma(B^0)$, where E is required to be $\mathcal{H}$ -invariant.

Here dR is the infinitesimalization of the representation R . The structure is **REGULAR** if its infinitesimalization $(B, \oplus_{j>0} \mathfrak{a}_j, dR, T)$ is regular.

In analogy with our observations in 12.1, we can associate with each transitive filtered Cartan geometry $(\mathcal{G}, D, \tau)$ an abstract Levi structure

$$(\mathcal{G}/\mathcal{G}^1, \text{gr}(A^1), R^D, \text{gr}_0(\tau))$$

that we call its **LEVI REDUCTION**. It is regular if $(\mathcal{G}, D, \tau)$ is. Here R^D is the representation of $\mathcal{G}/\mathcal{G}^1$ induced by the representation $\bar{D}$ of $\mathcal{G}$ on the Lie algebroid A of $\mathcal{G}$.

12.3. Morphisms. A **MORPHISM** of filtered Lie algebroids $q: A \rightarrow \tilde{A}$ is a Lie algebroid morphism respecting the filtrations: $q(A^j) \subset \tilde{A}^j$. A **MORPHISM** of filtered Lie groupoids $Q: \mathcal{G} \rightarrow \tilde{\mathcal{G}}$ is a Lie groupoid morphism whose infinitesimalization $q = dQ: A \rightarrow \tilde{A}$ is a morphism of firmly filtered Lie algebroids. (Infinitesimal) Weyl gauge transformations are examples of automorphisms of a filtered Lie groupoid (resp., Lie algebroid). We obtain the category of *firmly* filtered Lie algebroids or Lie groupoids by simply restricting the class of morphisms to those whose source and target are firmly filtered objects.

For simplicity, and as it will suffice for current purposes, we only define morphisms for filtered Cartan geometries that are isomorphisms. A **MORPHISM** of infinitesimal filtered Cartan geometries

$$(A, \tau, \nabla) \rightarrow (\tilde{A}, \tilde{\tau}, \tilde{\nabla})$$

is an isomorphism of underlying filtered Lie algebroids such that τ and ∇ push forward to $\tilde{\tau}$ and $\tilde{\nabla}$; since q is an isomorphism, pushing forward makes sense. A **MORPHISM** of global filtered Cartan geometries $(\mathcal{G}, \tau, D)$ and $(\tilde{\mathcal{G}}, \tilde{\tau}, \tilde{D})$ is an isomorphism of underlying filtered Lie groupoids that pushes τ and D to $\tilde{\tau}$ and $\tilde{D}$.

A morphism of abstract infinitesimal Levi structures

$$(B, \oplus_{j>0} \mathfrak{a}_j, R, T) \rightarrow (\tilde{B}, \oplus_{j>0} \tilde{\mathfrak{a}}_j, \tilde{R}, \tilde{T})$$

consists of a pair (q, ω) , where $q: B \rightarrow \tilde{B}$ is morphism of filtered Lie algebroids, covering some smooth map $f: M \rightarrow \tilde{M}$, and $\omega: \oplus_{j>0} \mathfrak{a}_j \rightarrow \oplus_{j>0} \tilde{\mathfrak{a}}_j$ a morphism of

graded vector bundles, such that the induced morphisms of graded vector bundles

$$(q, \omega)_*: \operatorname{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j) \rightarrow \operatorname{gr}(TM) \oplus \tilde{B}^0 \oplus (\oplus_{j>0} \tilde{\mathfrak{a}}_j)$$

$$(V, C, Z) \mapsto (\operatorname{gr}_0(Tf)(V), qC, \omega Z)$$

is: (i) a morphism of Lie algebra bundles, with respect to the brackets T and $\tilde{T}$; and (ii) an intertwining of the representations R and $\tilde{R}$. Here $Tf: TM \rightarrow TM$ is the tangent map of f , which will necessarily be filtration-preserving because $\# \circ q = Tf \circ \#$. A morphism of abstract Levi structures $(\mathcal{H}, \oplus_{j>0} \mathfrak{a}_j, R, T)$ and $(\tilde{\mathcal{H}}, \oplus_{j>0} \tilde{\mathfrak{a}}_j, \tilde{R}, \tilde{T})$ is a pair (Q, ω) , where $Q: \mathcal{H} \rightarrow \tilde{\mathcal{H}}$ is morphism of filtered Lie groupoids and $\omega: \oplus_{j>0} \mathfrak{a}_j \rightarrow \oplus_{j>0} \tilde{\mathfrak{a}}_j$ a morphism of graded vector bundles, such that: (i) (dQ, ω) is a morphism of the corresponding abstract infinitesimal Levi structures; and (ii) $(dQ, \omega)_*$ intertwines the (global) representations R and $\tilde{R}$.

As it turns out, for the applications we have in mind, only morphisms of infinitesimal or global Levi structures covering the identity on M will be relevant, in which case f above is just the identity.

12.4. Consequences of the semi-direct product structure. Beginning our analysis with the infinitesimal case, suppose (A, ∇, τ) is a transitive infinitesimal filtered Cartan geometry such that A^0 is the anchor kernel, and let ϵ be a τ -compatible gauge. According to Theorem 10.2, we may suppose, without loss of generality, that A is the semi-direct product of a transitive, filtered Lie algebroid B , with $|B| \leq 0$ and anchor B^0 , with a filtered Lie algebra bundle $\oplus_{j>0} \mathfrak{a}_j$, on which B is acting, an action that preserves the grading. Moreover, if we denote the action of B on $\oplus_{j>0} \mathfrak{a}_j$ by R , so that $A = B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$, then we know there exists a section E of B^0 such that:

$$[E, X] = 0 \quad \text{for all } X \in \Gamma(B)$$

$$R_E Y = jY \quad \text{for all } Y \in \Gamma(\mathfrak{a}_j), j > 0.$$

$$\epsilon = (E, 0) \in \Gamma(A).$$

The Weyl gauge determines a grading $A = \oplus_i \mathfrak{a}'_i$, where $\mathfrak{a}'_i$ is the kernel of $X \mapsto \tau(E, X) - iX: A^i \rightarrow A^{i+1}$.

In general, $\mathfrak{a}'_k \subset B \oplus 0$ does not hold for $k < 0$. However, noting that the filtration of $A = B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$ is defined by $A^i = B^i \oplus_R (\oplus_{j \geq \max\{1, i\}} \mathfrak{a}_j)$, it is not hard to see, from the identities above, that $\mathfrak{a}'_0 = B^0 \oplus 0$ and $\mathfrak{a}'_j = 0 \oplus \mathfrak{a}_j$ for $j > 0$. From this we may deduce that the splitting isomorphism $\Phi: \operatorname{gr}(A) \rightarrow A$ induced by the grading $A = \oplus_i \mathfrak{a}'_i$ has the form

$$\Phi: \operatorname{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j) \rightarrow B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$$

$$(V, C, Z) \mapsto (C + \tilde{\sigma}(V), Z - P(V)),$$

for some $P \subset \Gamma(\operatorname{gr}(T^*M) \otimes (\oplus_{j>0} \mathfrak{a}_j))$, and some vector bundle map $\tilde{\sigma}: \operatorname{gr}(TM) \rightarrow B$ satisfying

$$\tilde{\sigma}(\operatorname{gr}(TM)_k) \subset B^k \quad \text{and} \quad \operatorname{gr}_k(\# \tilde{\sigma}(V)) = V \quad \text{for all } V \in \operatorname{gr}(TM)_k, k < 0.$$

Here $\# : B \rightarrow TM$ denotes the anchor. This is just linear algebra and an unpacking of the implicit identification $\mathfrak{a}_k \cong \text{gr}(A)_k \cong \text{gr}(TM)_k$ for $k < 0$.

The composite $\sigma = \# \circ \tilde{\sigma}$ satisfies

$$\sigma(\text{gr}(TM)_k) \subset T^k M \text{ and } \text{gr}_k(\sigma(V)) = V \text{ for all } V \in \text{gr}(TM)_k, k < 0.$$

We call such $\sigma : \text{gr}(TM) \rightarrow TM$, which is necessarily an isomorphism, a SPLITTING of the filtered manifold M . The composite $s = \tilde{\sigma} \circ \sigma^{-1} : TM \rightarrow B$ is a right-inverse for the anchor $\# : B \rightarrow TM$, which we will call a SPLITTING OF THE ANCHOR OF B . We may now rewrite our formula for Φ as

$$\mathbf{S1.} \quad \Phi(V, C, Z) = (C + s(\sigma(V)), Z - P(V)).$$

By construction, the model data τ is the push-forward of $T := \text{gr}_0(\tau)$ under Φ :

$$\mathbf{S2.} \quad \tau(\Phi(\xi_1), \Phi(\xi_2)) = \Phi(T(\xi_1, \xi_2)),$$

for $\xi_1, \xi_2 \in \text{gr}(TM) \oplus B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$, and $(0, E, 0)$ is a fibre-wise grading section of $\text{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j)$, for the Lie bracket T . To complete our analysis, we derive a formula for the self-representation $\bar{\nabla}$ of A , which characterizes the connection ∇ , because A is transitive (Proposition 4.2). Since Proposition 11.2 applies, the grading $A = \oplus_i \mathfrak{a}'_i$ (there denoted $\oplus_i \mathfrak{a}_i$) is $\bar{\nabla}_{(X,0)}$ -invariant, for every section X of B . It follows that Φ intertwines some grading-preserving representation R of B on $\text{gr}(A) = \text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$ with the $\bar{\nabla}$ -representation of $B \oplus 0$ on A :

$$\bar{\nabla}_{(X,0)} \left(\Phi(V, C, Z) \right) = \Phi \left(R_X(V, C, Z) \right); \quad X \in B.$$

It's not hard to see that R on the right is the same thing as R^∇ , as defined in 12.1 above, under the identification $B \oplus 0 \cong A/A^1$. As a consequence we can write this as

$$(1) \quad \bar{\nabla}_{(X,0)} \left(\Phi(V, C, Z) \right) = \Phi(R_X V, [X, C], R_X Z),$$

where the “ R ” in $R_X Z$ has the meaning assigned to it in the beginning of the current discussion, and with

$$T(C, V) = R_C V, \quad T(C, Z) = R_C Z, \quad T(C, C') = [C, C'],$$

for all $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\text{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$.

On the other hand, for any $Y \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$, we have $\Phi(0, 0, Y) = (0, Y)$, so compatibility Axiom c2 leads to

$$(2) \quad \begin{aligned} \bar{\nabla}_{(0,Y)} \left(\Phi(V, C, Z) \right) &= \tau \left(\Phi(0, 0, Y), \Phi(V, C, Z) \right) = \Phi \left(T((0, 0, Y), (V, C, Z)) \right) \\ &= \Phi \left((0, 0, [Y, Z] - R_C Y) + T(Y, V) \right); \quad Y \in \Gamma(\oplus_{j>0} \mathfrak{a}_j). \end{aligned}$$

Combining this last with (1), we have, for any $(X, Y) \in \Gamma(B \oplus_R (\oplus_{j>0} \mathfrak{a}_j))$,

$$\mathbf{S3.} \quad \bar{\nabla}_{(X,Y)} \Phi(V, C, Z) = \Phi \left(R_X(V, C, Z) + T((0, 0, Y), (V, C, Z)) \right),$$

whose right-hand side can also be written as

$$\Phi\left((R_X V, [X, C], [Y, Z] + R_X Z - R_C Y) + T(Y, V)\right).$$

The above analysis leads to the characterizations of transitive filtered Cartan geometries given next.

12.5. Prolongation: the infinitesimal case. Suppose that $(B, \oplus_{j>0} \mathfrak{a}_j, R, T)$ is an abstract infinitesimal Levi structure. The structure T makes $\oplus_{j>0} \mathfrak{a}_j$ into a graded Lie algebra bundle and the PROLONGATION of the abstract infinitesimal Levi structure is the Lie algebroid semi-direct product $A = B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$. This is a firmly filtered Lie algebroid with $A^i = B^i \oplus_R (\oplus_{j \geq \max\{1, i\}} \mathfrak{a}_j)$.

Theorem (A characterization of transitive infinitesimal filtered Cartan geometries). *Let $(B, \oplus_{j>0} \mathfrak{a}_j, R, T)$ be an abstract infinitesimal Levi structure, P be an arbitrary section of $\text{gr}(T^*M) \otimes (\oplus_{j>0} \mathfrak{a}_j)$, $s: TM \rightarrow B$ any splitting of the anchor of B , and $\sigma: \text{gr}(TM) \rightarrow TM$ some splitting of the filtered manifold M . Then:*

- (1) *The map $\Phi: \text{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j) \rightarrow B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$ defined by S1 is invertible. Specifically, if $\pi: B \rightarrow B^0$ is defined by $\pi(X) := X - s(\#X)$, and if $(X, Y) = \Phi(V, C, Z)$, then $C = \pi X$, $V = \sigma^{-1}(\#X)$, and $Z = Y + P(V)$.*
- (2) *The prolongation $A = B \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$ is a transitive, infinitesimal, filtered Cartan geometry, with model data τ defined by S2 and Cartan connection ∇ uniquely determined by S3. The corresponding Levi reduction is isomorphic to $(B, \oplus_{j>0} \mathfrak{a}_j, R, T)$.*
- (3) *For any section E as in 12.1(6), $\epsilon := \Phi^{-1}(0, E, 0)$ is a Weyl gauge for A compatible with τ and for which $\Phi: \text{gr}(A) \rightarrow A$ is the corresponding splitting.*
- (4) *Every transitive, infinitesimal, filtered Cartan geometry (A, ∇, τ) , with anchor kernel A^0 , is, up to isomorphism, the prolongation of its Levi reduction, with Cartan connection ∇ and model data τ of the form just described.*

The section P will be called the (abstract) RHO TENSOR.

Proof. It is a triviality that Φ is injective. A dimension count implies Φ must be an isomorphism. A substitution verifies the rest of (1).

That τ , defined by S2, is model data follows from the properties of T posited in 12.1(5) and the definition of the semi-direct product Lie bracket for $B^0 \oplus_R (\oplus_{j>0} \mathfrak{a}_j)$.

The isomorphism Φ intertwines the representation R of B on $\text{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j)$ with some representation $\rho^{B \oplus 0}$ of $B \oplus 0 \subset A$ on A :

$$(5) \quad \rho_{(X,0)}^{B \oplus 0} \Phi(V, C, Z) = \Phi\left(R_X((V, C, Z))\right).$$

To apply Proposition 11.1 to A , with τ defined by S2, and $A' = A^1 = 0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$, we need to establish 11.1(1). Here this means verifying that

$$\rho_{(X,0)}^{B \oplus 0} \Phi(0, C, Z) = [(X, 0), \Phi(0, C, Z)],$$

for $X \in \Gamma(B)$, $C \in B^0$, $Z \in \Gamma(\oplus_{j>0} \mathfrak{a}_j)$, which is an easy computation following from (5), and the fact that the bracket on the right-hand side is the semi-direct product bracket. Similarly, since τ is defined by S2, the properties of T in 12.1(5) allow one to show that 11.1(3) also holds in the present case. We conclude that there exists a unique Cartan connection ∇ on A , compatible with τ , such that

$$\bar{\nabla}_{(X,Y)} \Phi(V, C, Z) = \rho_{(X,0)}^{B \oplus 0} \Phi(V, C, Z) + \tau((0, Y), \Phi(V, C, Z)).$$

The formula S3 follows from (5), S2, and the identity $(0, Y) = \Phi(0, 0, Y)$. The last part of (2), and (3), follow by construction. Part (4) of the theorem was already established in 12.4. $\square$

12.6. Prolongation: the global case. Suppose that $(\mathcal{H}, \oplus_{j>0} \mathfrak{a}_j, R, T)$ is an abstract Levi structure. The vector bundle $\oplus_{j>0} \mathfrak{a}_j$ gets an R -invariant graded Lie algebra bundle structure by restricting T . This bundle has a necessarily nilpotent Lie algebra model, and we let $\exp(\oplus_{j>0} \mathfrak{a}_j)$ denote the nilpotent Lie group bundle with simply connected fibres that integrates $\oplus_{j>0} \mathfrak{a}_j$. The restriction of R to a representation of $\mathcal{H}$ on $\oplus_{j>0} \mathfrak{a}_j$, by fibre-wise Lie algebra derivations, integrates to a representation of $\mathcal{H}$ on $\exp(\oplus_{j>0} \mathfrak{a}_j)$ by fibre-wise Lie group isomorphisms; denoting this by R also, we have $R_h(\exp(Y)) = \exp(R_h Y)$, for any arrow $h \in \mathcal{H}$ beginning at the base point of $Y \in \oplus_{j>0} \mathfrak{a}_j$, because $\exp$ is a diffeomorphism in this case.

The PROLONGATION of the Levi structure is the Lie groupoid semi-direct product $\mathcal{G} = \mathcal{H} \times_R \exp(\oplus_{j>0} \mathfrak{a}_j)$. This is a filtered Lie groupoid with Lie algebroid $B \oplus_{dR} (\oplus_{j>0} \mathfrak{a}_j)$, the prolongation of Levi structure's infinitesimalization.

Additionally, the representation ad^T of the nilpotent Lie algebra bundle $\oplus_{j>0} \mathfrak{a}_j$ on $\text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$ integrates to a representation of the Lie group bundle $\exp(\oplus_{j>0} \mathfrak{a}_j)$ on $\text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$. We denote this representation by Ad^T .

Theorem (A characterization of transitive filtered Cartan geometries).

Let $(\mathcal{H}, \oplus_{j>0} \mathfrak{a}_j, R, T)$ be an abstract Levi structure, B the Lie algebroid of $\mathcal{H}$, P an arbitrary section of $\text{gr}(T^*M) \otimes (\oplus_{j>0} \mathfrak{a}_j)$, $s: TM \rightarrow B$ any splitting of the anchor of B , and $\sigma: \text{gr}(TM) \rightarrow TM$ some splitting of the filtered manifold M . Then:

- (1) The map $\Phi: \text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j) \rightarrow B \oplus_{dR} (\oplus_{j>0} \mathfrak{a}_j)$ defined by S1 is invertible.
- (2) The prolongation $\mathcal{G} = \mathcal{H} \times_R \exp(\oplus_{j>0} \mathfrak{a}_j)$ is a transitive, filtered Cartan geometry, with model data τ defined by S2 and Cartan connection D uniquely determined by

$$\text{S3}'. \quad \bar{D}_{(h,k)} \Phi(V, C, Z) = \Phi(\text{Ad}_k^T(R_h V, \text{Ad}_h C, R_h Z)).$$

The corresponding Levi reduction is isomorphic to $(\mathcal{H}, \oplus_{j>0} \mathfrak{a}_j, R, T)$.

- (3) For any section E as in 12.2(5), $\epsilon = \Phi^{-1}(0, E, 0)$ is a Weyl gauge for $\mathcal{G}$ compatible with τ and for which $\Phi: \text{gr}(A) \rightarrow A$ is the corresponding splitting.

- (4) *Every transitive, filtered, Cartan geometry $(\mathcal{G}, D, \tau)$, with Lie algebroid A and anchor kernel A^0 , is, up to isomorphism, the prolongation of some abstract Levi structure, with Cartan connection D and model data τ of the form just described.*

Proof. The preceding infinitesimal result, Theorem 12.5, applies with R there replaced with dR . In particular, (1) is just a restatement of 12.5(1).

The isomorphism Φ intertwines the representation R of $\mathcal{H}$ on $\text{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j)$ with some representation $\rho^{\mathcal{H} \times \text{id}}$ on $\mathcal{H} \times \text{id} \subset \mathcal{G}$ on $A = B \oplus_{dR} (\oplus_{j>0} \mathfrak{a}_j)$:

$$(5) \quad \rho_{(h, \text{id})}^{\mathcal{H} \times \text{id}} \Phi(V, C, Z) = \Phi \left(R_h((V, C, Z)) \right).$$

To apply the global case of Proposition 11.1 to $\mathcal{G}$, with τ defined by S2, and $\mathcal{G}' = \text{id} \times \exp(\oplus_{j>0} \mathfrak{a}_j)$, we need to establish 11.1(4). Here this means verifying that

$$\rho_{(h, \text{id})}^{\mathcal{H} \times \text{id}} \Phi(0, C, Z) = \text{Ad}_{(h, \text{id})} \Phi(0, C, Z),$$

for each arrow $h \in \mathcal{G}$ beginning as some $m \in M$, $C \in B^0|_m$, $Z \in \oplus_{j>0} \mathfrak{a}_j|_m$. With the help of (5), and the definition of Ad for semi-direct products, this is readily checked. We conclude that there exists a unique Cartan connection D on $\mathcal{G}$, at least weakly compatible with τ , such that

$$(6) \quad \bar{D}_{(h, k)} \Phi(V, C, Z) = \text{Ad}_{(\text{id}, k)}^\tau \rho_{(h, \text{id})}^{\mathcal{H} \times \text{id}} \Phi(V, C, Z),$$

where Ad^τ is the representation of $\text{id} \times \exp(\oplus_{j>0} \mathfrak{a}_j) \subset \mathcal{G}$ on A integrating the representation ad^τ of $A^1 = 0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$ on A defined by $\text{ad}_{(0, Y)}^\tau W = \tau((0, Y), W)$. By the definition of τ , the isomorphism Φ intertwines the representation ad^τ of A^1 with the representation $Y \mapsto T((0, 0, Y), \cdot)$ of $\oplus_{j>0} \mathfrak{a}_j$. Consequently, as $\exp(\oplus_{j>0} \mathfrak{a}_j)$ has connected fibres, Φ intertwines the representations Ad^τ and Ad^T :

$$(7) \quad \text{Ad}_{(\text{id}, k)}^\tau \Phi(V, C, Z) = \Phi(\text{Ad}_k^T((V, C, Z))).$$

Combining (5), (6) and (7), we obtain S3'. The last part of (2), and (3), follow by construction.

It is easy to see that the infinitesimalization ∇ of D must be the Cartan connection delivered by Theorem 12.5. Since ∇ is *strongly* compatible with τ , D is also.

Turning to (4), let $(\mathcal{G}, D, \tau)$ be a transitive, filtered, Cartan geometry with Lie algebroid A and anchor kernel A^0 . By Theorem 10.3(8), we may suppose, without loss of generality, that $\mathcal{G} = \mathcal{H} \times_R \exp(\oplus_{j>0} \mathfrak{a}_j)$, for some filtered Lie groupoid $\mathcal{H}$ and graded Lie algebra bundle $\oplus_{j>0} \mathfrak{a}_j$, where $\mathcal{H}$ acts on $\oplus_{j>0} \mathfrak{a}_j$ by grading-preserving fibre-wise derivations, an action that integrates to an action of $\mathcal{H}$ on $\exp(\oplus_{j>0} \mathfrak{a}_j)$ by fibre-wise Lie group isomorphisms, both actions to be denoted R . Moreover, the Lie algebroid B of $\mathcal{H}$ will satisfy $B^j = 0$ for $j > 0$ and, by our hypotheses on A , B is transitive with B^0 being the anchor kernel.

Evidently, A is identifiable with $B \oplus_{dR} (\oplus_{j>0} \mathfrak{a}_j)$, which the infinitesimalization ∇ of D makes into a transitive, infinitesimal, filtered Cartan geometry. Consequently, the analysis of 12.4 applies, with R in that analysis replaced with dR . Define $\mathfrak{a}'_j$, $j > 0$, and Φ as in that analysis. Then τ can be recovered from $T = \text{gr}_0(\tau)$ according to S2, and, according to Proposition 11.2(3), the grading $A = \oplus_i \mathfrak{a}'_i$ (there denoted $\oplus_i \mathfrak{a}_i$) is $\bar{D}_{(h,\text{id})}$ -invariant, for every $h \in \mathcal{H}$. It follows that Φ intertwines some grading-preserving representation R of $\mathcal{H}$ on $\text{gr}(A) = \text{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$ with the $\bar{D}$ -representation of $\mathcal{H} \times \text{id}$ on A :

$$(8) \quad \bar{D}_{(h,\text{id})} \Phi(V, C, Z) = \Phi \left(R_h(V, C, Z) \right); \quad h \in \mathcal{H}.$$

Since $\Phi(0, C, Z) = (C, Z)$ and $\bar{D}$ is just the canonical representation on sections of A^0 , R must extend the canonical representation of $\mathcal{H}$ on $\text{gr}(A^0) = B^0 \oplus (\oplus_{j>0} \mathfrak{a}_j)$, which gives

$$(9) \quad \bar{D}_{(h,\text{id})} \Phi(V, C, Z) \Phi(R_h V, \text{Ad}_h C, R_h Z); \quad h \in \mathcal{H},$$

where the “ R ” in $R_h Z$ has the meaning given at the beginning of this discussion.

On the other hand, 12.4(2) implies

$$\bar{\nabla}_{(0,Y)} \Phi(\xi) = \Phi(\text{ad}_Y^T \xi), \quad Y \in \Gamma(\oplus_{j>0} \mathfrak{a}_j), \xi \in \Gamma(\text{gr}(A)).$$

Because $\exp(\oplus_{j>0} \mathfrak{a}_j)$ has simply-connected fibres, this intertwining by Φ of infinitesimal actions is an intertwining of their corresponding globalizations:

$$(10) \quad \bar{D}_{(\text{id},k)} \Phi(V, C, Z) = \Phi(\text{Ad}_k^T(V, C, Z)).$$

The formula S3' follows from (9) and (10) and the identity $(h, k) = (\text{id}, k)(h, \text{id})$ for the semi-direct product. $\square$

13. DEFORMING MODEL DATA AND CARTAN CONNECTIONS

We need a way to relate different choices of model data in a filtered Cartan geometry using transformations of the underlying Lie algebroid or integrating Lie groupoid. Proposition 13.1 below addresses this need.

13.1. ψ -deformations. Let A be a Lie algebroid. For each $\psi \in \Gamma(T^*M \otimes A)$, define $\psi_{TM} : TM \rightarrow TM$ by

$$(1) \quad \psi_{TM} V = V + \# \psi V,$$

and define $\text{Ad}_\psi : A \rightarrow A$ by

$$(2) \quad \text{Ad}_\psi X = X + \psi(\#X).$$

Here $\# : A \rightarrow TM$ denotes the anchor. The natural context giving rise to these definitions is provided in 13.3. For the moment, it suffices to observe that: (i) If ψ_{TM} is invertible, then so is Ad_ψ , with

$$(3) \quad \text{Ad}_\psi^{-1} = \text{Ad}_{-\psi \circ (\psi_{TM})^{-1}};$$

and (ii) If A is filtered and transitive and ψ is homogeneous of degree one, i.e., $\psi(A^k) \subset A^{k+1}$ for any k , then ψ_{TM} is invertible, because in that case ψ_{TM} is filtration-preserving with $\text{gr}_0(\psi_{TM}) = \text{id}$.

Lemma. Let A be a filtered Lie algebroid and $q: A \rightarrow A$ a vector bundle isomorphism covering the identity such that $q - \text{id}$ is homogeneous of degree $l \geq 1$. Then the push-forward $\tau' = q_*\tau$ of any model data τ on A compatible with the filtration is also compatible with the filtration, and satisfies $\text{gr}_0(\tau') = \text{gr}_0(\tau)$. Assuming A^0 is the anchor kernel of A , τ' is also model data.

Proof. Since q will be filtration-preserving, so is q^{-1} , which shows τ' is compatible with the filtration. If $qX = X + Y$, then $q^{-1}X = X - q^{-1}(Y)$, and we can therefore see that $q^{-1} - \text{id}$ is also homogeneous of degree l . The push-forward is defined by $\tau'(X, Y) = q\tau(q^{-1}X, q^{-1}Y)$, and, with the preceding observations in hand, checking $\text{gr}_0(\tau') = \text{gr}_0(\tau)$ is straightforward. The last statement holds because τ is model data and q, q^{-1} are filtration-preserving. $\square$

If τ is compatible with the filtration of a filtered Lie algebroid A , with anchor kernel A^0 , then, assuming ψ is homogeneous of degree $l \geq 1$, the lemma shows that $\tau' = (\text{Ad}_\psi)_*\tau$ is model data with $\text{gr}_0(\tau') = \text{gr}_0(\tau)$. Conversely, we have:

Proposition. Let A be a transitive filtered Lie algebroid with anchor kernel A^0 , and τ, τ' model data compatible with the filtration. Assume $T := \text{gr}_0(\tau)$ admits a fibre-wise grading section E , and that $\text{gr}_0(\tau') = T$. Then:

- (4) There exists a unique $\psi \in \Gamma(T^*M \otimes A)$, necessarily homogeneous of degree one, such that $\tau' = (\text{Ad}_\psi)_*\tau$.
- (5) If τ is compatible with some infinitesimal Cartan connection ∇ , then τ' is compatible with the infinitesimal Cartan connection ∇' defined implicitly by

$$\bar{\nabla}'_X Y = \text{Ad}_\psi(\bar{\nabla}_X(\text{Ad}_\psi^{-1} Y)).$$

- (6) If $\mathcal{G}$ is a Lie groupoid integrating A and τ is compatible with some Cartan connection D , then τ' is compatible with the Cartan connection D' defined implicitly by

$$\bar{D}'_g X = \text{Ad}_{\psi(gm)} \bar{D}_g \text{Ad}_{\psi(m)}^{-1} X,$$

for each arrow $g \in \mathcal{G}$ beginning at some $m \in M$, and each $X \in A|_m$, and whose infinitesimalization is ∇' above,

- (7) If $(\mathcal{G}, D, \tau)$ is a transitive filtered Cartan geometry, then $(\mathcal{G}, D', \tau')$ is another transitive filtered Cartan geometry with the same Levi reduction. An analogous statement holds in the infinitesimal case.

When $\tau' = (\text{Ad})_*\tau$, and D' and ∇' are defined (implicitly) by (6) and (5), then the primed quantities will be called the ψ -DEFORMATIONS of τ , D and ∇ . The proof of the proposition is given in 13.2–13.6 below. In 13.7 we connect ψ -deformations to Weyl gauge transformations.

13.2. The existence of deformations. The existence part of Proposition 13.1(4) has an elementary proof proceeding as follows. Let τ and τ' be model data as in the proposition and fix a lift $\epsilon \in \Gamma(A^0)$ through $A^0 \rightarrow A^0/A^1$ of E . Then, as we have described in 10.1, we get two corresponding splittings $\Phi, \Phi': \text{gr}(A) \rightarrow A$ of the Lie algebra bundles (A, τ) and (A, τ') . By definition, we have $\Phi_*T = \tau$ and $\Phi'_*T = \tau'$, and, because τ and τ' agree on A^0 (by the definition of model data) $\Phi' \circ \Phi^{-1}$ is the identity on A^0 . It follows from the transitivity of A that $\tau' = \Psi_*\tau$, where $\Psi := \Phi' \circ \Phi^{-1}$ satisfies $\Psi(X) = X + \psi(\#X) = \text{Ad}_\psi X$ for some vector bundle morphism $\psi: TM \rightarrow A$.

Also following from the definition of splittings, we have, for any $X \in \Gamma(A^k)$ and $k \in \mathbb{Z}$,

$$\begin{aligned} (1) \quad & X \equiv \Phi(\text{gr}_k(X)) \bmod A^{k+1}, \\ (2) \quad & \Phi'(\text{gr}_k(X)) \equiv X \bmod A^{k+1}. \end{aligned}$$

From (1) we conclude that $(\Phi' \circ \Phi^{-1})(X) \equiv \Phi'(\text{gr}_k(X)) \bmod A^{k+1}$, because $\Phi' \circ \Phi^{-1}$ is filtration-preserving. Combining this with (2), we conclude that $\Phi' \circ \Phi^{-1}(X) \equiv X \bmod A^{k+1}$, i.e., ψ is homogeneous of degree one.

13.3. The kernel of $J^1\mathcal{G} \rightarrow \mathcal{G}$. Let A be any integrable Lie algebroid. The sections ψ of $T^*M \otimes A$ for which ψ_{TM} are invertible are sections of an open neighborhood of the zero section of $T^*M \otimes A$ that we denote $\text{Aut}(TM, A)$. The proposition below shows that this set is a model for the kernel of the projection $J^1\mathcal{G} \rightarrow \mathcal{G}$ for *any* Lie groupoid $\mathcal{G}$ integrating A . In particular, it is a totally intransitive Lie groupoid (a bundle of Lie groups). We now describe the corresponding groupoid structure directly.

The zero section of $T^*M \otimes A$ constitutes the set of identity elements of $\text{Aut}(TM, A)$. Multiplication is given by

$$(1) \quad \psi\varphi = \psi + \varphi + \psi \circ \# \circ \varphi = \varphi + \psi \circ \varphi_{TM}.$$

Inversion is given by

$$\psi \mapsto -\psi \circ (\psi_{TM})^{-1}.$$

A convenient consequence we mention in passing is the formula

$$(2) \quad \psi\varphi^{-1} = (\psi - \varphi) \circ \varphi_{TM}^{-1}.$$

With respect to these groupoid structures, the map

$$\psi \mapsto \psi_{TM}: \text{Aut}(TM, A) \rightarrow \text{Aut}(TM)$$

is a morphism of totally intransitive Lie groupoids and the commutative kernel consists of those $\psi: TM \rightarrow A$ with image lying in the kernel of the anchor.

The reader will readily see that the maps defined by 13.3(1) and 13.3(2) define representations of $\text{Aut}(TM, A)$ on A and TM respectively. With respect to these representations the anchor is equivariant: $\# \text{Ad}_\psi X = \psi_{TM} \# X$.

Although the proposition below provides useful context, the proof of Proposition 13.1 will not actually require it. However, the result is needed in Appendix C.

Let $\mathcal{G}$ be any Lie groupoid with Lie algebroid A . Represent elements of $J^1\mathcal{G}$ by maps tangent to some local bisection of $\mathcal{G}$, i.e., by certain vector space morphisms $T_m M \rightarrow T_g \mathcal{G}$, where $g \in \mathcal{G}$ is some arrow beginning at some $m \in M$. Recall that the adjoint representation for a Lie groupoid $\mathcal{G}$ is actually a representation of $J^1\mathcal{G}$ on A [5], and that $J^1\mathcal{G}$ also acts naturally on TM . Both representations are denoted Ad , and with respect to these the anchor is equivariant.

Proposition ([5]). *The map $\psi \mapsto \psi^\wedge: \text{Aut}(TM, A) \rightarrow J^1\mathcal{G}$, defined by $\psi^\wedge V = V + \psi V$, is a Lie groupoid morphism and embedding, whose image is the kernel of the natural projection $J^1\mathcal{G} \rightarrow \mathcal{G}$.¹³ This map is equivariant with respect to the natural representations of both groupoids on A and TM : $\text{Ad}_{\psi^\wedge} V = \psi_{TM} V$, $\text{Ad}_{\psi^\wedge} X = \text{Ad}_\psi X$, for $V \in T_m M$, $X \in A|_m$, and $m \in M$.*

13.4. The deformed Cartan connections. In passing, let us explain how the deformed connections ∇' and D' appearing in Proposition 13.1 arise very naturally. Recall that a Cartan connection D is the n -plane field determined by some Lie groupoid right-inverse $S: \mathcal{G} \rightarrow J^1\mathcal{G}$ for the natural projection $J^1\mathcal{G} \rightarrow \mathcal{G}$. Given some $\psi \in \Gamma(\text{Aut}(TM, A))$, we may, according to the preceding proposition, view it as bisection of $J^1\mathcal{G}$ and obtain, via conjugation, a corresponding automorphism K_ψ of $J^1\mathcal{G}$, sending an arrow $\mu \in J_g^1\mathcal{G}$, beginning at some $m \in M$ with projection $g \in \mathcal{G}$, to $\psi(gm)^\wedge \mu (\psi(m)^\wedge)^{-1}$. It is easy to see that $S' := K_\psi \circ S: \mathcal{G} \rightarrow J^1\mathcal{G}$ is another right-inverse, defining a new Cartan connection D' . From the definition of the corresponding representations $\bar{D}$ and $\bar{D}'$, one immediately derives the formula in 13.1(6).

Now $\bar{\nabla}$ is the infinitesimalization of the representation $\bar{D}$, and it therefore follows from the formula in 13.1(6), and a simple computation, that the infinitesimalization $\bar{\nabla}'$ of $\bar{D}'$ is given by the formula in 13.1(5).

We now return to the proof of Proposition 13.1. Regardless of whether A is integrated by some Lie groupoid, the formula for $\bar{\nabla}'$ given in 13.1(5) must define an infinitesimal Cartan connection, by the characterization of such connections given in Proposition 4.2, which applies because we assume A is transitive. The main point here is that 13.1(5) indeed defines a representation. The details are not difficult and are left to the reader. It is a simple computation to show that $\tau' = (\text{Ad}_\psi)_* \tau$ must be $\bar{\nabla}'$ -invariant whenever τ is $\bar{\nabla}$ -invariant. Since Ad_ψ is the identity on A^0 , it is easy to see that compatibility condition c2 holds in the primed case, because it holds in the unprimed case, by hypothesis. So ∇' and τ' are compatible.

Similarly, the formula for $\bar{D}'$ given in 13.1(6) must define a global Cartan connection, by the characterization of such connections given in Proposition 4.1, and checking that $\tau' = (\text{Ad}_\psi)_* \tau$ must be $\bar{D}'$ -invariant (D' -compatible) whenever τ

¹³[5] adopts a different sign convention for the embedding and definition of $\text{Aut}(TM, A)$.

is $\bar{D}$ -invariant (D -compatible) is a simple computation. Since D' has ∇' as its infinitesimalization, the primed version of the compatibility condition c2 holds, because it holds for D , and thus D' and τ' are compatible.

This completes the proof of parts (5) and (6) of Proposition 13.1.

13.5. The uniqueness of ψ -deformations. Having seen that $\psi \mapsto \text{Ad}_\psi$ defines a representation of $\text{Aut}(TM, A)$ on A , to prove the uniqueness of ψ in Proposition 13.1(4), it evidently suffices to show that $(\text{Ad}_\psi)^*\tau = \tau$ implies $\psi = 0$, whenever ψ is homogeneous of degree one, because $\text{Ad}_0 = \text{id}$.

Let τ and E be as in Proposition 13.1 and $\psi: TM \rightarrow A$ be homogeneous of degree one. If we temporarily write $\hat{\psi}(X) = \psi(\#X)$, then $(\text{Ad}_\psi)^*\tau = \tau$ means

$$\tau(\hat{\psi}X, Y) + \tau(X, \hat{\psi}Y) + \tau(\hat{\psi}X, \hat{\psi}Y) = \hat{\psi}\tau(X, Y),$$

for all $X, Y \in \Gamma(A)$. In particular, if ϵ is any lift of E through $A^0 \rightarrow A^0/A^1$, then

$$(1) \quad \tau(\hat{\psi}X, \epsilon) = \hat{\psi}(\tau(X, \epsilon)); \quad X \in \Gamma(A).$$

For an inductive argument, suppose ψ (and therefore $\hat{\psi}$) is homogeneous of degree $l \geq 1$. Then (1) gives $T(\text{gr}_l(\hat{\psi})\xi, E) = \text{gr}_l(\hat{\psi})(T(\xi, E))$, for any $\xi \in \text{gr}(A)$. In particular, if $j < 0$ and $\xi \in \text{gr}(A)_j$, then $(j + l)\text{gr}_l(\hat{\psi})\xi = j\text{gr}_l(\hat{\psi})\xi$, which can only be true if $\text{gr}_l(\hat{\psi})\xi = 0$. Since this last is trivially true in the other cases $j \geq 0$, we conclude that $\hat{\psi}$, and hence ψ , is homogeneous of degree $l + 1$. Induction on l shows $\psi = 0$. This completes the proof of Proposition 13.1(4).

13.6. Levi reductions are unchanged by ψ -deformations. Conclusion 13.1(7) follows fairly easily from the definitions and Lemma 13.1, applied to $q = \text{Ad}_\psi$, and is left to the reader.

13.7. Weyl gauge transformations and ψ -deformations. If we use a Weyl gauge transformation to transform model data, then we can say something about the ψ -deformation realizing the same change:

Proposition. *Let A be a transitive filtered Lie algebroid with anchor kernel A^0 , τ model data compatible with the filtration, and assume $T := \text{gr}_0(\tau)$ admits a fibre-wise grading section E . Then for any $Z \in \Gamma(A^l)$, $l \geq 1$, there exists unique $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l , such that $(e^{-\text{ad}_Z})_*\tau = (\text{Ad}_\psi)_*\tau$. In that case, for any integer $k < 0$,*

$$(1) \quad \psi(\#X) \equiv \tau(Z, X) \bmod A^{k+l+1}; \quad X \in \Gamma(A^k).$$

Remark. After introducing the operator ∂ in 14.2, we will be able to express (1) as $\text{gr}_l(\psi) = \partial \text{gr}_l(Z)$.

Proof. By Proposition 13.1, there exists unique $\psi \in \Gamma(T^*M \otimes A)$ such that $(e^{-\text{ad}_Z})_*\tau = (\text{Ad}_\psi)_*\tau$, but *a priori* we know only that ψ is homogeneous of degree one. Fixing some lift ϵ through $A^0 \rightarrow A^0/A^1$ of E , we obtain corresponding splittings $A = \oplus_j \mathfrak{a}_j$ and $A = \oplus_j \mathfrak{a}'_j$ with respect to the Lie algebra bundle structures

τ and $\tau' := (\text{Ad}_\psi)_* \tau$, respectively. It is not too hard to see it suffices to prove (1) under the seemingly more restrictive hypothesis $X \in \mathfrak{a}_k$, in which case

$$(2) \quad \text{Ad}_\psi X \in \mathfrak{a}'_k.$$

Since $\text{Ad}_\psi(X) = X + \psi(\#X)$ and ψ is homogeneous of degree one, we may define $X_k := X$ and have $\psi(\#X) = \sum_{m \geq k+1} X_m$ and $\text{Ad}_\psi(X) = \sum_{m \geq k} X_m$, for some $X_{k+1} \in \mathfrak{a}_{k+1}$, $X_{k+2} \in \mathfrak{a}_{k+2}$, etc., finite in number. These preliminaries set up a simple but tedious computation, tracking the relevant degrees of homogeneity. For completeness, details now follow.

The requirement (2) gives

$$(3) \quad \tau'(\epsilon, \sum_{m \geq k} X_m) = k \sum_{m \geq k} X_m.$$

We have

$$\begin{aligned} e^{\text{ad}_Z} \epsilon &= \epsilon + [Z, \epsilon] + \frac{1}{2} [Z, [Z, \epsilon]] + \cdots \\ &= \epsilon + \tau(Z, \epsilon) + \frac{1}{2} \tau(Z, \tau(Z, \epsilon)) + \cdots. \end{aligned}$$

On the other hand, we may write $Z = Z_l + Z^{l+1}$, for some $Z_l \in \Gamma(\mathfrak{a}_l)$ and $Z^{l+1} \in \Gamma(A^l)$, and since $\tau(Z_l, \epsilon) = -lZ_l$, it's not hard to see that $e^{\text{ad}_Z} \epsilon = \epsilon - lZ + \delta$, for some $\delta \in \Gamma(A^{l+1})$.

Firmness of the filtration of A , and the expansion of e^{ad_Z} , ensure that

$$(4) \quad e^{\text{ad}_Z} Y \equiv Y \pmod{A^l},$$

where here $Y \in \Gamma(A)$ is completely arbitrary. Regarding the left-hand side of (3), we now compute

$$\begin{aligned} \tau'(\epsilon, \sum_{m \geq k} X_m) &= e^{-\text{ad}_Z} \tau(e^{\text{ad}_Z} \epsilon, e^{\text{ad}_Z} (\sum_{m \geq k} X_m)) \\ (5) \quad &= e^{-\text{ad}_Z} \tau(\epsilon - lZ + \delta, \sum_{m \geq k} X_m) + e^{-\text{ad}_Z} \tau(\epsilon - lZ + \delta, W), \end{aligned}$$

where $W := e^{-\text{ad}_Z} (\sum_{m \geq k} X_m) - \sum_{m \geq k} X_m$, which lies in A^l , by (4). In particular, the second term on the right-hand side of (5) must lie in A^l also. So, developing (5) further, we have

$$\begin{aligned} \tau'(\epsilon, \sum_{m \geq k} X_m) &\equiv e^{-\text{ad}_Z} \tau(\epsilon - lZ + \delta, \sum_{m \geq k} X_m) \pmod{A^l} \\ &\equiv e^{-\text{ad}_Z} \left(\sum_{m \geq k} mX_m - l \sum_{m \geq k} \tau(Z, X_m) + \sum_{m \geq k} \tau(\delta, X_m) \right) \pmod{A^l} \\ &\equiv \sum_{m \geq k} mX_m - l \sum_{m \geq k} \tau(Z, X_m) + \sum_{m \geq k} \tau(\delta, X_m) \pmod{A^l}, \end{aligned}$$

where in the last step we have again applied identity (4). Combining this with (3), we conclude

$$(6) \quad \sum_{m>k} (m-k)X_m \equiv l \sum_{m \geq k} \tau(Z, X_m) + \sum_{m \geq k} \tau(\delta, X_m) \mod A^l.$$

Since the terms on the right-hand side lie in A^{k+l} , this shows that $X_m = 0$ whenever $k+1 \leq m \leq k+l-1$. In particular, this shows that $\psi(\# \mathfrak{a}_k) \subset A^{k+l}$. Since $k < 0$ here is arbitrary, we have proven that ψ is homogeneous of degree l . But now we may rewrite (6) as

$$\sum_{m \geq l} mX_{m+k} \equiv l \sum_{m \geq k} \tau(Z, X_m) + \sum_{m \geq k} \tau(\delta, X_m) \mod A^l.$$

Since $l \geq l+k+1$, because $k < 0$, we have $A^l \subset A^{l+k+1}$, and it follows that

$$\begin{aligned} \sum_{m \geq l} mX_{m+k} &\equiv l \sum_{m \geq k} \tau(Z, X_m) + \sum_{m \geq k} \tau(\delta, X_m) \mod A^{l+k+1} \\ &\implies lX_{l+k} \equiv l\tau(Z, X_k) \mod A^{l+k+1} \\ &\implies \psi(\#X) \equiv \tau(Z, X) \mod A^{l+k+1}. \end{aligned}$$

□

14. CURVATURE UNDER ψ -DEFORMATIONS AND KOSTANT-HODGE THEORY

We now study certain chain and cochain complexes associated with bundles of reductive Lie algebras, given a bundle of parabolic subalgebras. This will provide us with: (i) language for concisely expressing what happens to model curvature when we deform the model data and Cartan connection defining a filtered Cartan geometry (14.2); and (ii) reduce obstructions to obtaining normalization of curvature to non-triviality of certain Lie algebra (co)homology (15.8).

14.1. The Kostant codifferential. Let A be any filtered Lie algebra bundle over M and denote the bracket by τ . The main examples we have in mind are: (i) ordinary filtered Lie algebras; and (ii) the Lie algebroid associated with any infinitesimal filtered Cartan geometry (A, ∇, τ) . For now, A need not be reductive, but we assume $\text{gr}(A)$ admits a fibre-wise grading section E , so that lifts $\epsilon \in \Gamma(A^0)$ of E define splitting isomorphisms $\Phi_\epsilon: \text{gr}(A) \rightarrow A$. We get this in case (i) if A is reductive and A^0 parabolic (Corollary A.4). In case (ii), we may take ϵ to be any Weyl gauge compatible with τ (11.2), which we have by hypothesis.

Recall from 9.3 that we have a natural chain complex

$$(1) \quad 0 \xleftarrow{\partial^\tau} A \xleftarrow{\partial^\tau} A^1 \otimes A \xleftarrow{\partial^\tau} \wedge^2 A^1 \otimes A \xleftarrow{\partial^\tau} \dots,$$

where the Kostant codifferential ∂^τ is defined by 9.3(2). In the proposition below, we will establish that the corresponding bundles of homology groups have realizations depending only on $T = \text{gr}_0(\tau)$, and are canonically *graded*, and not just canonically filtered.

Notice that the bracket $T = \text{gr}_0(\tau)$ makes $\text{gr}(A)$ into a representation of the Lie algebra bundle $\text{gr}(A^1)$ and we obtain a graded version of (1),

$$(2) \quad 0 \xleftarrow{\partial^*} \text{gr}(A) \xleftarrow{\partial^*} \text{gr}(A^1) \otimes \text{gr}(A) \xleftarrow{\partial^*} \wedge^2 \text{gr}(A^1) \otimes \text{gr}(A) \xleftarrow{\partial^*} \dots,$$

where ∂^* is defined by the same formula 9.3(2), with τ there replaced with T .

The filtration of A induces a filtration of each term in the complex (1); for example decomposable elements of $(\wedge^2 A^1 \otimes A)^l$ are of the form $Y \wedge Z \otimes X$, for some $Y \in A^i$, $Z \in A^j$ and $X \in A^k$ with $i, j \geq 1$ and $i + j + k = l$. Similarly, each term of the complex (2) is graded. The differential ∂^τ is filtration-preserving, while ∂^* is grading-preserving. There is no canonical chain map from complex (1) to (2). However, for each fixed $l \in \mathbb{Z}$, we have subcomplexes

$$(3) \quad 0 \xleftarrow{\partial^\tau} A^l \xleftarrow{\partial^\tau} (A^1 \otimes A)^l \xleftarrow{\partial^\tau} (\wedge^2 A^1 \otimes A)^l \xleftarrow{\partial^\tau} \dots$$

$$(4) \quad 0 \xleftarrow{\partial^*} \text{gr}(A)_l \xleftarrow{\partial^*} (\text{gr}(A^1) \otimes \text{gr}(A))_l \xleftarrow{\partial^*} (\wedge^2 \text{gr}(A^1) \otimes \text{gr}(A))_l \xleftarrow{\partial^*} \dots,$$

and gr is a chain map from (3) to (4).

We let $H_n(A^1, A)$, $H_n(\text{gr}(A^1), \text{gr}(A))$, $H_n(A^1, A)^l$, and $H_n(\text{gr}(A^1), \text{gr}(A))_l$ denote the degree n homologies of the complexes (1), (2), (3), and (4), understood as vector bundles over M . Because ∂^τ is filtration-preserving and ∂^* is grading-preserving, for each $l \in \mathbb{Z}$, the natural morphisms

$$\begin{aligned} H_n(A^1, A)^l &\rightarrow H_n(A^1, A) \\ H_n(\text{gr}(A^1), \text{gr}(A))_l &\rightarrow H_n(\text{gr}(A^1), \text{gr}(A)) \end{aligned}$$

are injective, and we see easily that this allows us to view $H_n(A^1, A)$ as a filtered vector bundle, and $H_n(\text{gr}(A^1), \text{gr}(A))$ as a graded vector bundle.

Each term in the complex (1) is a representation of A^1 , and ∂^τ is equivariant with respect to these representations, so that $H_n(A^1, A)$ is also an A^1 -representation. These integrate to representations of the bundle of simply-connected nilpotent Lie groups $\exp(A^1)$ integrating A^1 . If A is a Lie algebroid integrating some Lie groupoid $\mathcal{G}$ admitting a Weyl gauge (case (ii) above), we may identify $\exp(A^1)$ with a normal subgroupoid $\mathcal{G}^1 \subset \mathcal{G}$, by Proposition 10.3.

Proposition. *Let A be a filtered Lie algebra bundle such that $\text{gr}(A)$ admits a fibre-wise grading section E . Then:*

- (5) *The actions of A^1 and $\exp(A^1)$ on $H_n(A^1, A)$ are trivial.*
- (6) *For each lift $\epsilon \in \Gamma(A^0)$ of E , the corresponding splitting isomorphism $\Phi_\epsilon: \text{gr}(A) \rightarrow A$ induces a map*

$$\tilde{\Phi}_\epsilon: \wedge^n \text{gr}(A^1) \otimes \text{gr}(A) \rightarrow \wedge^n A^1 \otimes A$$

that is a chain map between the complexes (2) and (1).

(7) The corresponding isomorphism in homology

$$(\tilde{\Phi}_\epsilon)_*: H_n(\text{gr}(A^1), \text{gr}(A)) \rightarrow H_n(A^1, A)$$

is independent of the choice of ϵ . In particular, the filtered vector bundle $H_n(A^1, A)$ has a canonical splitting, i.e., is canonically graded.

Remark. If (A, ∇, τ) is an infinitesimal filtered Cartan geometry, then by (5), the representation $\bar{\nabla}$ of A on itself makes $H_n(A^1, A)$ into an A/A^1 -representation. If this is the infinitesimalization of a global filtered Cartan geometry $(\mathcal{G}, D, \tau)$, then the representation $\bar{D}$ of $\mathcal{G}$ on A makes $H_n(A^1, A)$ into $\mathcal{G}/\mathcal{G}^1$ -representation.

Lemma. Let Ad^τ denote the representation of $\exp(A^1)$ on A and let ϵ_1, ϵ_2 be two lifts of E . Then there exist a section k of $\exp(A^1)$ such that $\Phi_{\epsilon_2}(\xi) = \text{Ad}_{k(m)}^\tau \Phi_{\epsilon_1}(\xi)$, for any $\xi \in \text{gr}(A)|_m$, $m \in M$.

Proof. Applying Lemma 10.1(1) fibre-wise, there is a unique section k of the bundle of simply-connected nilpotent Lie groups $\exp(A^1)$ such that $\epsilon_2(m) = \text{Ad}_{k(m)}^\tau \epsilon_1(m)$. The desired result now follows from the fibre-wise application of 10.1(2). $\square$

Proof of proposition. In proving (5), it suffices to prove the claim for A^1 , since $\exp(A^1)$ has connected source-fibres. We need to show that A^1 maps the kernel of $\partial^\tau: \wedge^n A^1 \otimes A \rightarrow \wedge^{n-1} A^1 \otimes A$ into the image of $\partial^\tau: \wedge^{n+1} A^1 \otimes A \rightarrow \wedge^n A^1 \otimes A$. Following [12, p. 263], we see this follows from the identity

$$-\text{ad}_Y^\tau \omega = \partial^\tau(Y \wedge \omega) + Y \wedge \partial^\tau \omega; \quad \omega \in \Gamma(\wedge^n A^1 \otimes A), Y \in \Gamma(A^1).$$

To prove the identity, it suffices to prove it for decomposable ω . For example, if $\omega = Z \otimes X$ ($n = 1$), then we compute

$$\partial^\tau(Y \wedge \omega) = -Z \otimes \tau(Y, X) + Y \otimes \tau(Z, X) - \tau(Y, Z) \otimes X$$

$$Y \wedge \partial^\tau \omega = Y \wedge (-\tau(Z, X)),$$

$$\text{so that } \partial^\tau(Y \wedge \omega) + Y \wedge \partial^\tau \omega = -Z \otimes \tau(Y, X) - \tau(Y, Z) \otimes X = -\text{ad}_Y^\tau \omega.$$

See op. cited for the general argument.

The definition of $\tilde{\Phi}_\epsilon$ is clear. For example, we have $\tilde{\Phi}_\epsilon(Z_1 \wedge Z_2 \otimes X) = \Phi_\epsilon(Z_1) \wedge \Phi_\epsilon(Z_2) \wedge \Phi_\epsilon(X)$. To prove (6), it suffices to show that $\partial^\tau \tilde{\Phi}_\epsilon \omega = \tilde{\Phi}_\epsilon \partial^* \omega$ for all $\omega \in \wedge^{n+1} \text{gr}(A^1) \otimes \text{gr}(A)$ that is decomposable, which is readily accomplished: Take ω to be of the form $\text{gr}(Y_0) \wedge \dots \wedge \text{gr}(Y_n) \otimes \text{gr}(X)$, where, without loss of generality, we suppose $Y_0, \dots, Y_n$ and X each lie in some component $\mathfrak{a}_j$, in the decomposition $A = \oplus_i \mathfrak{a}_i$ associated with ϵ ($j \geq 1$ in the case of $Y_0, \dots, Y_n$). Part (7) now follows from preceding parts and the lemma above. $\square$

14.2. Model curvature under ψ -deformations. Under the same blanket assumptions as 14.1, consider the *cochain* complex

$$(1) \quad 0 \xrightarrow{\partial} \text{gr}(A) \xrightarrow{\partial} \text{gr}(A)_-^* \otimes \text{gr}(A) \xrightarrow{\partial} \wedge^2 \text{gr}(A)_-^* \otimes \text{gr}(A) \xrightarrow{\partial} \cdots,$$

where $\text{gr}(A)_- := \oplus_{j < 0} \text{gr}(A)_j$ is a Lie algebra subbundle, which accordingly acts on $\text{gr}(A)$. Over each point $m \in M$, (1) is the classical cochain complex associated with the representation ad^T of $\text{gr}(A)_-|_m$ on $\text{gr}(A)|_m$. Here ∂ is defined by

$$\begin{aligned} \partial\omega(V_1, \dots, V_{n+1}) &= \sum_{i=1}^{n+1} (-1)^{i+1} T(V_i, \omega(V_1, \dots, \hat{V}_i, \dots, V_{n+1})) \\ &\quad + \sum_{i < j} (-1)^{i+j} \omega(T(V_i, V_j), V_1, \dots, \hat{V}_i, \dots, \hat{V}_j, \dots, V_{n+1}), \end{aligned}$$

where hats indicate omission. So, for example, $\partial\omega(V_1, V_2) = T(V_1, \omega(V_2)) - T(V_2, \omega(V_1)) - \omega(T(V_1, V_2))$.

The formula shows that ∂ is grading-preserving if we equip $\wedge^n \text{gr}(A)_-^* \otimes \text{gr}(A)$ with the grading in which a section ω has degree l if it is homogeneous of degree l , by which we mean that $\omega(V_1, \dots, V_n) \in \Gamma(\text{gr}(A)_{l+i_1+i_2+\dots+i_n})$ whenever $V_j \in \Gamma(\text{gr}(A)_{i_j})$, for each $j \in \{1, \dots, n\}$.

Our interest in the cochain complex (1) is following:

Proposition. *Let (A, ∇, τ) be a transitive, infinitesimal, filtered Cartan geometry, and suppose that A^0 is the anchor kernel, so that $\text{gr}(A)_- \cong \text{gr}(TM)$ and ∂_n is a map*

$$\partial_n: \wedge^n \text{gr}(T^*M) \otimes \text{gr}(A) \rightarrow \wedge^{n+1} \text{gr}(T^*M) \otimes \text{gr}(A).$$

*For some $\psi \in \Gamma(T^*M \otimes A)$ that is homogeneous of degree $l \geq 1$, let ∇' and τ' be the ψ -deformations of ∇ and τ , as defined in 13.1, and let κ, κ' denote the model curvatures of the infinitesimal filtered Cartan geometries (A, ∇, τ) and (A, ∇', τ') , viewed as sections of $\wedge^2 T^*M \otimes A$. Then $\kappa' - \kappa$ is homogeneous of degree l with $\text{gr}_l(\kappa' - \kappa) = -\partial \text{gr}_l(\psi)$.*

Proof. It is not hard to see that the claim follows from the following two facts proven below:

- (2) The restriction of $\text{tor } \bar{\nabla}' - \text{tor } \bar{\nabla}$ to a section of $\wedge^2 \text{gr}(A)_-^* \otimes A$ is homogeneous of degree $l + 1$.
- (3) The restriction $r = \tau' - \tau$ to a section of $\wedge^2 \text{gr}(A)_-^* \otimes A$ is homogeneous of degree l , with $\text{gr}_l(r) = -\partial \text{gr}_l(\psi)$.

Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ arbitrary sections, with $j, k < 0$. Here are some preliminary observations: Since ψ is homogeneous of degree $l \geq 1$, $\psi_{TM}: TM \rightarrow TM$ is filtration-preserving, with $\text{gr}_0(\psi_{TM}) = \text{id} = \text{gr}_0(\psi_{TM}^{-1})$; $\bar{\nabla}_X$ and $\bar{\nabla}_Y$ are also filtration-preserving, because ∇ is required to be compatible with filtration, by the definition of infinitesimal filtered Cartan geometries; because $j, k < 0$, we always have $A^m \subset A^{m+j+1}$ and $A^m \subset A^{m+k+1}$ for any integer m ; because $l \geq 1$,

we always have $A^{m+l} \subset A^m$, for any m ; and $\text{gr}_0(\#): \text{gr}(A)_- \rightarrow \text{gr}(TM) \cong \text{gr}(A)_-$ is the identity.

On expanding the definition of $\bar{\nabla}'$ (Equation 13.1(5)) and the definition of Ad_ψ (Equation 13.1(2)) and its inverse (Equation 13.1(3)) one obtains

$$\bar{\nabla}'_X Y - \bar{\nabla}_X Y = \bar{\nabla}_X(\psi(\psi_{TM}^{-1}(\#Y))) + \psi(\#\bar{\nabla}_X Y) + \psi(\#\bar{\nabla}_X(\psi(\psi_{TM}^{-1}(\#Y)))).$$

By our preliminary observations, the terms on the right-hand side lie in A^{k+l} , A^{k+l} and A^{k+2l} , respectively, and accordingly,

$$\bar{\nabla}'_X Y - \bar{\nabla}_X Y \in \Gamma(A^{j+k+l}) \ni \bar{\nabla}'_Y X - \bar{\nabla}_Y X,$$

where the second part follows by symmetry. Claim (2) follows immediately.

Similarly expanding $\tau' = (\text{Ad}_\psi)_* \tau$, one derives

$$\begin{aligned} (4) \quad \tau'(X, Y) - \tau(X, Y) &= -\tau(\psi(\psi_{TM}^{-1}(\#X)), Y) - \tau(X, \psi(\psi_{TM}^{-1}(\#Y))) \\ &\quad + \tau(\psi(\psi_{TM}^{-1}(\#X)), \psi(\psi_{TM}^{-1}(\#Y))) \\ &\quad + \psi(\#\tau(X, Y)) \\ &\quad - \psi(\#\tau(\psi(\psi_{TM}^{-1}(\#X))), Y) - \psi(\#\tau(X, \psi(\psi_{TM}^{-1}(\#Y)))) \\ &\quad + \psi(\#(\psi(\psi_{TM}^{-1}(\#X)), \psi(\psi_{TM}^{-1}(\#Y)))). \end{aligned}$$

By our preliminary observations, the terms on the second and fourth lines lie in A^{j+k+2l} , while that on the last line lies in A^{j+k+3l} , which means both lie in $A^{j+k+l+1}$. The remaining terms lie in A^{j+k+l} , which shows that r is indeed homogeneous of degree l , and

$$\begin{aligned} \tau'(X, Y) - \tau(X, Y) &\equiv \\ &\quad - \tau(\psi(\psi_{TM}^{-1}(\#X)), Y) - \tau(X, \psi(\psi_{TM}^{-1}(\#Y))) + \psi(\#\tau(X, Y)) \pmod{A^{j+k+l+1}}, \end{aligned}$$

from which (3) now readily follows. $\square$

14.3. The graded form of the Bianchi identity. The next result will be used later to prove Theorem 9.4(3).

Proposition. *Let (A, ∇, τ) be a transitive, infinitesimal, filtered Cartan geometry, as in Proposition 14.2, and suppose κ is homogeneous of degree $l \geq 1$. Then $\partial \text{gr}_l(\kappa) = 0$.*

This result is the “graded” version of the Bianchi identity stated in 5.8, repeated here for convenience:

$$\begin{aligned} (1) \quad &(\bar{\nabla}_{X_3} \kappa)(\#X_1, \#X_2) + \kappa(\kappa(\#X_1, \#X_2), \#X_3) \\ &+ \kappa(\#\tau(X_1, X_2), \#X_3) + \tau(\kappa(\#X_1, \#X_2), X_3) + \text{terms cyclic in 1-2-3} = 0. \end{aligned}$$

Proof. Suppose $X_1, X_2, X_3 \in \Gamma(A)$ (which are arbitrary above) have *negative* filtration degrees, summing to m . Then, as $\bar{\nabla}$ is filtration-preserving, and as κ is homogeneous of degree $l \geq 1$, it follows that $(\bar{\nabla}_{X_3}\kappa)(\#X_1, \#X_2)$, and its cyclic permutations, have at most degree $m + l + 1$. The second term in (1) will have degree at most $m + 2l$, which likewise, cannot exceed $m + l + 1$. So,

$$\kappa(\#\tau(X_1, X_2), \#X_3) + \tau(\kappa(\#X_1, \#X_2), X_3) + \text{terms cyclic in 1-2-3} \equiv 0 \pmod{A^{m+l+1}}.$$

Evidently, both the terms on the left-hand have degree $m + l$, and from this the proposition now follows easily, noting that under our identification $\text{gr}(A)_- \cong \text{gr}(TM)$, $\text{gr}_0(\#)$ is just the projection $\text{gr}(TM) \oplus \text{gr}(A^0) \rightarrow \text{gr}(TM)$. $\square$

14.4. Hodge theory. We now assume A is the Lie algebroid of a global parabolic geometry $(\mathcal{G}, \tau)$, which, in particular, guarantees the existence of a $\bar{D}$ -invariant, non-degenerate, fibre-wise trace form B_ρ for τ , determined uniquely by its value on a single fibre; see 9.3. Note that $T = \text{gr}(\tau)$ will be reductive and that $|A| \leq m$ for some $m \in \mathbb{Z}$ (Proposition 7.1(1)). Furthermore, $(\text{gr}(A), T)$, $T := \text{gr}_0(\tau)$, inherits a non-degenerate trace-form as follows: For each fibre $A|_m$, choose a grading element for the reductive Lie algebra $(A|_m, \tau)$ and use the associated splitting to push the representation ρ of $A|_m$ to a representation of $\text{gr}(A|_m)$. Infinitesimal invariance of trace-forms and Lemma 10.1(2) show that the corresponding trace-form on $\text{gr}(A|_m)$ is independent of the choice of splitting. Moreover, for any Weyl gauge ϵ , we see that the corresponding identification $A \cong \text{gr}(A)$ will be an isomorphism, with respect to these trace-forms.

As we now explain, the existence of non-degenerate trace-forms NO WE NEED HERMITIAN POSITIVE DEFINITE TRACE FORMS means that ∂^* and ∂ enjoy a relationship analogous to that between the differential and codifferential operators of classical Hodge theory [32]. Here we are extending to reductive Lie algebra *bundles* observations about semisimple Lie *algebras* originating in Kostant [19].

Using the trace-form for T , we may identify $\text{gr}(A^1) = \text{gr}(A)_1 \oplus \text{gr}(A)_2 \oplus \cdots \oplus \text{gr}(A)_k$ with $\text{gr}(A)_-^* = (\text{gr}(A)_{-k} \oplus \cdots \oplus \text{gr}(A)_{-2} \oplus \text{gr}(A)_{-1})^*$. Under this identification, we can combine 14.2(1) and 14.1(2) into one diagram

$$(1) \quad 0 \xrightleftharpoons[\partial^*]{\partial} \text{gr}(A) \xrightleftharpoons[\partial^*]{\partial} \text{gr}(A^1) \otimes \text{gr}(A) \xrightleftharpoons[\partial^*]{\partial} \wedge^2 \text{gr}(A^1) \otimes \text{gr}(A) \xrightleftharpoons[\partial^*]{\partial} \cdots$$

As we shall see, the following non-trivial decomposition, together with the calculation summarised in Proposition 14.2, is what makes normalization of curvature in parabolic geometry possible:

$$(2) \quad \text{im } \partial_{n-1} \oplus \ker \partial_n^* = \boxed{\wedge^n \text{gr}(A^1) \otimes \text{gr}(A) = \ker \partial_n \oplus \text{im } \partial_{n+1}^*}.$$

As in Hodge theory, this follows from the existence (over each point $m \in M$) of an inner product with respect to which ∂_n is the adjoint of ∂_{n+1}^* , for any n . One obtains this result by applying Kostant's construction fibre-wise, replacing the Killing form used in Kostant's semisimple setting with the non-degenerate

trace-form. Since his argument only needs ad-invariance, it carries over verbatim to the reductive setting. For the semisimple case, see [12, §3.3].

As in Hodge theory, we can deduce from (2):

Theorem. *If $\square := \partial^* \partial + \partial \partial^*$, then*

$$(3) \quad \ker \square_n = \ker_n \partial_n \cap \ker_n \partial_n^*$$

$$(4) \quad \text{and} \quad \wedge^n \text{gr}(A^1) \otimes \text{gr}(A) = \text{im } \partial_{n-1} \oplus \ker \square_n \oplus \text{im } \partial_{n+1}^*.$$

In particular, the projection $\ker \partial_n \rightarrow H^n(\text{gr}(A)_-, \text{gr}(A))$ restricts to an isomorphism $\ker \square_n \cong H^n(\text{gr}(A)_-, \text{gr}(A))$, while the projection $\ker \partial_n^ \rightarrow H_n(\text{gr}(A^1), \text{gr}(A))$ restricts to an isomorphism $\ker \square_n \cong H_n(\text{gr}(A^1), \text{gr}(A))$.*

Here $H^n(\text{gr}(A)_-, \text{gr}(A))$ is the degree- n ∂ -cohomology of (1), which by the theorem, is isomorphic to the ∂^* -homology $H_n(\text{gr}(A^1), \text{gr}(A))$, which is canonically isomorphic to the ∂^τ -homology $H_n(A^1, A)$, by Proposition 14.1. That is, the (co)homologies of the chain complexes 14.1(1), 14.1(2) and 14.2(1) are all canonically isomorphic.

The next result is an immediate consequence of 14.1(7) and transitivity, which gives an identification $\text{gr}(A)_- \cong \text{gr}(TM)$. It connects us to the point of view in the principal bundle approach, but is also important because, also following from the work of [19], the cohomology groups $H^n(\mathfrak{g}_-, \mathfrak{g})$ have explicit computations, at least in the special semisimple case [12, §3.3].

Proposition. *Let $(\mathcal{G}, \tau)$ be a parabolic geometry with Lie algebroid A , and let $\mathfrak{g}$ be a parabolically filtered Lie algebra constituting a model for the semisimple Lie algebra bundle structure τ . Without loss of generality, we may suppose the filtration of $\mathfrak{g}$ comes from a grading (Proposition 7.1(2)), and let $\mathfrak{g}_- = \bigoplus_{j < 0} \mathfrak{g}_j$. Then the mutually isomorphic (co)homology vector bundles*

$$H_n(A^1, A), \quad H_n(\text{gr}(A^1), \text{gr}(A)), \quad H^n(\text{gr}(TM), \text{gr}(A))$$

are associated bundles for a representation of the vertex group of $\mathcal{G}/\mathcal{G}^1$ over some point $m_0 \in M$ on the Lie algebra cohomology $H^n(\mathfrak{g}_-, \mathfrak{g})$.

Remark. By Remark 7.6, the vertex group is reductive, its Lie algebra being identifiable with the reductive Levi quotient $\mathfrak{p}/\mathfrak{p}^\perp \cong \mathfrak{g}_0$, where $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$.

14.5. Tractor bundles. This section, not referenced in the sequel, is included to explain the relationship between the infinitesimal Cartan connections of the current approach, and the adjoint tractor connections of the principal bundle approach. Tractor bundles are relevant to the problem of constructing invariant differential operators, which is beyond the scope of the present exposition.

Let $(\mathcal{G}, D, \tau)$ be a transitive, filtered, Cartan geometry and ∇ the corresponding infinitesimal Cartan connection on the Lie algebroid A of $\mathcal{G}$. The compatibility

conditions c2 and C1 can be written

$$\begin{aligned} \mathrm{ad}_Y^\tau X &= \bar{\nabla}_Y X; & Y \in \Gamma(A^0), X \in \Gamma(A) \\ \bar{D}_g \mathrm{ad}_Y^\tau X &= \mathrm{ad}_{\bar{D}_g Y}^\tau \bar{D}_g X; & Y, X \in A|_m, m \in M, g \in \mathcal{G}. \end{aligned}$$

Here we understand g to be some arrow beginning at m .

More generally, a **TRACTOR BUNDLE** is any representation E of $\mathcal{G}$, together with some infinitesimal representation ad^τ of the Lie algebra bundle (A, τ) on E , satisfying

$$\begin{aligned} (1) \quad & \mathrm{ad}_Y^\tau \sigma = Y\sigma; & Y \in \Gamma(A^0), \sigma \in \Gamma(E) \\ (2) \quad & g \cdot \mathrm{ad}_Y^\tau \sigma = \mathrm{ad}_{\bar{D}_g Y}^\tau (g \cdot \sigma); & Y \in A|_m, m \in M, \sigma \in E, g \in \mathcal{G}. \end{aligned}$$

Here we use juxtaposition to denote the action of A on E as a Lie algebroid (the infinitesimalization of the representation of $\mathcal{G}$ on E). Note that (2) has the infinitesimalization

$$(3) \quad X \mathrm{ad}_Y^\tau \sigma = \mathrm{ad}_Y^\tau (X\sigma) + \mathrm{ad}_{\bar{\nabla}_X Y}^\tau \sigma; \quad X, Y \in \Gamma(A), \sigma \in \Gamma(E).$$

The significance of tractor bundles is that they come equipped with Koszul connections, called **TRACTOR CONNECTIONS**, well-defined by

$$\nabla_{\#X}^E \sigma = X\sigma - \mathrm{ad}_X^\tau \sigma.$$

Here $\#$ denotes anchor. With the help of (3), one computes

$$\mathrm{curv} \nabla^E(U, V)\sigma = -\mathrm{ad}_{\kappa(U, V)}^\tau \sigma.$$

Returning to the special case $E = A$, where ∇^A is known as the **ADJOINT TRACTOR CONNECTION**, we readily derive the formula

$$(4) \quad \nabla_V^A Y = \nabla_V Y + \kappa(V, \#Y),$$

which proves the following:

The infinitesimal Cartan connection is the adjoint tractor connection if and only if the model curvature vanishes.

For an explicit construction of tractor bundles, connecting them with the definition appearing in [12], we adopt the principal bundle point of view. Fix some arbitrary $m_0 \in M$, let $\Pi \subset \mathcal{G}$ denote all arrows beginning at m_0 , $P \subset \Pi$ the Lie group of arrows beginning and ending at m_0 , $\mathfrak{g}$ the semisimple Lie algebra $(A|_{m_0}, \tau)$, and $\mathfrak{g}^0$ the Lie algebra of P . Transitivity means Π will be a principal P -bundle. Now let V be a finite dimensional representation of P and suppose the corresponding infinitesimal representation $\mathfrak{g}^0 \rightarrow \mathfrak{gl}(V)$ is the restriction of a representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. Then, we have:

Proposition. *The associated bundle $E(\rho) = \Pi \times_P V$ is a tractor bundle, and all tractor bundles are of the form $E(\rho)$ for some $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ as above.*

Proof. For elements $g_1, g_2 \in \Pi$ and $\xi \in \mathfrak{g}$, let us write

$$\begin{aligned}\langle g_1, g_2 \rangle &= g_1 g_2^{-1} \in \mathcal{G} \\ \langle g_2, \xi \rangle &= \bar{D}_{g_2} \xi \in A.\end{aligned}$$

Transitivity means arbitrary elements of $\mathcal{G}$ and A are of this form, and we have

$$(5) \quad \bar{D}_{\langle g_1, g_2 \rangle} \langle g_2, \xi \rangle = \langle g_1, \xi \rangle.$$

If, for each $[g_2, v] \equiv (g_2, v) \bmod P \in E(\rho)$, we define

$$\text{ad}_{\langle g_2, \xi \rangle}^\tau [g_2, v] = [g_2, \rho_\xi v],$$

then, with the help of (5), it is easy to see that (1) and (2) hold. The proof of the second assertion is left to the reader. $\square$

15. PROOFS OF EXISTENCE, UNIQUENESS, AND EQUIVALENCE

This section proves results only stated in Section 9

15.1. All parabolic geometries admit compatible Weyl gauges. Suppose τ is the model data associated with a parabolic geometry (global or infinitesimal). Then, according to Lemma ??, $T = \text{gr}_0(\tau)$ is a semisimple Lie algebra bundle structure on $\text{gr}(A)$, where A is the underlying filtered Lie algebroid. By Lemma ??, we accordingly have a *unique* section E of $\text{gr}(A)_0$ that is a fibre-wise grading section of $(\text{gr}(A), T)$, what we shall call the CANONICAL GRADING SECTION. We prove the following after stating some consequences:

Proposition. *For any global or infinitesimal parabolic geometry, any lift ϵ of the canonical grading section E is a Weyl gauge compatible with the model data τ , and all Weyl gauges compatible with τ are lifts of this form.*

The existence of Weyl gauges, combined with Corollary 11.2, delivers:

Corollary A. *In a global or infinitesimal parabolic geometry, any Cartan connection compatible with the model data is automatically compatible with the Lie algebroid filtration.*

Having already established, in Theorem ??, that the (infinitesimal) Cartan connections associated with (infinitesimal) parabolic geometries are automatically strongly compatible with the model data, we obtain:

Corollary B. *Every (infinitesimal) parabolic geometry is an (infinitesimal) filtered Cartan geometry.*

Theorem 12.6, which describes all filtered Cartan geometries as semi-direct products, now gives:

Corollary C. *Every Lie groupoid $\mathcal{G}$ admitting the structure of a parabolic geometry $(\mathcal{G}, \tau)$, and in particular, every action groupoid $\mathcal{G} = G \times G/P$ of a generalized flag manifold, is isomorphic to the semi-direct product $\mathcal{G}/\mathcal{G}^1 \times_{R^D} \exp \operatorname{gr}(T^*M)$. Here D is a τ -compatible Cartan connection (defined by 4.1(1), in the flag manifold case) and $\mathcal{G}^1$ and the representation R^D have their meanings described in 8.1 and 9.1.*

Proof of proposition. This time we begin with the global case which is simpler. Let $(\mathcal{G}, \tau)$ be a parabolic geometry, M the base manifold, and D a Cartan connection compatible with τ and the filtration of the Lie algebroid A of $\mathcal{G}$. Since τ is $\bar{D}$ -invariant (D and τ are compatible) T is invariant with respect to the associated representation of $\mathcal{G}$ on $\operatorname{gr}(A)$, that we have also been denoting $\bar{D}$ (see 8.1). In particular, for any arrow $g \in \mathcal{G}$ beginning at some $m \in M$, $\bar{D}_g$ must map grading elements of $\operatorname{gr}(A)|_m$ to grading elements of $\operatorname{gr}(A)|_{gm}$, which, by the aforementioned uniqueness, means $\bar{D}_g E(m) = E(gm)$, i.e., E is $\bar{D}$ -invariant, which is the same as Ad -invariance, since $\bar{D}$ restricts to the canonical representation on A^0 . It follows that any lift ϵ of E through $A^0 \rightarrow A^0/A^1$ will be a Weyl gauge for $\mathcal{G}$. It is not hard to see that every Weyl gauge is of this form.

Turning to the infinitesimal case, let (A, τ) be an infinitesimal parabolic geometry and $\bar{\nabla}$ a Cartan connection compatible with τ and the filtration of A . Since τ is $\bar{\nabla}$ -invariant (compatibility) T is invariant with respect to the associated representation of A on $\operatorname{gr}(A)$, which we have also been denoting by $\bar{\nabla}$. Let $X \in \Gamma(A)$, $j \in \mathbb{Z}$ and $Y \in \Gamma(\operatorname{gr}(A)_j)$ be arbitrary. Then $T(E, Y) = jY$ and, applying $\bar{\nabla}_X$ to both sides, we obtain, courtesy of the $\bar{\nabla}$ -invariance of T ,

$$(1) \quad T(\bar{\nabla}_X E, Y) + T(E, \bar{\nabla}_X Y) = j \bar{\nabla}_X Y.$$

On the other hand, the grading of $\operatorname{gr}(A)$ is $\bar{\nabla}$ -invariant (the filtration of A is $\bar{\nabla}$ -invariant) so that $\bar{\nabla}_X Y \in \Gamma(\operatorname{gr}(A)_j)$, implying $T(E, \bar{\nabla}_X Y) = j \bar{\nabla}_X Y$. From (1) we now conclude that $T(\bar{\nabla}_X E, Y) = 0$. From the arbitrariness of our choice of j and Y above, we conclude that $\bar{\nabla}_X E$ is fibre-wise in the center of a semisimple Lie algebra, and consequently vanishes. As $\bar{\nabla}$ restricts to the canonical representation on $\operatorname{gr}(A^0)$, this shows that E is ad -invariant. Accordingly, any lift ϵ of E through $A^0 \rightarrow A^0/A^1$ will be a Weyl gauge for A . It is not hard to see that every Weyl connection is of this form. $\square$

15.2. Axiom P3 holds everywhere if it holds at a point.

Proposition. *If M is connected then, in the global groupoid case, and in the case of an integrable Lie algebroid, it suffices to check that Axiom P3 of 7.5 holds over a single point $m \in M$.*

Proof. We begin with the global case. Assume P3 for parabolic geometries holds at $m_1 \in M$ and let $m_2 \in M$ be arbitrary. By transitivity, there exists an arrow $g \in \mathcal{G}$ from m_1 to m_2 . If D is a Cartan connection on $\mathcal{G}$ compatible with τ , then compatibility means $\bar{D}_g$ is a Lie algebra isomorphism between $(A|_{m_1}, \tau|_{m_1})$ and

$(A|_{m_2}, \tau|_{m_2})$. It is also filtration-preserving, because D is automatically compatible with the filtration (Corollary B above). It follows that P3 holds at m_2 .

In the integrable, infinitesimal case, we may suppose A is integrated by some Lie groupoid $\mathcal{G}$ with simply connected source fibres, so that the representation $\bar{\nabla}$ integrates to a representation $\bar{D}$, which must leave τ and the filtration of A invariant, because $\bar{\nabla}$ does. We can now apply the preceding arguments verbatim. $\square$

15.3. On the Lie subgroupoid $\mathcal{G}^1$ integrating A^1 .

Proposition. *Let $(\mathcal{G}, \tau)$ be a parabolic geometry, and let $\mathcal{G}^1 \subset \mathcal{G}$ denote the source-connected subgroupoid integrating A^1 . Then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups; it is also a normal subgroupoid of $\mathcal{G}$, so that $\mathcal{G}/\mathcal{G}^1$ is a Lie groupoid, also transitive.*

Proof. The first result claim follows from Proposition 10.3(3), the second from Theorem 10.3(4) and the existence of Weyl gauges. $\square$

15.4. The existence of parabolic geometries with given Levi reduction. Corollary B above shows that that following result is a special case of Theorem 12.6(2):

Theorem (Existence). *Every Levi structure $(\mathcal{H}, T, R)$ is, up to isomorphism, the reduction of some parabolic geometry $(\mathcal{G}, \tau)$. If the Levi structure is regular, then the parabolic geometry can be chosen to be regular.*

15.5. Curvature under automorphisms. As remarked in 9.3, FIX ME we can use a multiple of a non-degenerate trace-form on $\text{gr}(A)$ the Killing form to view model curvature as a section of $\wedge^2 A^1 \otimes A$, which appears in a natural chain complex with differential ∂^τ defined by ??(2). This alternative representation of the curvature transforms naturally under automorphisms:

Theorem. *Let $(\mathcal{G}, \tau)$ be any parabolic geometry and $Q: \mathcal{G} \rightarrow \mathcal{G}'$ a morphism of filtered Lie groupoids, in the sense of 12.3. Then $(\mathcal{G}', \tau')$ is a parabolic geometry, where $\tau' = q_*\tau$ and $q := dQ: A \rightarrow A'$ is the infinitesimalization of Q . In particular, (A', τ') is an infinitesimal parabolic geometry.*

More generally, for any infinitesimal parabolic geometry (A, τ) , and filtration-preserving Lie algebroid morphism $q: A \rightarrow A'$, we have: (i) (A', τ') is an infinitesimal parabolic geometry; (ii) τ' is regular if and only if τ is regular; and (iii) The corresponding model curvatures, κ and κ' , when expressed as sections of $\wedge^2 A^1 \otimes A$ and $\wedge^2 (A')^1 \otimes A'$, are q -related, i.e., $\kappa' = q_\kappa$, where $q_*(Z_1 \wedge Z_2 \otimes X) := q_*Z_1 \wedge q_*Z_2 \otimes q_*X$. In particular, if τ is normal, then so is τ' .*

Proof. The very last assertion follows from the others, for it is easy to see, from the definition 9.3(2) of ∂^τ , that $\partial^{\tau'} \circ q_* = q_* \circ \partial^\tau$. These other claims follow readily from the Proposition 5.7 and Theorem ??, with the exception of the transformation rule $\kappa' = q_*\kappa$. To prove it, remember that the identification $T^*M \cong A^1$, needed to view

curvature as a section of $\wedge^2 A^1 \otimes A$, depends on the model data τ , while in the case of $T^*M' \cong (A')^1$, it depends on τ' . With this in mind, we see then that the rule follows from 5.7(3) and the lemma below. $\square$

Lemma. Let $\iota: A^1 \rightarrow T^*M$ denote the isomorphism induced by the τ -Killing form, B (or some fixed multiple thereof), and $\iota': (A')^1 \rightarrow T^*M'$ the isomorphism induced by the τ' -Killing form, B' (or the same fixed multiple thereof). Then the following diagram commutes:

$$\begin{array}{ccc} A^1 & \xrightarrow{q} & (A')^1 \\ \iota \downarrow & & \downarrow \iota' \\ T^*M & \xrightarrow{(Tf^{-1})^*} & T^*M \end{array} .$$

Here $f: M \rightarrow M'$ is the map covered by $q: A \rightarrow A'$ and $(Tf^{-1})^*$ denotes the adjoint of Tf^{-1} .

Proof of lemma. It is easy to see the choice of multiplicative constant is immaterial, so suppose it is one. Now $B'(X, Y) = B(q^{-1}X, q^{-1}Y)$, because $\text{ad}^{\tau'} = q \circ \text{ad}^\tau \circ q^{-1}$. On the other hand, ι is defined implicitly by $\langle \iota X, \#Y \rangle = B(X, Y)$, for arbitrary sections X and Y of A^1 and A , respectively, and we have an analogous characterization for ι' . We now compute, for arbitrary $X \in A^1|_m$, $Y \in A|_m$, $m \in M$,

$$\begin{aligned} \langle \iota' qX, \#Y \rangle &= B'(qX, Y) = B(X, q^{-1}Y) = \langle \iota X, \#q^{-1}Y \rangle \\ &= \langle \iota X, (Tf^{-1})(\#Y) \rangle = \langle (Tf^{-1})^* \iota X, \#Y \rangle. \end{aligned}$$

$\square$

15.6. Normalizing model curvature. We turn now to the proof of the following central result:

Theorem (Normalization). *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry. Then there exists **normal**, model data τ' such that $(\mathcal{G}, \tau')$ is a regular parabolic geometry with the same Levi reduction.*

Let A be the Lie algebroid of $\mathcal{G}$.

Lemma A. Suppose $\sigma \in \Gamma(\wedge^n A^1 \otimes A)$ and $\partial^\tau \sigma$ is homogeneous of degree l . Then there exists $\tilde{\sigma} \in \Gamma(\wedge^n A^1 \otimes A)$, homogeneous of degree l , such that $\partial^\tau \tilde{\sigma} = \partial^\tau \sigma$.

Proof. Any lift ϵ of the canonical grading section E of $\text{gr}(A)$ defines a splitting $A = \bigoplus_{j \in \mathbb{Z}} A_j$ for the filtration of A , which in turn gives us a grading of $C^n := \wedge^n A^1 \otimes A$, which is naturally filtered; specifically, C_m^n is all finite sums of expressions of the form $Z_1 \wedge \dots \wedge Z_n \otimes X$, where $Z_{i_j} \in A_j$ ($i_j \geq 1$), $X \in A_k$ ($k \in \mathbb{Z}$), and $i_1 + \dots + i_n + k = m$. In particular, writing $\sigma = \sum_m \sigma_m$ (finite sum) for some $\sigma_m \in C_m^n$, we have $\sum_m \partial^\tau \sigma_m \in (C^{n-1})^l$, by hypothesis. But from this it follows

that $\partial^\tau \sigma_j = 0$ for all $j < l$, because ∂^τ maps C_m^n to C_m^{n-1} , for all m . We may therefore take $\tilde{\sigma} = \sum_{j \geq l} \sigma_j$. $\square$

Notation. If τ' is also model data on A , and if $\tau' = (\text{Ad}_\psi)_* \tau$, for some $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l , then we will write $\tau \stackrel{l}{\sim} \tau'$.

Lemma B. If $\sigma \in \Gamma(\wedge^n A^1 \otimes A)$ and $\tau' \stackrel{l}{\sim} \tau$, then $\partial^{\tau'} \sigma - \partial^\tau \sigma$ is homogeneous of degree $l + 1$

Proof. Expanding the definition of Ad_ψ (Equation 13.1(2)) it is not difficult to see that $\tau' \stackrel{l}{\sim} \tau$ implies that $\tau' - \tau$ is homogeneous of degree l . The lemma follows easily from this fact and the definition of ∂^τ (Equation 9.3(2)). $\square$

Proof of theorem. Recall that NORMAL means $\partial^\tau \kappa = 0$. To prove existence of normal model data it suffices to prove the following:

If $\partial^\tau \kappa$ is homogeneous of degree $l \geq 1$, then there exists model data τ' with $\tau' \stackrel{l}{\sim} \tau$ such that $\partial^{\tau'} \kappa$ is homogeneous of degree $l + 1$, where κ' is the model curvature of τ' .

For then the theorem follows from an induction on l , an induction anchored by the fact that κ (and hence $\partial^\tau \kappa$) is homogeneous of degree one (Proposition ??).

So, suppose $\partial^\tau \kappa$ is homogeneous of degree $l \geq 1$. Recall that gr is a chain map between the chain complexes 14.1(3) and 14.1(4), so that we have the identity

$$(1) \quad \partial^* \circ \text{gr}_l = \text{gr}_l \circ \partial^\tau.$$

Identifying T^*M with A^1 using the Killing form of τ , there exists, by Lemma A, $\tilde{\kappa} \in \Gamma(\wedge^2 T^*M \otimes A)$, homogeneous of degree l , such that $\partial^\tau \tilde{\kappa} = \partial^\tau \kappa$. From the boxed equation in 14.4(2), there exists $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l , such that $\text{gr}_l(\tilde{\kappa}) - \partial \text{gr}_l(\psi)$ lies in $\ker \partial^*$, so that $\partial^* \text{gr}_l(\tilde{\kappa}) = \partial^* \partial \text{gr}_l(\psi)$. Applying (1) to the left-hand side, we can write this as

$$(2) \quad \text{gr}_l(\partial^\tau \kappa) = \partial^* \partial \text{gr}_l(\psi),$$

which eliminates the auxiliary variable $\tilde{\kappa}$. On the other hand, if we take $\tau' = (\text{Ad}_\psi)_* \tau$, then Proposition 14.2 tells us that $\kappa' - \kappa$ is homogeneous of degree l with

$$(3) \quad \text{gr}_l(\kappa' - \kappa) = -\partial \text{gr}_l(\psi).$$

Now as both $\kappa' - \kappa$ and $\partial^\tau \kappa$ are homogeneous of degree l , so is $\partial^\tau \kappa'$. But, applying ∂^* to both sides of (3) and applying (1) again on the left, we obtain $\text{gr}_l(\partial^\tau \kappa' - \partial^\tau \kappa) = -\partial^* \partial \text{gr}_l(\psi)$, which can accordingly be written

$$\text{gr}_l(\partial^\tau \kappa') = \text{gr}_l(\partial^\tau \kappa) - \partial^* \partial \text{gr}_l(\psi)$$

Substituting (2), we obtain $\text{gr}_l(\partial^\tau \kappa') = 0$ which, on account of Lemma B, shows $\text{gr}_l(\partial^{\tau'} \kappa') = 0$, as required.

By construction, the normal model data τ' is a ψ -deformation of τ . Keeping Theorem ?? in mind, it follows, as a special case of Proposition 13.1(7), that (A', τ') is an infinitesimal parabolic geometry with the same Levi reduction as (A, τ) , and in the global case that $(\mathcal{G}, \tau')$ is a parabolic geometry with the same Levi reduction as $(\mathcal{G}, \tau)$. $\square$

15.7. Harmonic curvature as an obstruction to symmetry. Recall that for model curvature to vanish, it suffices that the harmonic curvature vanishes (Corollary 9.4). The proof of this result rested on Theorem 9.4(3), restated here for convenience:

Proposition. *Let κ be the model curvature of a regular and normal parabolic geometry $(\mathcal{G}, \tau)$. Then if κ is homogeneous of $l \geq 1$ and $(\kappa_H)_l = 0$, then κ is homogeneous of degree $l + 1$.*

Proof. Choose any Weyl gauge ϵ to get a splitting $\tilde{\Phi}_\epsilon: \wedge^2 \text{gr}(A^1) \otimes \text{gr}(A) \rightarrow \wedge^2 A^1 \otimes A$, as in Proposition 14.1. Then, as κ is homogeneous of degree l , we may write $(\tilde{\Phi}_\epsilon)^* \kappa = \sum_{i \geq l} \kappa_i$, for some sections κ_i of $(\wedge^2 \text{gr}(A^1) \otimes \text{gr}(A))_i$ satisfying $\partial^* \kappa_i = 0$, because $\partial^* \kappa = 0$. Since $(\kappa_H)_l = 0$, we can see that the homology class $[\kappa_l]$ must vanish. This follows from the proposition just cited and the definition of the natural grading of $H_2(A^1, A)$ given there. However, $\partial \text{gr}_l(\kappa) = 0$, by Proposition 14.3, i.e., $\partial \kappa_l = 0$, so that in fact $\kappa_l \in \ker \partial_2 \cap \ker \partial_2^* = \ker \square_2$. But, according to Theorem 14.4, the map $\omega \rightarrow [\omega]: \ker \partial_2^* \rightarrow H_2(\text{gr}(A^1), \text{gr}(A))$ is, when restricted to $\ker \square_2$, injective. Since $[\kappa_l] = 0$, so then $\kappa_l = 0$, and we conclude κ is homogeneous of degree $l + 1$. $\square$

15.8. Uniqueness up to Weyl gauge transformations.

Theorem (Uniqueness). *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry and let $\mathfrak{g}$ be a Lie algebra model for τ , identified with a graded reductive Lie algebra, as detailed at the end of 9.4. Assume $H^1(\mathfrak{g}_-, \mathfrak{g})^1 = 0$. Then if $(\mathcal{G}, \tau')$ is second parabolic geometry with $\text{gr}_0(\tau') = \text{gr}_0(\tau)$ and both τ and τ' are normal, then $\tau' = q_* \tau$ for some infinitesimal Weyl gauge transformation q .*

Proof. Write $\tau_1 := \tau$. We will construct a sequence of infinitesimal Weyl gauge transformations $q_1, q_2, q_3, \dots$, such that for any $l > 1$, $\tau' \stackrel{l}{\sim} \tau_l$ (see Notation 15.6) where $\tau_l := (q_{l-1})_* \tau_{l-1}$. Then, for sufficiently large k , we will have $\tau' = \tau_k$, because any $\psi \in \Gamma(T^*M \otimes A)$ of sufficiently high a degree of homogeneity vanishes, making Ad_ψ the identity. In that case, we will have $\tau' = q_* \tau$, where $q = q_{k-1} \circ q_{k-2} \circ \dots \circ q_1$, and the theorem will be proven, because normality is preserved by Weyl gauge transformations (Theorem 15.5), as is regularity, because we will have $\text{gr}_0(\tau') = \text{gr}_0(\tau)$, by Lemma 13.1.

So, for an induction, suppose τ_l is model data such that

- (1) $\tau' \stackrel{l}{\sim} \tau_l,$
- (2) τ_l is regular with normal curvature $\kappa_l,$

If $l = 1$, (1) holds, by Proposition 13.1(4); (2) holds by hypothesis. This anchors our induction. Our remaining task is to construct an infinitesimal Weyl gauge transformation q_l such that $\tau_{l+1} := (q_l)_* \tau_l$ satisfies the above with l replaced with $l + 1$.

Now (1) is shorthand for the existence of some $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l , such that $\tau' = (\text{Ad}_\psi)_* \tau_l$. By Proposition 14.2, $\kappa' - \kappa_l$ will be homogeneous of degree l , with

$$(3) \quad \text{gr}_l(\kappa' - \kappa_l) = -\partial \text{gr}_l(\psi).$$

Since τ_l is normal, $\partial^{\tau_l} \kappa' = \partial^{\tau_l}(\kappa' - \kappa_l) - \partial^{\tau_l} \kappa_l = \partial^{\tau_l}(\kappa' - \kappa_l)$ must be homogeneous of degree l . Since $\partial^{\tau'} \kappa' = 0$, it follows from Lemma B, that $\text{gr}_l(\partial^{\tau_l} \kappa') = 0$. Applying ∂^* to both sides of (3), using the normality of κ_l , and the identity $\partial^* \circ \text{gr}_l = \text{gr}_l \circ \partial^{\tau_l}$, we obtain $\text{gr}_l(\partial^{\tau_l} \kappa') = -\partial^* \partial \text{gr}_l(\psi)$, and whence $\partial^* \partial \text{gr}_l(\psi) = 0$. So $\partial \text{gr}_l(\psi) = 0$, by the direct sum decomposition on the left of 14.4(2). But the hypothesis on vanishing cohomology and Proposition 14.4 show that $H^1(\text{gr}(TM), \text{gr}(A))_l = 0$, so that there exists $Z \in \Gamma(A^l)$ such that

$$(4) \quad \text{gr}_l(\psi) = \partial \text{gr}_l(Z).$$

Define $q_l := e^{-\text{ad}_Z}$, and $\tau_{l+1} = (q_l)_* \tau_l$. Applying Proposition 13.7, there exists $\tilde{\psi} \in \Gamma(T^*M \otimes A)$, homogeneous of degree l , such that $\tau_{l+1} = (\text{Ad}_{\tilde{\psi}})_* \tau_l$ and $\tilde{\psi}(\#X) \equiv \tau_l(Z, X) \bmod A^{k+l+1}$, whenever $X \in A^k$. This last can be expressed compactly as $\text{gr}_l(\tilde{\psi}) = \partial \text{gr}_l(Z)$. By (4) we therefore have $\text{gr}_l(\psi - \tilde{\psi}) = 0$, i.e., $\psi - \tilde{\psi}$ is homogeneous of degree $l + 1$. Viewing $\psi, \tilde{\psi}$ as sections of the totally intransitive groupoid $\text{Aut}(TM, A)$, we conclude, using the product formula 13.3(2), that $\psi \tilde{\psi}^{-1}$ is also homogeneous of degree $l + 1$, which is relevant because

$$\tau' = (\text{Ad}_\psi)_* \tau_l = (\text{Ad}_\psi)_* (\text{Ad}_{\tilde{\psi}^{-1}})_* \tau_{l+1} = (\text{Ad}_{\psi \tilde{\psi}^{-1}})_* \tau_{l+1},$$

which shows $\tau' \stackrel{l+1}{\sim} \tau_{l+1}$. The new model curvature τ_{l+1} is normal and regular because τ_l is normal and regular, and these properties are preserved by infinitesimal Weyl gauge transformations. $\square$

15.9. Prolongation as functor. Morphisms of filtered Lie algebroids and Lie groupoids have been described in 12.3 (where morphisms of abstract Levi structures were also defined). As it suffices for current purposes, we only define morphisms of (non-abstract) Levi structures that cover the identity on M . So, a MORPHISM of infinitesimal Levi structures (B, R, T) and $(\tilde{B}, \tilde{R}, \tilde{T})$ consists of a

morphism $q: B \rightarrow \tilde{B}$ of filtered Lie algebroids covering the identity such that the induced morphisms of graded vector bundles

$$\begin{aligned} q_*: \operatorname{gr}(TM) \oplus B^0 \oplus \operatorname{gr}(T^*M) &\rightarrow \operatorname{gr}(TM) \oplus \tilde{B}^0 \oplus \operatorname{gr}(T^*M) \\ (V, C, Z) &\mapsto (V, qC, Z) \end{aligned}$$

is: (i) a morphism of Lie algebra bundles, with respect to the brackets T and $\tilde{T}$; and (ii) an intertwining of the infinitesimal representations R and $\tilde{R}$. Note that (i) implies that T and $\tilde{T}$ have the same restrictions to $\operatorname{gr}(TM)$ and $\operatorname{gr}(T^*M)$. A MORPHISM of Levi structures $(\mathcal{H}, R, T)$ and $(\tilde{\mathcal{H}}, \tilde{R}, \tilde{T})$ is a morphism $Q: \mathcal{H} \rightarrow \tilde{\mathcal{H}}$ of filtered Lie groupoids such that: (i) dQ is a morphism of the corresponding infinitesimal Levi structures; and (ii) $(dQ)_*$ intertwines the *global* representations R and $\tilde{R}$.

The PROLONGATION of a morphism q of abstract infinitesimal Levi structures as above is the morphism of firmly filtered Lie algebroids

$$\begin{aligned} B \oplus_R \operatorname{gr}(T^*M) &\rightarrow \tilde{B} \oplus_{\tilde{R}} \operatorname{gr}(T^*M) \\ (X, Y) &\mapsto (qX, Y). \end{aligned}$$

The PROLONGATION of a morphism Q of global Levi structures as above is the morphism of firmly filtered Lie groupoids,

$$\begin{aligned} \mathcal{H} \times_R \exp \operatorname{gr}(T^*M) &\rightarrow \tilde{\mathcal{H}} \times_{\tilde{R}} \exp \operatorname{gr}(T^*M) \\ (h, k) &\mapsto (Q(h), k). \end{aligned}$$

With these definitions, one readily establishes the following facts:

Proposition. *Abstract prolongation is a functor from the category of infinitesimal or global Levi structures to the category of firmly filtered Lie algebroids, resp., Lie groupoids.*

Corollary. *If two infinitesimal or global Levi structures are isomorphic, then so are their corresponding prolongations.*

15.10. The equivalence problem. Recall that we can associate a graded semisimple Lie algebra model $\mathfrak{g}$ with any parabolic geometry or Levi structure, and that vanishing of $H(\mathfrak{g}_-, \mathfrak{g})^1$ depends only on the isomorphism class in each case.

Theorem (Equivalence). *The map that sends a regular normal parabolic geometry over M to its corresponding Levi reduction, induces a bijection Ξ between: (i) isomorphism classes of regular normal parabolic geometries over M satisfying $H(\mathfrak{g}_-, \mathfrak{g})^1 = 0$; and (ii) isomorphism classes of regular Levi structures over M satisfying $H(\mathfrak{g}_-, \mathfrak{g})^1 = 0$.*

Proof. The surjectivity of Ξ follows immediately from the Existence Theorem 15.4 and Normalization Theorem 15.6. To prove injectivity, suppose that γ and $\tilde{\gamma}$ are isomorphism classes of parabolic geometries that have representatives $(\mathcal{G}, \tau)$ and $(\tilde{\mathcal{G}}, \tilde{\tau})$ with isomorphic Levi reductions, $\mathcal{L}$ and $\tilde{\mathcal{L}}$. Our task is to show that $\gamma = \tilde{\gamma}$.

By Corollary 15.9, the prolongations $\text{prol}(\mathcal{L})$ and $\text{prol}(\tilde{\mathcal{L}})$ of $\mathcal{L}$ and $\tilde{\mathcal{L}}$ are isomorphic, as filtered Lie groupoids. So there exists a Lie groupoid isomorphism $Q: \text{prol}(\tilde{\mathcal{L}}) \rightarrow \text{prol}(\mathcal{L})$ such that $q := dQ$ is filtration-preserving.

On the other hand, every regular parabolic geometry is isomorphic, as a filtered Lie groupoid, to the prolongation of its Levi reduction (Theorem 12.1(4), specialized to the parabolic case) so that $\mathcal{G} \cong \text{prol}(\mathcal{L})$ and $\tilde{\mathcal{G}} \cong \text{prol}(\tilde{\mathcal{L}})$. We push the model data τ and $\tilde{\tau}$ to $\text{prol}(\mathcal{L})$ and $\text{prol}(\tilde{\mathcal{L}})$; denoting these by the same symbols τ and $\tilde{\tau}$, Theorem 15.5 shows that $(\text{prol}(\mathcal{L}), \tau)$ and $(\text{prol}(\tilde{\mathcal{L}}), \tilde{\tau})$ are also regular normal parabolic geometries. By construction, they are representatives of γ and $\tilde{\gamma}$ respectively.

According to Theorem 15.5, $\tau' := Q_*\tilde{\tau}$ is regular normal model data on $\text{prol}(\mathcal{L})$ that makes $(\text{prol}(\mathcal{L}), \tau')$ into a parabolic geometry. By the Uniqueness Theorem 15.8, there exists an infinitesimal Weyl gauge transformation $r: A \rightarrow A$ such that $\tau' = r_*\tau$. But every infinitesimal Weyl gauge transformation is integrated by a global Weyl gauge transformation, i.e., $r = dR$, for some Weyl gauge transformation $R: \mathcal{G} \rightarrow \mathcal{G}$, which is, in particular, an isomorphism $(\mathcal{G}, \tau) \rightarrow (\mathcal{G}, \tau')$. But then, by construction, $Q^{-1} \circ R$ is an isomorphism $(\text{prol}(\mathcal{L}), \tau) \rightarrow (\text{prol}(\tilde{\mathcal{L}}), \tilde{\tau})$, which shows $\gamma = \tilde{\gamma}$. □

APPENDIX A. REDUCTIVE LIE ALGEBRAS AND THEIR PARABOLIC SUBALGEBRAS

Sections A.1–A.4 below are adapted from [10], to which the reader is referred for omitted proofs. The results there are established by rather direct methods the reader will not readily find in standard texts, which tend to emphasize semisimple Lie algebras and algebraically closed base fields.

In this appendix all Lie algebras are finite-dimensional over a field of characteristic zero.

A.1. Nilpotency. A subalgebra $\mathfrak{m} \subset \mathfrak{g}$ is a **NIL SUBALGEBRA** if $\rho(X)$ is nilpotent for any representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. The **NILPOTENT RADICAL** of $\mathfrak{g}$, denoted here by $\text{nil}(\mathfrak{g})$, is the largest nil subalgebra that is also an ideal of $\mathfrak{g}$. See [10] for other characterizations.¹⁴

Remark. It can be shown that if $\mathfrak{g} = \mathfrak{gl}(V)$, or $\mathfrak{g}$ is any perfect (e.g., semisimple) subalgebra of $\mathfrak{gl}(V)$, then $\rho(X)$ is nilpotent for any representation of $\mathfrak{g} \ni X$ if and only if X is nilpotent as an automorphism of V . On the other hand, Engel's theorem shows that any subalgebra $\mathfrak{g} \subset \mathfrak{gl}(V)$ consisting of exclusively nilpotent automorphisms is a Lie algebra of strictly upper triangular matrices in some basis. These facts are frequently useful for identifying nil subalgebras in examples.

¹⁴For perfect Lie algebras, such as semisimple ones, the nilpotent radical coincides with the so-called *nilradical*, sometimes denoted $\text{nil}(\mathfrak{g})$ also, but it may be different more generally.

A.2. Reductive Lie algebras and gradings thereof. A Lie algebra is REDUCTIVE if it satisfies (1), (2), or (3) below.

Proposition (Theorem 2.20 of [10]). *The following are equivalent:*

- (1) $\text{nil}(\mathfrak{g}) = 0$.
- (2) $\mathfrak{g}$ admits a faithful, completely reducible representation.
- (3) There exists a non-degenerate trace-form B_ρ on $\mathfrak{g}$.

Moreover, for any Lie algebra $\mathfrak{g}$, the quotient $\mathfrak{g}/\text{nil}(\mathfrak{g})$ always satisfies these conditions, and is called the REDUCTIVE LEVI QUOTIENT of $\mathfrak{g}$.

Recall here that a TRACE-FORM B_ρ is defined by $B_\rho(X, Y) = \text{trace}(\rho_X \circ \rho_Y)$, for some representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, and this is always ad-invariant; B_{ad} is the KILLING FORM, which is non-degenerate when $\mathfrak{g}$ is semisimple (a direct sum of simple ideals) by Cartan's criterion [10, Proposition 2.19].

Theorem (Theorem 2.20 of [10]). *If $\mathfrak{g}$ is reductive, then:*

- (4) $\mathfrak{g} = Z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$ and this decomposition is orthogonal, for any non-degenerate trace-form.
- (5) $[\mathfrak{g}, \mathfrak{g}]$ is semisimple.
- (6) A derivation $D \in \text{End}(\mathfrak{g})$ is of the form $D(X) = [Y, X]$, for some $Y \in \mathfrak{g}$ (i.e., is inner) if and only if D vanishes on $Z(\mathfrak{g})$, in which case Y is uniquely determined by requiring $Y \in [\mathfrak{g}, \mathfrak{g}]$.

We will later need:

Lemma. Let $\mathfrak{g}$ be a graded reductive Lie algebra and B any ad-invariant symmetric bilinear form. Then:

- (7) With respect to B , we have $\mathfrak{g}_j \perp \mathfrak{g}_k$ whenever $j + k \neq 0$.
- (8) If B is the Killing-form, then its kernel $\mathfrak{g}^\perp$ is exactly $Z(\mathfrak{g})$.

If $Z(\mathfrak{g}) \subset \mathfrak{g}_0$, then all of the following also hold:

- (9) There exists a unique grading element $E \in \mathfrak{g}_0 \cap [\mathfrak{g}, \mathfrak{g}]$.
- (10) If $\mathfrak{g}^\perp \subset \mathfrak{g}_0$ (e.g., B is the Killing form) then $\mathfrak{g}_j$ and $\mathfrak{g}_{-j}$ are B -dual, for all $j \neq 0$, and in particular $|\mathfrak{g}| = m$ for some $m \geq 0$ (see Notation 3.1).
- (11) If B is non-degenerate, then $\mathfrak{g}_j$ and $\mathfrak{g}_{-j}$ are B -dual for all $j \in \mathbb{Z}$, and $|\mathfrak{g}| = m$, for some $m \geq 0$.

Proof. (7): For any $X \in \mathfrak{g}_j$ and $Y \in \mathfrak{g}_k$, we have $(j+k)B(X, Y) = B([E, X], Y) + B(X, [E, Y]) = 0$, by ad-invariance of B .

(8): Suppose that B is the Killing form. Then $\text{ad}_X = 0$ for any $X \in Z(\mathfrak{g})$, which shows $Z(\mathfrak{g}) \subset \mathfrak{g}^\perp$. To prove the reverse inclusion, note that for any $X, Y \in \mathfrak{g}$, the derivations ad_X , ad_Y , and hence $\text{ad}_X \circ \text{ad}_Y$, send $Z(\mathfrak{g})$ to 0 and $[\mathfrak{g}, \mathfrak{g}]$ to itself. It follows from (4) that $B(X, Y) = B_{\text{ss}}(X_{\text{ss}}, Y_{\text{ss}})$, where B_{ss} is the Killing form on $[\mathfrak{g}, \mathfrak{g}]$, while X_{ss} and Y_{ss} are the projections of X and Y along $Z(\mathfrak{g})$ onto $[\mathfrak{g}, \mathfrak{g}]$, which is semisimple, according to (5). In particular, B_{ss} is non-degenerate, and therefore $X \in \mathfrak{g}^\perp$ only if $X_{\text{ss}} = 0$, i.e., $X \in Z(\mathfrak{g})$.

(9): Define $D(X) = jX$, for $X \in \mathfrak{g}_j$. Then D is a derivation vanishing on $\mathfrak{g}_0 \supset Z(\mathfrak{g})$. By (6), there exists a unique $E \in [\mathfrak{g}, \mathfrak{g}]$ such that $D(X) = [E, X]$, and $E \in \mathfrak{g}_0$ because $[E, E] = 0$.

(10) & (11): Suppose $\mathfrak{g}^\perp \subset \mathfrak{g}_0$. Let $X \in \mathfrak{g}_j$ and suppose $B(X, Y) = 0$ for all $Y \in \mathfrak{g}_{-j}$. Then in fact, $B(X, Y) = 0$ for all $Y \in \mathfrak{g}$, by (7), i.e., $X \in \mathfrak{g}^\perp$. So (11) is clear. If $\mathfrak{g}^\perp \subset \mathfrak{g}_0$, then $X \in \mathfrak{g}_j \cap \mathfrak{g}_0$ and so (10) holds. $\square$

A.3. Parabolic subalgebras. Recall that a subalgebra $\mathfrak{p} \subset \mathfrak{g}$ is PARABOLIC if it contains the normalizer of a maximal nil subalgebra [10].

Theorem (Theorem 3.4 of [10]). *Let $\mathfrak{g}$ be reductive and fix an arbitrary non-degenerate trace-form. Then the following are equivalent:*

- (1) $\mathfrak{p} \subset \mathfrak{g}$ is parabolic.
- (2) $\mathfrak{p}^\perp \subset [\mathfrak{g}, \mathfrak{g}]$ and $\mathfrak{g}^\perp$ is nilpotent.
- (3) $\mathfrak{p}^\perp$ is a nil subalgebra of $\mathfrak{g}$.
- (4) $\mathfrak{p}^\perp \subset \mathfrak{p}$ and $N_{\mathfrak{g}}(\mathfrak{p}) = \mathfrak{p}$.
- (5) $\mathfrak{p}^\perp = \text{nil}(\mathfrak{p})$.

In particular, (5) shows that $\mathfrak{p}^\perp$ does not depend on the choice of trace-form. Evidently, the theorem continues to hold if “trace-form” is replaced by “non-zero scalar multiple of a trace-form”.

Proposition A.6 characterizes parabolics in terms of gradings.

Remark. If $\mathfrak{g}$ is reductive and the base field is algebraically closed, then the normalizers of minimal subalgebras coincide with the Borel (maximal solvable) subalgebras. However, Borel subalgebras play no role here.

A.4. Canonical filtrations. When $\mathfrak{g}^1 \subset \mathfrak{g}^0 \subset \mathfrak{g}$ and $\mathfrak{g}^1$ is an ideal of $\mathfrak{g}^0$ then we have the PRE-FILTRATION associated with $(\mathfrak{g}^0, \mathfrak{g}^1)$, introduced in 7.1.

Proposition. *Fix a non-degenerate trace-form on a reductive Lie algebra $\mathfrak{g}$ and let $\mathfrak{p} \subset \mathfrak{g}$ a parabolic subalgebra.*

- (1) *The pre-filtration of $\mathfrak{g}$ associated with $(\mathfrak{p}, \mathfrak{p}^\perp)$, which is independent of the choice of form, is well-defined and is even a bona fide Lie algebra filtration, with $|\mathfrak{g}| = m$ for some $m \geq 0$ (see Notation 3.1). This is the CANONICAL FILTRATION associated with $\mathfrak{p}$.*
- (2) *In the canonical filtration, $(\mathfrak{g}^j)^\perp = \mathfrak{g}^k$, whenever $j + k = 1$.*

Proof. Since $\mathfrak{p}$ is parabolic, $\mathfrak{p}^\perp$ is an ideal of $\mathfrak{p}$, by Theorem A.2. In particular Lemma 7.1 applies with $\mathfrak{g}^0 := \mathfrak{p}$ and $\mathfrak{g}^1 := \mathfrak{p}^\perp$, so that the pre-filtration associated with $(\mathfrak{p}, \mathfrak{p}^\perp)$ is well-defined. The triviality $(\mathfrak{g}^0)^\perp = \mathfrak{g}^1$ anchors a straightforward inductive argument that establishes (2). Since $\mathfrak{p}^\perp$ is nilpotent, there is $m \geq 0$ such that $\mathfrak{g}^m \neq 0$ but $\mathfrak{g}^{m+1} = 0$. In that case $\mathfrak{g}^{-m} = (\mathfrak{g}^{m+1})^\perp = \{0\}^\perp = \mathfrak{g}$, while $\mathfrak{g}^{-m+1} = (\mathfrak{g}^m)^\perp \neq \mathfrak{g}$. This shows the pre-filtration is a filtration and $|\mathfrak{g}| = m$. $\square$

Theorem (Proposition 3.9 of [10]). *Let $\mathfrak{g}$ be a reductive Lie algebra. Then, with respect to the canonical filtration associated with any parabolic subalgebra $\mathfrak{p} \subset \mathfrak{g}$, $\text{gr}(\mathfrak{g})$ is a reductive Lie algebra with centre lying in $\text{gr}(\mathfrak{g})_0$.*

While a proof of the theorem is given in [10], some details were not immediately obvious to us, and we accordingly provide them below.

Proof. Let B_ρ be a non-degenerate trace form on $\mathfrak{g}$. By (2), we have $\mathfrak{g}^j \perp \mathfrak{g}^k$, whenever $j + k = 1$, which allows us to drop B_ρ to an invariant quadratic form Q on $\text{gr}(\mathfrak{g})$ satisfying $Q(\text{gr}(\mathfrak{g}_j), \text{gr}(\mathfrak{g}_k)) = 0$ for $j + k \neq 0$ and $Q(\text{gr}_j(X), \text{gr}_{-j}(Y)) = B_\rho(X, Y)$ for $X \in \mathfrak{g}^j$ and $Y \in \mathfrak{g}^{-j}$, $j \in \mathbb{Z}$. The stronger condition (2) ensures Q is non-degenerate. To show $\text{gr}(\mathfrak{g})$ is reductive, it therefore suffices to show $Q = B_{\bar{\rho}}$ for some representation $\bar{\rho}$ of $\text{gr}(\mathfrak{g})$. If we can find a filtration of V that makes ρ into a *filtered* representation, i.e., such that

$$(3) \quad \rho_{\mathfrak{g}^j}(V^k) \subset V^{j+k},$$

for all $j, k \in \mathbb{Z}$, then we can take $\bar{\rho}$ to be the corresponding representation of $\text{gr}(\mathfrak{g})$ on $\text{gr}(V)$ [10, Proposition 2.14].

To construct such a filtration of V , first define $V^j = 0$, for $j > 0$, and let $V^0 = \{v \in V \mid \rho_{\mathfrak{g}^1}(v) = 0\}$. Now $V^0 \neq 0$, by Engel's theorem [10, Lemma 2.7], because $\mathfrak{g}^1 = \text{nil}(\mathfrak{p})$ acts nilpotently under ρ . For $j \leq -1$, we inductively define $V^j = \{v \in V \mid \rho_{\mathfrak{g}^1} v \subset V^{j+1}\}$. Since $\mathfrak{g}^1$ acts nilpotently on V , it acts nilpotently on every space V/V^k , $k \in \mathbb{Z}$, and so V^k is a proper subset of V^{k-1} by Engel's theorem, and $\cup_j V^j = V$, by finite-dimensionality.

By definition, (3) holds for $j = 1$ and all $k \in \mathbb{Z}$. For $j \geq 2$, we have $\mathfrak{g}^j = [\mathfrak{g}^1, \mathfrak{g}^{j-1}]$, so a simple induction shows that (3) holds for all $j \geq 2$ and $k \in \mathbb{Z}$. Finally, supposing $j \leq 0$, we have (3) if $k \geq -j + 1$, because $V^k = 0$ in that case. So, to deal with the $j \leq 0$ case, we proceed with an induction, supposing (3) holds whenever $k + j \geq m$, for some $m \geq 1$, and show it must also hold if $k + j \geq m - 1$. So, supposing, $X \in \mathfrak{g}^j$ and $v \in V^k$, with $j + k \geq m - 1$, we need to show $\rho_{X^j} v \in V^{j+k}$, which means showing, for any $X^1 \in \mathfrak{g}^1$, that $\rho_{X^1} \rho_{X^j} v \in V^{j+k+1}$. But by the induction hypothesis, we will have $\rho_{[X^1, X^j]} v \in V^{j+k+1}$, because $[X^1, X^j] \in \mathfrak{g}^{j+1}$ and $(j+1) + k \geq m$, while $\rho_{X^j} \rho_{X^1} v \in V^{j+k+1}$ because $\rho_{X^1} v \in V^{k+1}$ and $j + (k+1) \geq m$. So the desired result follows from the identity $\rho_{X^1} \rho_{X^j} v = \rho_{[X^1, X^j]} v + \rho_{X^j} \rho_{X^1} v$.

If an element of $\text{gr}(\mathfrak{g})$ is central, then its homogeneous components must be central. So suppose $\text{gr}_j(X) \neq 0$ is central, for some $X \in \mathfrak{g}^j$ and, initially, suppose $j \geq 1$. Then

$$(4) \quad \mathfrak{g}^j = [\mathfrak{g}^1, \mathfrak{g}^{j-1}].$$

In case $j \geq 2$, this follows from the definition of $\mathfrak{g}^j$. In case $j = 1$, this follows because $[\mathfrak{g}^1, \mathfrak{g}^0] = ([\mathfrak{p}^\perp, \mathfrak{p}]^\perp)^\perp = N_{\mathfrak{g}}(\mathfrak{p}^\perp)^\perp = N_{\mathfrak{g}}(\mathfrak{p})^\perp = \mathfrak{p}^\perp = \mathfrak{g}^1$. The fourth equality holds because parabolic subalgebras are self-normalizing (Theorem A.2(4)). Regarding the rest of the argument, note that ad-invariance of the trace-form

guarantees $N_{\mathfrak{g}}(\mathfrak{p}) \subset [\mathfrak{p}, \mathfrak{p}^\perp]^\perp = N_{\mathfrak{g}}(\mathfrak{p}^\perp)$ for *any* subspace $\mathfrak{p} \subset \mathfrak{g}$, so that the reverse inclusion $N_{\mathfrak{g}}(\mathfrak{p}^\perp) \subset N_{\mathfrak{g}}(\mathfrak{p})$ follows from $\mathfrak{p}^{\perp\perp} = \mathfrak{p}$, which we have by non-degeneracy. The point of (4) is that it shows $\mathrm{gr}(\mathfrak{g})_j \subset [\mathrm{gr}(\mathfrak{g}), \mathrm{gr}(\mathfrak{g})]$. By Theorem A.2(4), $\mathrm{gr}(\mathfrak{g})_j$ must intersect the centre of $\mathrm{gr}(\mathfrak{g})$ trivially, because $\mathrm{gr}(\mathfrak{g})$ is reductive, as we have already shown. So $\mathrm{gr}_j(X) = 0$, a contradiction, and therefore $j \leq 0$.

Since $\mathrm{gr}_j(X)$ is central, we have, in particular,

$$(5) \quad [X, \mathfrak{g}^1] \subset \mathfrak{g}^{j+2}.$$

If $j = -1$, then we conclude $X \in N_{\mathfrak{g}}(\mathfrak{g}^1) = N_{\mathfrak{g}}(\mathfrak{p}^\perp) = \mathfrak{p} = \mathfrak{g}^0$ by preceding arguments. But again this implies the contradiction, $\mathrm{gr}_j(X) = 0$. If $j \geq -2$, then (5) implies $X \in \mathfrak{g}^{j+1}$, by the definition of $\mathfrak{g}^{j+1}$, and again the disallowed $\mathrm{gr}_j(X) = 0$ follows. The only remaining possibility is $j = 0$, which proves the stated property of the centre. $\square$

Corollary. *Let $\mathfrak{g}$ be a reductive Lie algebra filtered canonically by some parabolic subalgebra $\mathfrak{p}$. Then the associated graded $\mathrm{gr}(\mathfrak{g})$ admits a unique grading element $E \in [\mathrm{gr}(\mathfrak{g}), \mathrm{gr}(\mathfrak{g})]$, so that $\mathfrak{g}$ and $\mathrm{gr}(\mathfrak{g})$ are isomorphic Lie algebras, and the parabolic $\mathfrak{p}$ contains $Z(\mathfrak{g})$.*

Proof. The theorem and Lemma A.2(9) establish the existence of the grading element E . According to arguments recalled in 10.1, any lift $\epsilon \in \mathfrak{p} = \mathfrak{g}^0$ of E defines a Lie algebroid isomorphism $\Phi_\epsilon: \mathrm{gr}(\mathfrak{g}) \rightarrow \mathfrak{g}$. Since $\mathrm{gr}(\mathfrak{g})_0$ contains the centre, by the theorem, so must $\mathfrak{p}$, because of the isomorphism. $\square$

A.5. Lifting trace-forms. Recall that a filtration of $\mathfrak{g}$ is a PARABOLIC FILTRATION if $\mathfrak{g}^0$ is a parabolic subalgebra and the filtration is the associated canonical filtration.

Proposition. *If $\mathfrak{g}$ is parabolically filtered, then each trace-form B on $\mathrm{gr}(\mathfrak{g})$ (or non-zero scalar multiple thereof) has a NATURAL LIFT $B^\wedge$ to a (scalar multiple of a) trace-form on $\mathfrak{g}$, non-degenerate if B is non-degenerate. Here naturality means that for any filtration-preserving isomorphism $q: \mathfrak{g} \rightarrow \mathfrak{h}$ between parabolically filtered reductive Lie algebras, we have*

$$(1) \quad q_*(B^\wedge) = (\mathrm{gr}_0(q)_*B)^\wedge,$$

where $q_*b(X, Y) := b(q^{-1}X, q^{-1}Y)$.

Proof. We define $B^\wedge = (\Phi_\epsilon)_*B$, where $\Phi_\epsilon: \mathrm{gr}(\mathfrak{g}) \rightarrow \mathfrak{g}$ is the splitting associated with some lift $\epsilon \in \mathfrak{g}^0$ of a grading element $E \in [\mathrm{gr}(\mathfrak{g}), \mathrm{gr}(\mathfrak{g})]$ of $\mathrm{gr}(\mathfrak{g})$, whose existence and uniqueness is guaranteed by Corollary A.4. To see that $B^\wedge$ does not depend on the choice of lift ϵ , we apply Lemma 10.1(2) and use the Ad-invariance of $B^\wedge$ under the representation of $G^1 = \exp(\mathfrak{g}^1)$, which follows from the ad-invariance of $B^\wedge$. If $B = B_\rho$, then it is not hard to see $B^\wedge = B_{\rho \circ \Phi_\epsilon^{-1}}$, so trace-forms are lifted to trace-forms. The general case of scalar multiples is a trivial extension.

Since $\mathrm{gr}_0(q): \mathrm{gr}(\mathfrak{g}) \rightarrow \mathrm{gr}(\mathfrak{h})$ is a Lie algebra isomorphism, $\mathrm{gr}_0(q)(E)$ is evidently the unique grading element of $\mathrm{gr}(\mathfrak{h})$ lying in $[\mathrm{gr}(\mathfrak{h}), \mathrm{gr}(\mathfrak{h})]$, which has $q(\epsilon)$ as a lift. So, by definition, we have

$$(2) \quad (\mathrm{gr}_0(q)_* B)^\wedge = (\Phi_{q(\epsilon)})_*(\mathrm{gr}_0(q)_* B).$$

To prove (1) it suffices to show it holds when evaluated on arbitrary pairs $X \in \mathfrak{g}_j$ and $Y \in \mathfrak{g}_k$, where we grade $\mathfrak{g}$ using ϵ . Unpacking the definition of the splittings Φ_ϵ and $\Phi_{q(\epsilon)}$, this follows readily from (2). $\square$

A.6. Parabolics in terms of gradings. According to the following, if $Z(\mathfrak{g}) \subset \mathfrak{g}_0$, then any grading of reductive $\mathfrak{g}$ immediately defines a parabolic subalgebra. However, the filtration corresponding to the grading need not be the canonical one defined by the parabolic:

Proposition. *Suppose $\mathfrak{g}$ is reductive. Then a subalgebra $\mathfrak{p}$ is parabolic if and only if there exists a grading of $\mathfrak{g}$ such that:*

$$(1) \quad Z(\mathfrak{g}) \subset \mathfrak{g}_0 \text{ and } \mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j.$$

In that case the corresponding filtration, defined by $\mathfrak{g}^j := \bigoplus_{i \geq j} \mathfrak{g}_i$, is the canonical filtration associated with $\mathfrak{p}$ if and only if one of the following equivalent conditions holds:

$$(2) \quad \mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \mathfrak{g}^3 \supset \cdots \text{ is the lower central series of } \mathfrak{g}^1.$$

$$(3) \quad \mathfrak{g}_1 \text{ generates } \bigoplus_{j \geq 1} \mathfrak{g}_j = \mathfrak{g}^1.$$

$$(4) \quad \mathfrak{g}_{-1} \text{ generates } \bigoplus_{j \leq -1} \mathfrak{g}_j.$$

For any parabolic $\mathfrak{p} \subset \mathfrak{g}$, a grading satisfying (1)–(4) exists.

Proof. The equivalence of (2) and (3) is not hard to see. When the ground field is $\mathbb{C}$, the equivalence of (3) and (4) is a known consequence of structure theory in that case (not developed here) but I don't have a more direct, field-agnostic proof yet.

Suppose $\mathfrak{p}$ is parabolic and we filter $\mathfrak{g}$ with the canonical filtration defined by $\mathfrak{p}$. Then Corollary A.4 allows us to identify $\mathfrak{g}$ with the associated graded $\mathrm{gr}(\mathfrak{g})$ using some lift of the grading element $E \in [\mathrm{gr}(\mathfrak{g}), \mathrm{gr}(\mathfrak{g})]$ and we have $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$. Any element of $Z(\mathfrak{g})$ must commute with $E \in \mathfrak{g}_0$, so $Z(\mathfrak{g}) \subset \mathfrak{p}$ already forces $Z(\mathfrak{g}) \subset \mathfrak{g}_0$.

Conversely, suppose $\mathfrak{g}$ is a graded reductive Lie algebra with $Z(\mathfrak{g}) \subset \mathfrak{g}_0$ and define $\mathfrak{p} := \bigoplus_{j \geq 0} \mathfrak{g}_j$. Fix an arbitrary non-degenerate trace-form. Then, to show that $\mathfrak{p}$ is parabolic, it suffices, according to Theorem A.3, to show that $\mathfrak{p}^\perp$ is nilpotent and $\mathfrak{p}^\perp \subset [\mathfrak{g}, \mathfrak{g}]$. Now suppose $j + k = 1$. Then, writing $\mathfrak{g}^j := \bigoplus_{j' \geq j} \mathfrak{g}_{j'}$, we claim

$$(5) \quad \mathfrak{g}^j = (\mathfrak{g}^k)^\perp.$$

Indeed, for any $j' \geq j$ and $k' \geq k$, we have $j' + k' \neq 0$, so that $\mathfrak{g}_{j'} \perp \mathfrak{g}_{k'}$, by Lemma A.2(7), and consequently $\mathfrak{g}^j \subset (\mathfrak{g}^k)^\perp$. But, counting dimensions, it's easy to see that $\mathfrak{g}^j$ and $\mathfrak{g}^k$ have complementary dimensions, if $\dim(\mathfrak{g}_i) = \dim(\mathfrak{g}_{-i})$ for

all i , which is true, by Lemma A.2(11). So (5) holds. In particular, taking $j = 1$ and $k = 0$, we have $\mathfrak{g}^1 = \mathfrak{g}^\perp$, which shows that $\mathfrak{p}^\perp$ is nilpotent. Also, as $\mathfrak{g}_0 \subset \mathfrak{p}$, we have $\mathfrak{p}^\perp \subset \mathfrak{g}_0^\perp \subset Z(\mathfrak{g})^\perp = [\mathfrak{g}, \mathfrak{g}]$, by Theorem A.2(4).

It remains to show the filtration defined by the grading is the canonical filtration associated with $\mathfrak{p}$, under the additional assumption (2) (say). As the canonical filtration also satisfies (5) for $j + k = 1$ (Proposition A.4(2)) it suffices to show the filtration components of degree j coincide, for $j \geq 0$. For $j = 0$, this is tautological, and we just established the $j = 1$ case. For the canonical filtration, $\mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \mathfrak{g}^3 \supset \dots$ is the lower central series, by definition, which finishes the proof. $\square$

A.7. Parabolic subalgebras are separating. The notion of a SEPARATING or INFINITESIMALLY SEPARATING subalgebra $\mathfrak{p} \subset \mathfrak{g}$ has been introduced in 5.5.

Proposition. *Every parabolic subalgebra $\mathfrak{p}$ of a reductive Lie algebra $\mathfrak{g}$ is separating, and hence also infinitesimally separating.*

A proof of this result was obtained after David Calderbank pointed out a proof for the infinitesimal case, which is an immediate corollary.

Proof. Suppose $\phi \in \text{Aut}(\mathfrak{g})$ fixes elements of $\mathfrak{p}$. By Proposition A.6, we may suppose $\mathfrak{g}$ is graded with $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$ and $Z(\mathfrak{g}) \subset \mathfrak{g}_0$. Supposing $j \geq 1$, it suffices to show ϕ fixes elements of $\mathfrak{g}_{-j}$. By uniqueness of the grading element E , we must have $\phi(E) = E$, which already shows that ϕ must map $\mathfrak{g}_{-j}$ to itself. Suppose $Y \in \mathfrak{g}_j$ is arbitrary, and let B denote the Killing form. Then as $\phi \in \text{Aut}(\mathfrak{g})$, we must have $B(\phi X, \phi Y) = B(X, Y)$, i.e., $B(\phi X, Y) = B(X, Y)$, because ϕ fixes elements of $\mathfrak{g}_j$, as $j \geq 1$. So, by Lemma A.2(10), we conclude that $\phi X = X$. $\square$

A.8. The infinitesimal order of a parabolic subalgebra. In 6.3 we defined the INFINITESIMAL ORDER of Lie algebra modulo a subalgebra. This section is dedicated to the proof of the following:

Theorem. *Let $\mathfrak{g}$ be a reductive Lie algebra and $\mathfrak{p} \subset \mathfrak{g}$ parabolic. Equip $\mathfrak{g}$ with the canonical filtration corresponding to $\mathfrak{p}$. If a non-zero ideal of $\text{gr}(\mathfrak{g})$ lies in $\text{gr}(\mathfrak{g})_0$, then the infinitesimal order of $\mathfrak{g}$ modulo $\mathfrak{p}$ is ∞ . Otherwise it is 2.*

So the order is 2, provided it is finite, but it cannot be finite unless $\mathfrak{g}$ is semisimple and $\mathfrak{g} \neq \mathfrak{p}$. It is always finite if $\mathfrak{g} \neq \mathfrak{p}$ is simple. All this follows immediately, because $\mathfrak{g} \cong \text{gr}(\mathfrak{g})$ (Corollary A.4).

Example (a semisimple Lie algebra with infinite order). Let $\mathfrak{g}$ be the direct sum $\mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$ with standard basis $\{H_\alpha, X_\alpha, X_{-\alpha}, H_\beta, X_\beta, X_{-\beta}\}$ and give $\mathfrak{g}$ the filtration whose non-zero components are $\mathfrak{g}_{-1} = \langle X_{-\alpha} \rangle$, $\mathfrak{g}_0 = \langle X_{-\beta}, H_\alpha, H_\beta, X_{-\beta} \rangle$, and $\mathfrak{g}_1 = \langle X_\alpha \rangle$. Then as $|\mathfrak{g}| = 1$, the filtration is the canonical filtration associated with $\mathfrak{p} = \mathfrak{g}_0 \oplus \mathfrak{g}_1$ (Proposition A.6) and $0 \oplus \mathfrak{sl}(2, \mathbb{C})$ is an ideal of $\mathfrak{g}$ lying in $\mathfrak{g}_0$.

Proving the theorem will require some preparations. Call a graded Lie algebra $\mathfrak{g}$ **LEFT-RIGHT SYMMETRIC** if $\dim \mathfrak{g}_j = \dim \mathfrak{g}_{-j}$ for all j . A graded *reductive* Lie algebra is always left-right symmetric, provided $Z(\mathfrak{g}) \subset \mathfrak{g}_0$ (Lemma A.2(10)), which holds in particular if $\mathfrak{g}$ is semisimple.

Lemma A. Let $\mathfrak{g}$ be a graded reductive Lie algebra with $Z(\mathfrak{g}) \subset \mathfrak{g}_0$. Then every ideal $I \subset \mathfrak{g}$ is of the form $I = \bigoplus_{j \in \mathbb{Z}} I_j$, where $I_j = I \cap \mathfrak{g}_j$. In particular, every such ideal inherits a grading from $\mathfrak{g}$, with respect to which it is left-right symmetric.

Proof. We give a simple extension of [12, Proposition 3.1.2(2), p. 214] to the reductive case. Since $Z(\mathfrak{g}) \subset \mathfrak{g}_0$, there exists a grading element E (Lemma A.2(9)), so that the filtration is just the decomposition of $\mathfrak{g}$ by eigenspaces of ad_E . It follows that the projection onto any one component $\mathfrak{g}_j$, $j \in \mathbb{Z}$, can be expressed as a polynomial in ad_E . But any such polynomial maps any ideal into itself. So, decomposing an arbitrary element $X \in I$ as a sum of homogeneous components, $X = X_m + \dots + X_{-m}$, each component X_j , being a projection of X onto $\mathfrak{g}_j$, is also in I . The first claim is established.

Since $\mathfrak{g}$ is reductive, every ideal is a direct sum of some subalgebra of $Z(\mathfrak{g})$, left-right symmetric because $Z(\mathfrak{g}) \subset \mathfrak{g}_0$, and a semisimple ideal of $\mathfrak{g}$, necessarily left-right symmetric, as already noted above. $\square$

Lemma B. Let $\mathfrak{g}$ be a reductive Lie algebra and $\mathfrak{p} \subset \mathfrak{g}$ a parabolic subalgebra, and give $\mathfrak{g}$ the canonical filtration associated with $\mathfrak{p}$. Then the pair $(\mathfrak{g}, \mathfrak{p})$ is effective, in the sense of 6.3, if and only if $\text{gr}(\mathfrak{g})$ contains no non-zero ideals lying in $\text{gr}(\mathfrak{g})_0$.

Proof of lemma. Identify $\mathfrak{g}$ with $\text{gr}(\mathfrak{g})$ using the canonical filtration defined by $\mathfrak{p}$ (Corollary A.4). Let $I \subset \mathfrak{g}$ be an ideal lying in $\mathfrak{p} \cong \bigoplus_{j \geq 0} \mathfrak{g}_j$. By Lemma A, I is left-right symmetric with respect to a filtration compatible with that of $\mathfrak{g}$, so that only $I \subset \mathfrak{g}_0$ is possible. The claim now follows from the definition of effectiveness. $\square$

Applying Proposition 6.3, we conclude that the Lie algebra $\mathfrak{g}$ in the theorem has finite infinitesimal order modulo $\mathfrak{p}$ if and only if $\text{gr}(\mathfrak{g})$ contains no non-zero ideals lying in $\text{gr}(\mathfrak{g})_0$. To compute the order in this case, where $\mathfrak{g}$ is necessarily semisimple, requires further preparations.

Proposition. Let $\mathfrak{g}$ be a graded semisimple Lie algebra such that:

- (1) $\mathfrak{g}_1$ generates $\bigoplus_{j \geq 1} \mathfrak{g}_j$, or one of the equivalent conditions in Proposition A.6 holds.
- (2) $\mathfrak{g}$ contains no non-zero ideals lying in $\mathfrak{g}_0$.

Then $[\mathfrak{g}_j, \mathfrak{g}_{\pm 1}] = \mathfrak{g}_{j \pm 1}$ whenever $|j| \leq |\mathfrak{g}|$.

Proof. To prove the special case $[\mathfrak{g}_{-1}, \mathfrak{g}_1] = \mathfrak{g}_0$ we follow [12, p. 214]. Let $I := [\mathfrak{g}_{-1}, \mathfrak{g}_1] \bigoplus_{j \neq 0} \mathfrak{g}_j$, which we claim is an ideal. Indeed, it suffices to show $[\mathfrak{g}_{-j}, \mathfrak{g}_j] \subset [\mathfrak{g}_{-1}, \mathfrak{g}_1]$ for any $j \geq 0$. But, we can get this by induction, for supposing this holds

for some specific j , we have $\mathfrak{g}_{j+1} = [\mathfrak{g}_j, \mathfrak{g}_1]$, by (1), and compute

$$\begin{aligned} [\mathfrak{g}_{-j-1}, \mathfrak{g}_{j+1}] &= [\mathfrak{g}_{-j-1}, [\mathfrak{g}_j, \mathfrak{g}_1]] \subset [\mathfrak{g}_j, [\mathfrak{g}_{-j-1}, \mathfrak{g}_1]] + [\mathfrak{g}_1, [\mathfrak{g}_j, \mathfrak{g}_{-j-1}]] \\ &\subset [\mathfrak{g}_j, \mathfrak{g}_{-j}] + [\mathfrak{g}_1, \mathfrak{g}_{-1}] \subset [\mathfrak{g}_1, \mathfrak{g}_{-1}]. \end{aligned}$$

Since $\mathfrak{g}$ is semisimple, I has a complementary ideal, which by Lemma A, lies in $\mathfrak{g}_0$, and is accordingly trivial, by (2). Therefore $[\mathfrak{g}_{-1}, \mathfrak{g}_1] = \mathfrak{g}_0$.

To prove the general case, we adapt a proof for simple $\mathfrak{g}$ given in [33, Lemma 3.2, p. 437]. We prove only that $[\mathfrak{g}_j, \mathfrak{g}_1] = \mathfrak{g}_{j+1}$, as $[\mathfrak{g}_j, \mathfrak{g}_{-1}] = \mathfrak{g}_{j-1}$ will follow from symmetry, as $\mathfrak{g}_1$ generates $\oplus_{j \geq 1} \mathfrak{g}_j$ if and only if $\mathfrak{g}_{-1}$ generates $\oplus_{j \leq -1} \mathfrak{g}_j$.

First, we suppose that $\mathfrak{g}$ is simple. Let $m = |\mathfrak{g}|$ and define $\mathfrak{a}_{-m} = \mathfrak{g}_{-m}$ and, inductively, $\mathfrak{a}_{j+1} = [\mathfrak{g}_1, \mathfrak{a}_j]$, $j \geq -m$. A simple induction using Jacobi's identity shows that $\mathfrak{a}_j$ is $\text{ad}(\mathfrak{g}_0)$ -invariant for all $j \geq -m$. Armed with this and the previously established result $[\mathfrak{g}_{-1}, \mathfrak{g}_1] = \mathfrak{g}_0$, a similar inductive argument shows that $[\mathfrak{g}_{-1}, \mathfrak{a}_j] \subset \mathfrak{a}_{j-1}$. It follows that $\mathfrak{a} := \oplus_{|j| \leq m} \mathfrak{a}_j$ is invariant under elements of $\text{ad}(\mathfrak{g}_0)$, $\text{ad}(\mathfrak{g}_{-1})$, and $\text{ad}(\mathfrak{g}_1)$ (this last, by construction). Since $\mathfrak{g}_1$ generates $\oplus_{j \geq 1} \mathfrak{g}_j$ (by assumption) and $\mathfrak{g}_{-1}$ generates $\oplus_{j \leq -1} \mathfrak{g}_j$ (by Proposition A.6), it follows that $\mathfrak{a}$ is an ideal, and so $\mathfrak{a} = \mathfrak{g}$, by simplicity. Therefore $[\mathfrak{g}_j, \mathfrak{g}_1] = \mathfrak{g}_{j+1}$ for all $j \geq -m$.

In the semisimple case, we can write $\mathfrak{g} = \oplus_{I \in \mathcal{I}} I$, for some finite set $\mathcal{I}$ of simple ideals. Each simple ideal gets a filtration compatible with that of $\mathfrak{g}$ (Lemma A) and by the result just established, $[I_j, I_1] = I_{j+1}$, $j \geq m$. Since elements from distinct simple ideals commute, $[\mathfrak{g}_j, \mathfrak{g}_1] = [\oplus_{I \in \mathcal{I}} I_j, \oplus_{I \in \mathcal{I}} I_1] = \oplus_{I \in \mathcal{I}} [I_j, I_1] = \oplus_{I \in \mathcal{I}} I_{j+1} = \mathfrak{g}_{j+1}$. $\square$

Frequently cited but weaker forms of the following corollary do not suffice for current purposes.

Corollary. *Let $\mathfrak{g}$ be a graded semisimple Lie algebra satisfying the conditions of the proposition, and let i, j be integers such that:*

- (3) i, j and $i + j$ are all less than or equal to $|\mathfrak{g}|$ in absolute value.
- (4) i or $i + j$ lies in $\{-1, +1\}$.

Then the only $X \in \mathfrak{g}_j$ for which $[\mathfrak{g}_i, X] = 0$ is 0.

Remark. In particular, taking $i = -1$ and $0 \leq j \leq |\mathfrak{g}|$, we see that $\text{ad}: \mathfrak{p} \rightarrow \mathfrak{gl}(\mathfrak{g}_{-1})$ will be a faithful representation of the parabolic subalgebra $\mathfrak{p} = \oplus_{j \geq 0} \mathfrak{g}_j$.

Proof. Let B be any non-degenerate trace-form (e.g., the Killing form). Then, under the stated hypothesis,

$$\begin{aligned} [\mathfrak{g}_i, X] = 0 &\Rightarrow B([\mathfrak{g}_i, X], \mathfrak{g}_{-i-j}) = 0 \Rightarrow B(X, [\mathfrak{g}_i, \mathfrak{g}_{-i-j}]) = 0 \\ &\Rightarrow B(X, \mathfrak{g}_{-j}) = 0 \Rightarrow X = 0. \end{aligned}$$

The third implication follows from the proposition, while the last from B -duality of $\mathfrak{g}_j$ and $\mathfrak{g}_{-j}$ (Lemma A.2(11)). $\square$

To finish the proof of the theorem, suppose $\mathfrak{g}$ is semisimple and that $\mathfrak{g}$ contains no non-zero ideal lying in $\text{gr}(\mathfrak{g})_0$, under the canonical filtration associated with $\mathfrak{p}$.

Then $m := |\mathfrak{g}| \geq 1$, for otherwise $\mathrm{gr}(\mathfrak{g})$ is an ideal of itself lying in $\mathrm{gr}(\mathfrak{g})_0 = \mathrm{gr}(\mathfrak{g})$. Identifying $\mathfrak{g}$ with $\mathrm{gr}(\mathfrak{g})$ using Corollary A.4, $\mathfrak{g}$ satisfies the hypotheses of the above proposition. We will need the following special cases of its corollary:

- (5) Whenever $0 \leq i \leq m-1$ and $X \in \mathfrak{g}_i$, $[\mathfrak{g}_{-i-1}, X] = 0 \Rightarrow X = 0$.
- (6) Whenever $X \in \mathfrak{g}_m$, $[\mathfrak{g}_{-m+1}, X] = 0 \Rightarrow X = 0$.
- (7) If $X \in \mathfrak{g}_1$ and $[\mathfrak{g}_{-1}, X] = 0$, then $X = 0$.

In the notation of 6.3, with $\mathfrak{h} := \mathfrak{p}$, we have $\mathfrak{a}^1(\mathfrak{g}) := \{X \in \mathfrak{p} \mid [\mathfrak{g}, X] \subset \mathfrak{p}\}$. If $X \in \mathfrak{a}^1(\mathfrak{g})$, we may write $X = X_0 + X_1 + \dots + X_m$, for some $X_j \in \mathfrak{g}_j$. If $0 \leq i \leq m-1$, then $[\mathfrak{g}_{-i-1}, X] \cap \mathfrak{g}_{-1} = [\mathfrak{g}_{-i-1}, X_i]$, which must accordingly vanish, as $\mathfrak{g}_{-1} \cap \mathfrak{p} = 0$. So, in fact, $X_i = 0$, by (5). Since $0 \leq i \leq m-1$ is arbitrary, this shows $\mathfrak{a}^1(\mathfrak{g}) \subset \mathfrak{g}_m$. The reverse inclusion is trivial.

From the above, we have $\mathfrak{a}^2(\mathfrak{g}) := \{X \in \mathfrak{g}_m \mid [\mathfrak{g}, X] \subset \mathfrak{g}_m\}$. Let $X \in \mathfrak{g}_m$. Then $[\mathfrak{g}_{-m+1}, X_m]$ lies in $\mathfrak{g}_1$ and must accordingly vanish, as $\mathfrak{g}_1 \cap \mathfrak{g}_m = 0$, if we additionally suppose $m \geq 2$. In that case, $X = 0$, by (6). On the other hand, if $m = 1$, $[\mathfrak{g}_{-1}, X]$ lies in $\mathfrak{g}_0$, and so must vanish, as $\mathfrak{g}_0 \cap \mathfrak{g}_m = 0$ in that case. By (7), we still conclude $X = 0$. So in any event, $\mathfrak{a}^2(\mathfrak{g}) = 0$, and whence the infinitesimal order is 2.

APPENDIX B. THE SPACE OF SPLITTINGS OF A FILTERED LIE ALGEBRA

In this section we prove Lemma 10.1, which applies to arbitrary graded Lie algebras, not just reductive ones. While a known result, we have given a formulation tailored for current purposes. For convenience, here is a restatement:

Lemma. Let $\mathfrak{g}$ be a filtered Lie algebra and let $\epsilon \in \mathfrak{g}^0$ be a lift of some grading element $E \in \mathrm{gr}(\mathfrak{g})$. Let G^0 be *any* Lie group integrating $\mathfrak{g}^0$, and let $G^1 \subset G^0$ be the connected Lie subgroup with Lie algebra $\mathfrak{g}^1$. Then:

- (1) Under the adjoint action, G^1 acts freely and transitively on the affine space $\epsilon + \mathfrak{g}^1 \subset \mathfrak{g}^0$, and consequently on E -splittings of $\mathfrak{g}$. In particular, G^1 is simply connected (i.e., $G^1 \cong \exp(\mathfrak{g}^1)$) so that the infinitesimal adjoint action of $\mathfrak{g}^1$ on $\mathfrak{g}$ has a unique globalization $\mathrm{Ad}: G^1 \rightarrow \mathrm{Aut}(\mathfrak{g})$ extending the adjoint action just mentioned.
- (2) For any $\epsilon_1, \epsilon_2 \in \epsilon + \mathfrak{g}^1$, we have $\Phi_{\epsilon_2} = \mathrm{Ad}_g \circ \Phi_{\epsilon_1}$, where $g \in G^1$ is the unique element for which $\epsilon_2 = \mathrm{Ad}_g \epsilon_1$.

Proof. Note first that the filtration property ensures that the action of $\mathfrak{g}^1$ on $\mathfrak{g}^0$ is filtration-preserving, and the corresponding action on $\mathrm{gr}(\mathfrak{g}^0)$ is trivial. The same must be true for adjoint action of G^1 , as G^1 is connected. In particular $\epsilon + \mathfrak{g}^1$ must be G^1 -invariant, for any lift $e \in \mathfrak{g}^0$ of E . Suppose that the action of G^1 on $\epsilon + \mathfrak{g}^1$ is infinitesimally free. Then, by a dimension count, this action is infinitesimally transitive, and so transitive, since G^1 is connected. Infinitesimal freeness means the stabilizer subgroup H for the action is discrete, so that $G^1 \rightarrow G^1/H \cong \epsilon + \mathfrak{g}^1$ is a covering space. Since $\epsilon + \mathfrak{g}^1$ is simply connected, H is trivial and the remaining claims in (1) follow.

To show that the action of G^1 on $e + \mathfrak{g}^1$ is infinitesimally free, for each $X \in \mathfrak{g}^1$, let $X^\dagger$ denote the infinitesimal generator of X , a vector field on $e + \mathfrak{g}^1$. We have $X^\dagger(e') = [X, e']$, for $e' \in e + \mathfrak{g}^1$. Supposing $X^\dagger(e') = 0$, write $X = \sum_{j>0} X_j$ for some $X_j \in \mathfrak{g}_j$, where $\mathfrak{g} = \oplus_j \mathfrak{g}_j$ is the E -splitting corresponding to e' . Then we conclude that $\sum_{j>0} jX_j = 0$, from which it follows that $X = 0$.

To prove (2), observe that Φ_e is characterized among all isomorphisms $\text{gr}(\mathfrak{g}) \rightarrow \mathfrak{g}$ by the requirement

$$[\Phi_e(X), e] = j\Phi_e(X) \quad \text{for all } X \in \text{gr}_j(\mathfrak{g}), j \in \mathbb{Z}.$$

Taking $e = e_1$ and applying Ad_g to both sides, we obtain $[\text{Ad}_g \Phi_{e_1} X, \text{Ad}_g e_1] = j \text{Ad}_g e_1$, i.e.,

$$[(\text{Ad}_g \circ \Phi_{e_1})X, e_2] = j e_2.$$

So, by the aforementioned characterization, we conclude $\text{Ad}_g \circ \Phi_{e_1} = \Phi_{e_2}$. $\square$

APPENDIX C. CARTAN CONNECTIONS FROM REPRESENTATIONS

In this section we prove 4.1, restated here for convenience:

Proposition. *Suppose $\mathcal{G}$ is a Lie groupoid with a transitive Lie algebroid A . Then for every representation ρ of $\mathcal{G}$ on A , extending the canonical representation of $\mathcal{G}$ on the anchor kernel, there exists a unique Cartan connection D on $\mathcal{G}$ such that $\bar{D} = \rho$.*

First note that the kernel of the canonical projection $J^1\mathcal{G} \rightarrow \mathcal{G}$ is a totally intransitive Lie groupoid whose form we can show is independent of the particular integration $\mathcal{G}$ of A . This kernel is denoted $\text{Aut}(TM, A)$ and described further in Section 13. For present purposes, the following properties suffice:

Lemma. *If the Lie algebroid A of a Lie groupoid $\mathcal{G}$ is transitive, then:*

- (1) *The adjoint representation of $J^1\mathcal{G}$ on A restricts to representation of $\text{Aut}(TM, A) \subset J^1\mathcal{G}$ that is faithful.*
- (2) *For any $m \in M$ and each linear map $L: A|_m \rightarrow A|_m$, restricting to the identity on the anchor kernel, and whose corresponding factoring $T_m M \rightarrow T_m M$ through the anchor $\#: A \rightarrow TM$ is an isomorphism, there exists a unique $\psi \in \text{Aut}(TM, A)|_m$ such that $\text{Ad}_\psi = L$.*

Proof of lemma. Part (1) follows from Proposition 13.3, while part (2) follows from 13.1(2) and transitivity. $\square$

Proof of proposition. We seek a Lie groupoid morphism $S: \mathcal{G} \rightarrow J^1\mathcal{G}$ that is a right-inverse for the projection $\pi: J^1\mathcal{G} \rightarrow \mathcal{G}$ and satisfies

$$(3) \quad \text{Ad}_{S(g)} = \rho_g \quad g \in \mathcal{G}.$$

To that end, let $\tilde{S}: U \rightarrow \pi^{-1}(U)$ be any smooth, local, right inverse for π , defined on some open neighborhood $U \subset \mathcal{G}$ (not necessarily respecting groupoid multiplication). Then, for each arrow $g \in U$ beginning at some $m \in M$, $L = \text{Ad}_{\tilde{S}(g)}^{-1} \circ \rho_g$

satisfies the condition in (2), because ρ extends the adjoint representation of $\mathcal{G}$ on the anchor kernel of A . Accordingly, there exists unique $\psi(g) \in \text{Aut}(TM, A)$ such that $\text{Ad}_{\psi(g)} = \text{Ad}_{\tilde{S}(g)}^{-1} \circ \rho_g$, so that (3) holds if we define $S(g) = \tilde{S}(g)\psi(g)$. It is not hard to see that $S: U \rightarrow J^1\mathcal{G}$ is smooth. Moreover, from the uniqueness, it is clear that $S(g)$ does not actually depend on the choice of local right inverse $\tilde{S}$, which means we can collate local definitions of S as just described into a global map $S: \mathcal{G} \rightarrow J^1\mathcal{G}$. From (3), we will have $\text{Ad}_{S(g)S(h)} = \text{Ad}_{S(gh)}$ for all multipliable $g, h \in \mathcal{G}$. Using (1) it is not hard to see that $S(gh) = S(g)S(h)$ must follow, because $S(g)S(h)$ and $S(gh)$ share a common image under the projection $J^1\mathcal{G} \rightarrow \mathcal{G}$, namely gh . A similar argument shows that S is uniquely determined by requirement (3). □

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